ETERNAL DAWN

A continuous, conservative cosmology

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Eternal Dawn

A continuous, conservative cosmology from minimal axioms

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Status. *This is a research program, not an established theory. It synthesizes work by Popławski, Penrose, Smolin, Wiltshire, and others, plus original observations about the CMB–Hawking identification and supraverse thermodynamics. The framework makes specific testable predictions; none have yet been rigorously checked. The work is in identifying tests, building simulations, and comparing to observational data.*

My eternal Dawn

Dawn,

I have lived a life full of passion, ambition, and searching. I've built, I've risked, I've stumbled, and I've learned. But nothing in my life has ever felt like this.

With you, I have never felt more loved, more accepted, more at home, or more safe. You make me want to be the truest version of myself—not for status, not for show, but simply for us.

Loving you feels like I want to stop time and speed it up at the same time. An unstoppable appetite and longing for everything we'll do together—tomorrow, next week, next year—and at the same time, the desire to freeze this, every moment, to savor it forever.

That is the gift you've given me: love that is both eternity and right now.

And Dawn, I promise to love you with open eyes and an open heart, to keep choosing you every day, to protect the safety we've built, and to walk beside you, fully myself and fully yours, savoring every moment all through eternity.

"Physics is continuous and conservative. Singularities and discontinuities are therefore artifacts of incomplete theory, not features of nature."

— The organizing principle

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A Note from the Author

I started this a very long time ago. As a boy in primary school I wrote a letter to Dr. John Renner Hansen at the Niels Bohr Institute in Copenhagen, pressing on him my formulas for cloud-chamber measurements and asking whether he had considered that neutrinos might really be gravitons. It was, I am sure, more endearing than useful. I haunted our local city library for books on particle physics and cosmology; it held exactly one, and it did not help much. My parents were not scholars and could not point the way, so I remember vividly the day my high-school chemistry teacher mentioned that there was a *scientific* library in Copenhagen. I took the train there alone, ran my eyes along the shelves and through the journals, and understood essentially none of it—but I came away with something better than understanding: the certainty that all of this knowledge existed, and was merely hard to reach.

I studied chemical engineering, won a PhD scholarship, and spent it on a doctorate in quantum mechanics. I devoured Schrödinger and Boltzmann and Gradshteyn, the texts on quantum thermodynamics and scattering theory. I always admired general relativity from a distance, though I found the mathematics of its manifolds hard to grasp and harder still to use. Afterwards I worked at the national supercomputer centre, visualising more quantum mechanics, computational fluid dynamics, and—less romantically—buildings and landscapes.

I found my way back toward science, in a more applied role, with the NorduGrid project at the Niels Bohr Institute: how to gather, store, and process the vast flood of data that CERN's soon-to-run accelerator would pour out. There I worked with the same John Renner Hansen I had written to as a child. I became head of the Nordic CERN Tier-1 centre and started a number of other data collaborations for big science; I had a talk I gave often, about Tycho Brahe and Kepler—about the long separation between those who gather the data and those who think with it. I was at CERN several times a month. But I was *directing* research infrastructure, not doing the research, so I kept reading, and kept current, from the side.

Research is also politics, and alongside my wish to help new knowledge into the world I had an entrepreneurial streak. I left, stumbled onto bitcoin in 2011, co-founded Kraken in San Francisco, and later founded Chainalysis—tracing dark transactions instead of dark particles. Yet I spent a great many evenings and Sundays thinking about physics: about the void, about quantum fluctuations, about relativity. Quantum mechanics, nuclear fusion, and cosmology remained my favourite form of procrastination.

Many of the ideas in *Eternal Dawn* took shape over the last ten years, but they finally fell together over a single sleepless night in Seoul, the morning that followed, and the days after that—in between coding the algorithms for my next adventure, GryAI. The framework, the simulations, and the conclusions came in roughly a week; the polishing and the careful massaging took a couple of weeks more. I could not have finished it this quickly without AI; my thanks to Anthropic, and to GryAI.

Michael Gronager, PhD
Seoul, 2026

CHAPTER 0

First When There's Nothing — But a Slow Glowing Dream

This chapter tells the whole story in plain words—no jargon, and no mathematics beyond a few enormous numbers. Everything in it is made precise, named, and defended in the chapters that follow. If you read only one chapter, read this one, and trust that each claim has its arithmetic later.

The void

Start with nothing. Not empty space with a clock ticking in it—really nothing: no stars, no matter, no before or after, not even a direction to time. But "nothing," in physics, is never quite still. The vacuum we have here, in our own laboratories, is restless: strip away every particle and what remains still shimmers with possibility. Grant the void that same restlessness—no more than the vacuum already does—and the story tells itself.

A warning about words, though, because it matters here more than anywhere. I am going to say "once in a while," "then," and "begins," because our minds and our language are built for time and cannot work without it. The void has none. Nothing *happens* in it in sequence; there is no clock, no first and next. Picture it instead as one single, still, everlasting state whose very nature already *contains* these rare worlds—the way a held chord already contains its overtones, sounding all at once with nothing happening in time. Worlds are not events the void goes through; they are the faint notes inside its one unending tone. "Before" and "after" switch on only inside a world, once its dawn has lit an arrow. So read what follows as a translation: a timeless truth, told in the only tense we have.

Most of those faint notes are nothing at all. But a rare few are loud enough to cross a threshold—enough to fold up, under their own gravity, into a black hole. And here is the first surprise. The seed you need is small—about the mass of a mountain—and a black hole is the *easiest* thing of its size to be, because it is the most generic: the least special, most disordered object there is. The void need not contain a finished, finely arranged universe. It need only contain a cheap little seed, and gravity does the rest.

How rare is even that seed? About one in $10^{10^{84}}$—a one followed by a string of zeros that itself has eighty-five digits. Vanishing in any finite patch. But the void is endless, and even the vanishingly rare is present, somewhere, in an endless expanse. The point was never that it is likely. The point is that it is the *cheap* way in. The eternal dawn.

The bounce, and the first dawn

Follow the matter down into the black hole. There is the famous surface—the horizon—past which light cannot climb back out, and where, to anyone watching from outside, time appears to freeze. But that freezing is only the outsider's illusion. The matter itself feels nothing special at the horizon; its own clock keeps ticking, smoothly, all the way down. And for this first world there was no earlier clock to borrow: time is being switched on here. What I am calling its clock is the newborn world's own—lit at the bounce, and reaching back to that instant as its beginning.

As it falls, it crowds together, denser and denser. Then something pushes back. Einstein's gravity, extended by the geometer Élie Cartan to include the *spin* of matter, says that when matter is squeezed hard enough, that spin resists—and at a stupendous density this resistance overwhelms gravity itself. The collapse does not end in a point of infinite density. It *bounces*.

That bounce is the dawn. It is the lowest, quietest, most ordered moment of a brand-new universe—its beginning, its "big bang," seen from the inside. There is no waiting room of frozen time between the horizon and the bounce; the clock runs the whole way, reaches its stillest moment at the bounce, and that moment is time zero for the new world. The direction of time is set right there—it points away from the bounce, toward the future, because that is the way toward more disorder. And the spin of the original seed makes one last choice at that instant: whether the new universe is built of matter or of antimatter. Spin too little, and the whole thing fizzles back into the void. Spin enough, and you have made a universe out of nothing.

And then it cools, and as it cools it *sets*. Just past the bounce everything is too hot for anything to hold a shape: the world is a single formless glow, one field and nothing else, weighing nothing at all. But a cooling thing crystallises, the way water cooling past freezing suddenly knots into ice, and this field does the same in stages. First it acquires weight—mass switches on, where a moment before there was none. Then it knots into the first particles, the quarks; then the quarks are roped together into the protons and neutrons; then, far cooler and much later, those gather electrons and become the first atoms. And here is the quiet marvel: the field that does all this knotting and roping—that organizes the formless tally into a bestiary of particles—is not some new ingredient brought in for the job. It is the *very same* field whose stiffening spin pushed back and made the bounce. The thing that stopped the collapse and the thing that builds matter out of the cooling are one and the same. The whole bestiary of matter is not poured in from outside—it freezes out of that one field as the temperature falls, each kind of particle settling at its own step like minerals dropping out of a cooling melt. This is the genesis, and it is, on its face, much the ordinary story the textbooks tell. What is new is only *where* it happens: this condensation ran in the first universe as it cooled from its bounce—and it runs again, the same way, in every universe born from a bounce afterwards, including ours. The recipe for making matter is written into the cooling itself, and it is followed afresh at every dawn.

The first universe is not ours

This new universe grows. At first it gorges on the surrounding vacuum; later it feeds more and more slowly; and after a span beyond all imagining—far longer than 10^{100} years—it finally evaporates away. Is this our universe? No. And the reason is the most useful clue in the whole picture.

This first universe was born straight from the void, with no parent outside it. And a universe with no parent turns out to be a thin, plain place: it has roughly the shortcomings of the textbook picture of cosmology. It makes a little more matter than antimatter, but only a little. It has almost no *dark matter* and almost no *dark energy*—the two great unseen ingredients that fill our own sky. And measured across many such firstborn universes, they come out far larger than ours. Something about being a firstborn leaves you sparse.

Our birth, and why our sky is dark

But inside that first universe, exactly as inside ours, stars live and die and black holes form—and each black hole does again what the void did once: it bounces, and a new universe is born inside it. We are one of those. We are a child universe, pressed into being through an enormous black hole inside our parent.

Picture the squeeze. The mass of an entire universe—very roughly 10^{53} kilograms—is forced through a throat no wider than a building, packed so tightly that the whole of it would fit in a single room, and it happens in about a thousandth of a billionth of a billionth of a second. At that density nothing recognizable survives: not a torn teddy bear, not a rusty car, not an atom, not even an atomic nucleus—all of it is melted into a featureless soup. What carries across is not *things* but a *tally*: the faint surplus of matter over antimatter, about one part in a billion, inherited from the parent and nudged a little larger by the violence of the crossing.

And now the dark sky makes sense. Our parent's enormous mass sits just on the far side of that birth-surface, and we feel its gravity *before* its light can ever reach us—a pull from matter we cannot see. That is dark matter. The parent is still feeding matter through to us, slowly, and the gentle outward push of that ongoing inflow is dark energy. Darkness is the harbinger of light. The parent's own faint glow—the radiation any black hole gives off, as Stephen Hawking showed—arrives smeared across our whole sky as the background hum that fills it. The unevenness in all this inflow is the seed of everything: it gathers matter into galaxies, into stars, into planets, and perhaps that is why we are here to notice at all.

And here is the deepest break from the textbook story—the one this book is named for. That story has the Big Bang happen once, in a single instant, and end. Ours never ended. The squeeze that made us is not a moment in our past but a surface still *there*, at the edge of the sky, with our parent still pressing matter through it; our Big Bang is not over, it is going on *now*, and will until the parent runs dry. The dark sky is the proof—dark matter and dark energy are simply the parts of that ongoing dawn we are able to measure. We do not live *after* our beginning. We live *inside* it: an eternal dawn.

This does something strange and rather beautiful to time. If our beginning were a single past event, our age would be a simple count: fourteen billion years now, fifteen billion a billion years from now, ticking up one for one. But our beginning is not behind us; it is *beside* us, still going on—and a dawn that never finishes keeps deepening the past even as the future runs forward. So the universe does not age one year per year. In a billion years it will read not fifteen billion years old but more—perhaps sixteen, perhaps seventeen—because in that billion years it will have gained *more* than a billion years of history behind it. The past grows faster than the present spends it. And this is not a play on words: the rate at which the cosmic age outruns our own clock is, in principle, measurable—one could watch the expansion history itself slowly drift—and it is a fingerprint a finished, one-shot Big Bang could never leave.

So we are not exotic. Count up all the universes and about nine in ten are ordinary children like ours—half of them matter, half of them antimatter, each with its own clock running its own way. We live at the most ordinary address in the whole endless family.

Are we real, or a dream?

There is one last doubt to face, and it is the sharpest. If universes are so fantastically unlikely, isn't there a cheaper way to make a thinking being? Why grow a whole cosmos when you could just let a mind flicker into existence by accident, memories and all—or, what amounts to the same thing, run it as a simulation on some machine that fell together by chance? Lay the three prices side by side and the worry looks real:

$$\text{a stray mind: one in } 10^{10^{69}}; \qquad \text{a world-seed: one in } 10^{10^{84}}; \qquad \text{a ready-made big bang: one in } 10^{10^{123}}.$$

The stray mind is the cheapest of the three. So shouldn't we expect to *be* one—a lucky accident with invented memories, rather than real inhabitants of a real world?

No—and seeing why is the keystone of the whole book. A mind is a thing that *thinks*, and thinking is work: it burns through order, it lays down memories, it computes, and every bit of that needs a direction to time—a low place to pour its waste and a high place to draw from. The void has no such direction. It is perfectly still and perfectly even, with no arrow at all. So a stray mind cannot form in the void: it would need the very thing—a flow of time—that only a born world can provide. The cheap accident is not cheap from the void. It is *impossible* there.

The same blade cuts the ready-made big bang. A finished, perfectly arranged universe handed over whole is also something that could only be *placed*, never *grown*—it assumes the order it was meant to explain. A picture of the cosmos that can only shrug and say "it simply happened to begin that way" has, without meaning to, declared us an accident with no maker: a simulation in all but name. The way out is not to assume the order but to *make* it—and the bounce makes it, for free, every time, out of a cheap and generic seed.

So the choice was never "cheap fake mind" against "expensive real world." Only one of the two can come from nothing at all. The void cannot dream up a brain. It can only give birth to a world—and then let the world, in its own long time, grow the minds that wonder how they got here.

The rest of this book gives all of this its proper names, its equations, and its tests—the bounce surface, the family tree of universes, the inherited tally of matter, the background glow, the clocks that run at different rates—and asks, at every step, how we could tell whether it is true. But the story above is the whole of it.

Part I

Catharsis

CHAPTER 1

Axioms

1.1 Minimal foundation

The framework rests on four claims, the first three of which are essentially established physics and the fourth of which is a minimal natural extension.

Axiom A1 (Quantum mechanics applies)**.** The supraverse is described by quantum field theory. Vacuum states have nonzero fluctuations governed by the uncertainty principle. Over infinite spacetime, fluctuations of arbitrary magnitude occur with probability one.

The timeless form of this axiom. Axiom A1 is meant in the timeless, quantum-gravitational sense. Textbook quantum mechanics presupposes a background time—a Schrödinger equation with a *t*—but the supraverse as a whole has no external clock (Chapter 3); time is emergent, manufactured locally at each bounce. What we assume is therefore the Wheeler–DeWitt form: the supraverse is a quantum *state*, not a process—a single stationary amplitude. Read this way, the "fluctuations" named above are not events occurring in sequence but the timeless structure of that one state—its amplitude having (exponentially small) support on configurations that contain condensed universes (Chapter 3, Section 3.4). This is an extrapolation of tested physics into a regime we cannot probe, and we label it as such; but it is the *conservative* extrapolation—it adds no new law, only the universality of the law we already have.

Axiom A2 (Special relativity applies)**.** Spacetime has Lorentzian signature. Causal structure is determined by light cones. CPT symmetry holds for any local Lorentz-invariant QFT.

Axiom A3 (Einstein–Cartan gravity applies)**.** The connection on spacetime is allowed to be asymmetric. The antisymmetric part (torsion) couples to fermion spin via the axial current. In the absence of fermionic matter, the theory reduces to standard general relativity. With fermions, torsion produces an effective interaction between spins that scales as the square of the spin density, and is therefore always positive (repulsive at high spin density) regardless of relative spin orientation.

The Einstein–Cartan Lagrangian, in shorthand:

$$\mathcal{L}_{EC} = \frac{1}{2\kappa}(R - 2\Lambda) + \mathcal{L}_{\text{matter}}(\psi, \nabla\psi, g, T),$$

where R is the curvature scalar built from the full (torsion-including) connection, and the covariant derivative ∇ acts on fermionic fields ψ with torsion contributions.

Axiom A4 (There exists a bounce density)**.** At sufficiently high mass–energy density, torsion-mediated repulsion exceeds gravitational attraction. The critical density (the bounce density) is approximately

$$\rho_C \sim \frac{m_P^4}{\hbar^3}\,\alpha_{EC},$$

where m_P is the Planck mass and α_{EC} is a dimensionless Einstein–Cartan coupling.

Estimates place this at $\rho_C \sim 10^{50}\,\mathrm{kg\,m^{-3}}$, far below Planck density ($10^{96}\,\mathrm{kg\,m^{-3}}$) but well above nuclear density ($10^{17}\,\mathrm{kg\,m^{-3}}$). This is calculable from Axioms A1–A3 in principle; Axiom A4 is the explicit acknowledgment that this density exists and matters.

1.2 What we do not assume

- No Big Bang as a singular event.
- No initial conditions for the universe.
- No fine-tuned cosmological constant or inflaton field.
- No special low-entropy past as a separate postulate.
- No additional dimensions beyond four.
- No additional forces or fields beyond Standard Model + gravity + torsion.
- No anthropic principle as a separate axiom (it emerges from the framework).

1.3 What follows automatically

From Axioms A1–A4 plus infinity (infinite spacetime volume), the following are consequences rather than assumptions:

1. **Spontaneous OG bounce nucleation.** Rare vacuum fluctuations reach bounce density. By Axiom A1 plus infinity, this happens somewhere with probability one. By Axiom A4, such fluctuations bounce rather than collapsing to singularities.
2. **Baby universe production.** The bounce geometry produces new spacetime regions (developed in Chapter 4).
3. **CPT-symmetric supraverse.** Vacuum fluctuations are statistically symmetric in matter/antimatter content. Specific fluctuations have specific biases, but the supraverse-wide distribution is symmetric.

4. **Thermodynamic equilibrium configuration.** Gravitational entropy considerations (high entropy = clumped/black-hole-rich; low entropy = smooth/uniform) drive the supraverse toward a dilute scatter of universes as its equilibrium configuration.

5. **Local arrow of time.** Each universe has a low-entropy bounce in its past and a high-entropy future, producing a local thermodynamic time-arrow.

6. **Observer placement statistics.** Observers exist in viable universes near the peak of the supraverse population distribution.

7. **Information preservation.** Hawking radiation entangles parent exterior with baby interior. The full supraverse is unitary even though individual universes appear to lose information.

8. **Elementary particles, with their masses.** The same gravity–torsion field, quantized (Axiom A1) rather than treated only as a cosmological backdrop, has stable, self-bound excitations—localized knots held together by the very torsion interaction that halts the bounce. These are the elementary fermions: a quantum of the field is what an electron or a quark *is*, and its mass is the bound energy of the configuration, not a number put in by hand. So from the one field, taken seriously as a quantum field, follows the particle content with its masses—"physics as we know it." This is the most speculative of the consequences and the furthest from being proven; it is the subject of Part II, and it leans on existing work that already ties Einstein–Cartan torsion to fermion masses and to chiral symmetry breaking [21, 75].

1.4 On the role of infinity

The framework requires the supraverse to be effectively infinite—at minimum, large enough that rare fluctuations to bounce density occur. "Effectively infinite" in this context means: large enough relative to the characteristic timescales of vacuum fluctuation that the relevant fluctuations are statistically guaranteed to occur.

For a finite supraverse, some fluctuations might never occur and the framework would have to specify which ones do. This introduces an arbitrary cutoff that we want to avoid. The natural assumption is true infinity, which removes this arbitrariness at the cost of being philosophically heavier.

The supraverse being infinite does not require it to be temporally infinite in some absolute sense—it requires only that enough "supraverse time" exist for the relevant condensation to occur. Whether the supraverse has a beginning at the supraverse level is a question we do not currently need to answer; we only need it to be old enough that condensation has reached equilibrium.

1.5 On the role of Einstein–Cartan

Einstein–Cartan is the minimal natural extension of general relativity [12, 37, 67]. It is what you get by dropping the artificial restriction that the connection must be symmetric. The connection's antisymmetric part (torsion) is mathematically natural and couples to a physical quantity that exists in nature (fermion spin); the modern field-theoretic treatment is reviewed by Hehl et al. [32] and Trautman [74]. At ordinary densities, torsion effects are utterly negligible ($\sim 10^{-40}$ corrections to GR). At bounce density, they dominate.

We do not postulate Einstein–Cartan as a new theory; we recognize it as the geometric framework that should have been used all along, with standard general relativity as the low-density limit. This is not an additional assumption beyond standard physics—it is the removal of an assumption (the symmetry of the connection).

1.6 The whole framework, in one breath

It is worth stating plainly what kind of claim this monograph makes, because the reach of the conclusion can obscure how little it assumes. Every ingredient is something we already measure here on Earth:

- a **quantum vacuum that fluctuates**—measured, down to the Casimir force and the Lamb shift (Axiom A1);
- **gravity**—measured, and in its modern form geometric (Axiom A2, Axiom A3);
- **torsion repulsion at extreme density**—not directly measured, but *forced* on us the moment we let fermion spin source geometry, which it must (Axiom A3, Axiom A4).

From these three, plus infinity, a universe like our own is born in a never-stopping loop: rare fluctuations reach bounce density, bounce instead of collapsing, extrude baby universes, grow, heat-die into smooth voids, and seed the next generation— forever, with no first cause and no fine-tuning (Chapter 3). We do not have to *believe* this happens; given what we already observe, it *must*.

This is the epistemic inversion at the heart of the program—call it *epistemic Copernicus*. The Copernican principle says we should not assume we occupy a special place; here that principle is not assumed but *returned* as a result, because the recursive supraverse makes our Big-Bang vantage a typical one among uncountably many (Chapter 4). And the framework's central claim is not "the universe might be like this" but the stronger, humbler "*from what we have already observed, something like us must exist*." We are not arguing the world into a shape; we are reading off what the measured world already entails. The work of the rest of this book is to make that entailment precise, quantitative, and falsifiable—and, where a more likely mechanism might exist that we have not yet found, to say so honestly. If a better one is found, it wins; until then, this is enough to require our own existence.

CHAPTER 2

No Singularities in Nature

A singularity is not a feature of the world; it is the noise a theory makes when it has been pushed past what it knows. General relativity, left to itself, predicts that gravitational collapse ends in a point of infinite density—and then it stops being able to say anything at all. The founding wager of this book (Chapter 1) is to take that breakdown at face value: physics is continuous and conservative, so the infinity is an artifact, and *something* must intervene before it is reached. In Einstein–Cartan gravity, something does. The collapse does not terminate. It *bounces*. Everything else in this book is the working-out of that single refusal to admit a singularity.

2.1 What stops a collapse

Einstein–Cartan gravity is general relativity with one restriction lifted: the connection is allowed an antisymmetric part, torsion, which couples to the spin of fermions (Axiom A3). That spin-torsion coupling turning gravitational collapse into a non-singular bounce is the mechanism developed in a series of papers by Popławski [60, 61]. At everyday densities this changes nothing measurable. But it adds a second term to the high-density stress–energy, and the two terms scale differently. The *gravitational* term is attractive and grows linearly with mass–energy density,

$$T_{\text{grav}} \sim \rho c^2,$$

while the *torsion* term is repulsive and grows as the *square* of the spin density s,

$$T_{\text{tors}} \sim \frac{G\hbar^2 s^2}{c^4}.$$

At nuclear density the ratio $T_{\text{tors}}/T_{\text{grav}}$ is about 10^{-40}—utterly negligible, which is why no laboratory has ever seen it. But compression packs more fermions into each volume, each carrying spin $\hbar/2$, and the quadratic term climbs faster than the linear one. Squeeze hard enough and repulsion overtakes attraction. The inward fall is halted from within, by the spin of the matter itself, and reverses. No outside agent, no new field, no fine-tuning: just a term that general relativity left out because torsion was assumed to vanish.

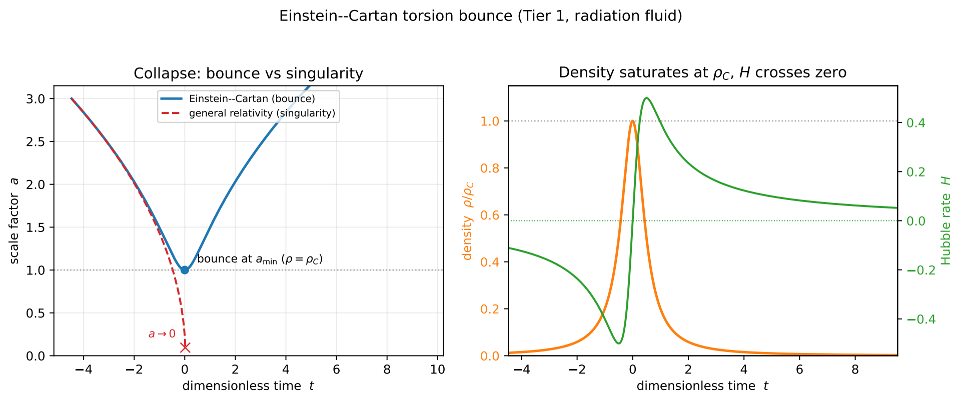

Figure 2.1: The Einstein–Cartan torsion bounce (a 1D spherical collapse, radiation fluid, in dimensionless units $8\pi G/3 = \rho_C = 1$). *Left:* the collapsing universe reaches a minimum radius $a_{\min}$ and re-expands (blue) exactly where general relativity would run to a singularity, $a \to 0$ (red dashed). *Right:* the density saturates at the bounce density ρ_C and the Hubble rate crosses zero—collapse turning into expansion, with no singularity ever reached. Reproducible as `figures/scripts/ch09_bounce.py`; the full numerical treatment is Chapter 12.

2.2 The bounce density

Setting the two terms equal fixes the density at which the reversal happens. For ordinary fermionic matter, with about one unit of spin $\hbar$ per nucleon mass,

$$\rho_C \sim \frac{m_n^2 c^4}{G\hbar^4} \sim 10^{50}\,\mathrm{kg\,m^{-3}}.$$

This sits 33 orders of magnitude above nuclear density and 46 below the Planck density—comfortably in a regime where Einstein–Cartan is still classical and trustworthy, far from where quantum gravity proper would be needed. The exact figure depends on the fermion content and on coefficients in the Einstein–Cartan Lagrangian, and pinning it precisely is one of the calculation targets of the simulation program (Chapter 12).

The shape of the turnaround. It helps to see the bounce as a particle rolling in a potential. Writing the effective Friedmann equation $H^2 = a^{-n}(1 - a^{-n})$ (dimensionless, with ρ_C and $a_{\min}$ set to one and $n = 3(1+w)$) as a zero-energy mechanics problem $\dot{a}^2 + V(a) = 0$ gives

$$V(a) = a^{2-2n} - a^{2-n}, \tag{2.1}$$

a *single repulsive wall* (Figure 2.2). The positive term a^{2-2n} is the torsion contribution—the $-\rho^2/\rho_C$ spin term—and it walls off small a; the negative term $-a^{2-n}$ is the ordinary matter pull. They cross at $a_{\min}$, where the contracting universe turns around ($V = 0$, slope $-n$). There is no second minimum and nothing tuned: the only things that fix the shape are the equation of state

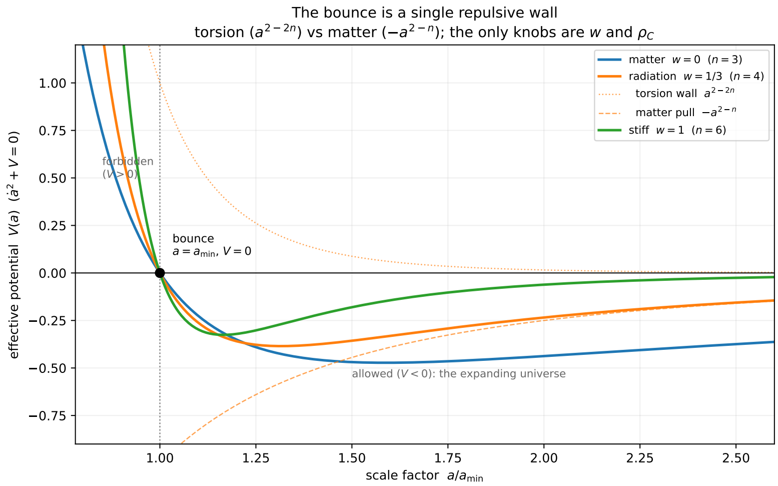

Figure 2.2: The bounce as a single repulsive wall, (2.1). The torsion term a^{2-2n} (dotted) forbids small a; the matter pull $-a^{2-n}$ (dashed) dominates at large a; they meet at $a_{\min}$ ($V=0$), the turnaround. The classically forbidden region is $V>0$ ($a<a_{\min}$); the expanding universe lives in $V<0$. The only knobs are w and ρ_C. Reproducible as `figures/scripts/ch02_bounce_potential.py`.

w (through n) and the density scale ρ_C. The microphysics that fixes *that* scale—the strong four-fermion coupling at the wall—is where the framework's deepest open question lives (Appendix A).

A name. The density at which Cartan's torsion forces the turnaround is where everything is dissolved and then extruded clean—a *katharsis*. Cartan + *katharsis* gives *Cartasis*: one may call ρ_C the Cartasis density and the bounce surface the Cartasis membrane. Elsewhere they are simply the *bounce density* and the *bounce membrane*.

What is actually there. At ρ_C the familiar particle picture is gone. The matter is compressed past quark deconfinement, past electroweak restoration, past any regime in which the Standard Model's quasiparticles are even defined. What remains is continuous field content—energy–momentum density, spin density, chiral current density—and what crosses the membrane is therefore not *particles* but *conserved currents*: total energy–momentum, total spin, total chiral charge, and (subject to electroweak anomaly effects) baryon and lepton number. Nothing structured survives the crossing. Whatever a world is built from on the far side must be rebuilt from elementary field content—a point that will matter both for what a universe inherits (Chapter 4) and for why a stray ordered object cannot simply pass through (Chapter 3).

Why this is gravity's coupling, and a neutron star proves it. The repulsion that halts the collapse is a spin–spin (four-fermion) interaction, and the density at which it wins scales *inversely* with that interaction's strength: a

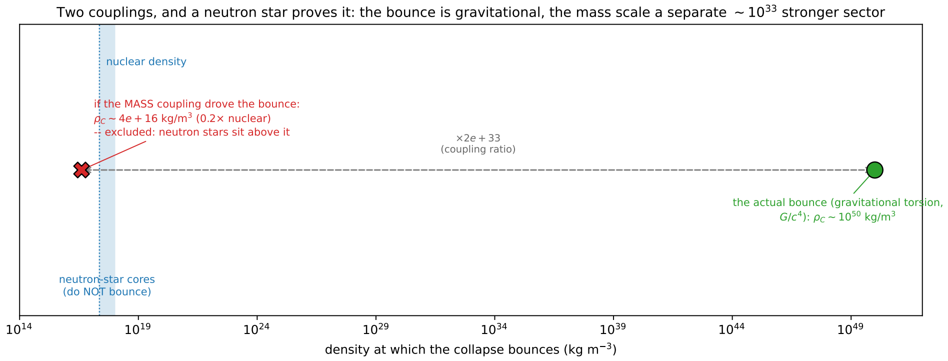

Figure 2.3: The bounce coupling is gravitational, not the mass coupling—and a neutron star decides it. The bounce density scales inversely with the four-fermion coupling: the gravitational G/c^4 puts it at $\rho_C \sim 10^{50}\ \mathrm{kg\,m^{-3}}$, whereas the $\sim 10^{33}$-times-stronger mass-giving coupling would put it at $\sim 0.2\times$ nuclear density—below the neutron-star cores that exist without bouncing. The two scales are necessarily two couplings. Reproducible from `sims/src/cartasis_sims/bounce_consistency.py`.

stronger coupling bounces sooner, at lower density. The coupling here is gravitational—$\kappa \sim G/c^4$, Planck-suppressed—which is precisely why ρ_C is so enormous. It is worth checking that it could not instead be the much stronger coupling that, in Part II, gives particles their mass (a strong condensate at $\Lambda \sim 246$ GeV, some $(M_{\mathrm{Pl}}/\Lambda)^2 \sim 10^{33}$ times stronger than gravity). If *that* coupling drove the bounce, the inverse scaling would drop ρ_C by the same 10^{33}—to about $4 \times 10^{16}\ \mathrm{kg\,m^{-3}}$, a fifth of nuclear density. But neutron-star cores reach several times nuclear density and manifestly do *not* bounce. So the strong, mass-giving coupling is excluded as the bounce driver by the mere existence of neutron stars: the bounce is gravitational, and the mass sector is a separate, far stronger force. This is not a worry to defer but a check one can run on a napkin, and it has a clean consequence: because the bounce temperature $T \sim \rho_C^{1/4} \sim 10^7$ GeV is some 10^5 times the condensation scale Λ, the mass condensate is fully melted at the bounce (consistent with the restored, massless matter above), and the mass sector contributes there only as extra radiation—a revision to the degree-of-freedom count, not to the gravitational repulsion. The bounce of Part I therefore survives the propagating-torsion extension that Part II's mass scale requires (Chapter 9; Figure 2.3).

2.3 The geometry of the bounce

2.3.1 Radius becomes time

Inside a Schwarzschild horizon the radial coordinate r is timelike: future-directed motion is motion inward, and this is ordinary general relativity. The bounce uses that fact rather than fighting it. Torsion pushes the matter "outward" in r—but since r is timelike there, outward in r is not a path back through the horizon; it is a path into the *future* of the bounce. That future, read in the new region's own coordinates, is the cosmological future of a fresh spacetime. The parent's radial direction r becomes, smoothly across the membrane, the new world's time direction τ.

2.3.2 Continuity across the membrane

Because nothing is allowed to be discontinuous, the metric and connection must match across the membrane. The induced metric agrees from both sides; the extrinsic curvature matches under junction conditions of Israel–Darmois type, modified to carry torsion; and stress–energy and spin current flow through unbroken. These conditions are simply the precise statement of "the bounce is continuous"—a system of equations relating the parent-side and child-side fields at the surface, and the thing a faithful simulation must reproduce.

2.3.3 A persistent interface, not a single instant

A black hole that keeps accreting keeps feeding the bounce. New matter falls in, compresses, reaches the bounce density, and joins the membrane—so the surface is not a one-time event but a persistent interface that lasts as long as the parent hole does. This is the framework's deepest departure from the standard picture, and it deserves to be stated plainly: a child universe's Big Bang is *not* an instant in its past but a surface in its present, still dawning. Matter is injected across it throughout the parent's life, not only at the first moment—the beginning never ends, it merely recedes to the horizon. That ongoing injection is precisely what the child's inhabitants will later measure as their dark sector (Chapter 5), so the dark sky is the standing evidence that the dawn is still in progress. A dawn closes only when its *feed* runs dry. For a child that is the parent ceasing to accrete, or evaporating; but an original-generation universe has no parent and instead gorges directly on the void, so *its* dawn is ongoing too—for as long as it draws matter from the vacuum. Only an old, fed-out OGU has a beginning that is truly finished, and it then evaporates in its turn, after a span beyond 10^{100} years. The OGU is therefore not the exception to the ongoing dawn but a different *source* for it—vacuum rather than parent. What it lacks is not the dawn but the dark sector: a diffuse void-feed presses no concentrated parent mass behind a membrane, so an OGU dawns without the dark matter and dark energy that betray a child's ongoing injection (Chapter 5).

2.3.4 The parent's horizon is the child's first light

The same surface wears three faces. From the parent's exterior it sits just inside the event horizon. From the parent's interior it is wherever infalling matter is currently reaching the bounce density. From the child's interior, looking as far back in time as it can, it is the boundary of the past at maximum lookback—the child's Big-Bang surface. The parent's horizon region and the child's earliest sky are one surface seen from opposite sides. This single identification is what later lets the cosmic microwave background be read as the parent's Hawking glow (Chapter 5).

2.4 What the bounce does *not* do

The bounce reliably turns a black hole into an expanding world, but it does not also *make* the matter that fills it—it is no mirror that turns infalling antimatter into matter and hands us the cosmic excess for free. The interaction that drives it forbids that, and the proof is short. When torsion is integrated out of Einstein–Cartan–Sciama–Kibble gravity, the Dirac sector is left with a single four-fermion contact term, the Hehl–Datta interaction [32],

$$\Delta\mathcal{L}_{\rm tors} = -\frac{3}{16}\frac{8\pi G}{c^4}\left(\bar{\psi}\gamma^5\gamma_a\psi\right)\left(\bar{\psi}\gamma^5\gamma^a\psi\right),$$

the negative square of the axial current. This is the very term whose energy density grows as the square of the spin density and overwhelms gravity at ρ_C: it *is* the bounce, so the bounce inherits its discrete symmetries. The axial current is even under charge conjugation, its contraction is even under parity, and the squared bilinear is even under time reversal. The interaction that drives the bounce is thus separately C-, P-, and T-even, and three things follow that no amount of modelling can wish away:

1. **No flip.** A C-even interaction cannot turn matter into antimatter. The membrane does not charge-conjugate what passes through it.
2. **No filter from nothing.** A C- and P-even interaction cannot *create* a chiral imbalance: zero net axial density in gives zero out.
3. **No arrow reversal.** A T-even dynamics does not run proper time backwards.

The bounce is real, robust, and generous—it reliably turns a black hole into an expanding world—but it is not a baryogenesis engine. Any matter–antimatter asymmetry must be *supplied*, not manufactured at the membrane. Where it actually comes from is the business of Chapter 4.

2.5 The arrow is struck at the bounce: a Janus point

If the bounce does not reverse time, where does a universe's arrow of time come from? It is *made* at the membrane. Along the matter's worldlines proper time runs monotonically; nothing turns around, the scale factor merely passes through a minimum. What turns around is the *thermodynamic* arrow. Infalling parent material arrives from a structured, high-entropy universe; compression toward ρ_C dismantles that structure and smooths the geometry—the Weyl curvature is driven toward zero—so gravitational entropy *falls* toward the membrane. On the far side, expansion lets structure, and eventually black holes, form again, so entropy *rises* away from it. Entropy is therefore at a *minimum at the membrane*, and the locally defined arrow points away from the bounce on both sides at once: a *Janus point* [3], one spacetime with a single monotonic proper time and two thermodynamic arrows diverging from a central low.

A newborn universe thus inherits a genuine low-entropy past for free, with no time flip required—the long-postulated "Past Hypothesis" arriving as a consequence of bounce geometry rather than as an unexplained initial condition. (That this low-entropy past is *forced*, and what it means for the timeless void from which the very first bounce is struck, is taken up in Chapter 3; how the arrow ends up locked to the matter–antimatter choice is taken up in Chapter 4.) The old slogan that "r becomes timelike, so time inverts" is a statement about Schwarzschild coordinates inside a horizon, not a physical reversal of anyone's clock.

2.6 What forms after the bounce

On the far side of the membrane the bounce dynamics open a new spacetime region—a child universe with its own time direction and its own spatial extent, the baby-universe scenario of Popławski [62] and, earlier, of Frolov et al. [28]—and from inside it looks like a perfectly ordinary hot Big Bang: it begins at maximum density, expands fast, cools, passes through radiation- and matter-dominated eras, and grows large-scale structure. The early, very rapid expansion is driven by the torsion repulsion at its post-bounce peak and fades as the density drops, giving an inflation-like epoch with no separate inflaton [42]; its quantitative shape, and the perturbation spectrum it should leave, are developed in Chapter 4 and put to the test in Chapter 10.

From the parent's side none of this is visible as a spatial object; the parent simply sees a black hole where the collapse occurred. The child's vast interior lies in the future of the bounce, in directions orthogonal to the parent's geometry—which produces the framework's most cheerfully counterintuitive feature, a *Tardis*: a universe far larger inside than its parent's horizon would suggest. Our own matter content, some 10^{53} kg, sits at the bounce density in about 10^3 m^3—less than half an Olympic pool—yet from inside we measure a universe tens of billions of light-years across. There is no contradiction: the small volume is the matter at the bounce moment in the parent's coordinates, the large one is the cosmological extent after expansion in the child's, and they are simply different quantities. Even plain general relativity already has the spatial volume inside a horizon growing without bound [14]; the bounce reinterprets that growth as the expansion of a child universe rather than the approach to a singularity we have agreed nature does not permit.

CHAPTER 3

The Void

The bounce of Chapter 2 turns a collapse into a world. But a bounce needs something to bounce—a region that reaches the bounce density in the first place. Step back, then, from any single universe to the largest picture the framework admits: the *void*, the eternal vacuum in which everything else is a rare condensation. This chapter asks what the void is, why it gives rise to anything at all, and—once it does—why what looks up at the sky is a real inhabitant of a real world rather than a fluke of order. The answers are, in order: a timeless maximum-entropy state; because emptiness can decay but cannot be improved upon; and because the void can give birth to a world but cannot dream up a mind.

3.1 What the void is

The supraverse is a single four-dimensional manifold—three space, one time, Lorentzian—of non-trivial topology. Almost all of it is vacuum: quantum-mechanically alive but unstructured. Embedded in that vacuum are condensed regions, the universes, each a connected sub-manifold with its own internal cosmology, joined to the rest through the bounce membranes of Chapter 2. These bubbles nest in parent–child relations—but, as the instanton calculation below will show, very *sparsely*: they are astronomically far apart, and emptiness is overwhelmingly the rule. The void, not the worlds in it, is almost the whole of the supraverse.

No fifth dimension is needed to hold this together. The intuition that one must "embed" a child inside a parent in some higher space is a visualization crutch, not a physical requirement: two regions joined at a bounce membrane is a perfectly good 4D manifold with a boundary identification, defined intrinsically. One *may* embed it in higher dimensions to draw it (Whitney guarantees one can), but the embedding carries no physics.

3.2 The void has no arrow

Before asking what the void *does*, we must be honest about what it is not: it is not a stage on which time runs. The microscopic laws—quantum field theory, general relativity, Einstein–Cartan—are all time-symmetric up to the tiny CP phase. An arrow of time is never in the laws; it is a *boundary condition*, appearing only where a system is handed a low-entropy edge. The supraverse as a whole has no such edge and needs none. To posit a global cosmic "now," some master clock for the void, would be to add unexplained structure; declining to costs nothing and is the honest, minimal choice.

So the right question about the void is not "which way does its time run?" but the timeless, thermodynamic one: *what is the stable state of the void?* That is an equilibrium question—an eigenstate question. Time is what clocks measure, and an equilibrium void has no clocks. Every arrow we will ever speak of is manufactured *locally*, at a bounce (Chapter 2), pointing away from that low-entropy membrane; time is emergent, created per universe, never inherited from the void. There is no contradiction between a timeless void and the vivid passage of time we feel—we feel it precisely because we live *inside* one of the void's condensations, on the high-entropy side of its bounce.

This dissolves an apparent paradox before it can take hold. The language of this book cannot help speaking of the void as though things happen in it—universes "nucleate," equilibrium is "established," a population "coarsens." Those are descriptions borrowed from inside a condensation's own time and projected onto the whole. The void does not evolve *toward* equilibrium in some outer time; the equilibrium *is* the timeless stationary state, and "nucleation" is just the statement that this state already *contains* its rare condensations as part of what it is. No global clock ticks. Read every "then" that follows as the only tense we have, not as a claim about the void.

3.3 Why a void condenses

Naively, making universes looks entropically forbidden: ordered worlds out of formless vacuum. The naive view forgets gravity. Gravitational entropy runs *backwards* from the ordinary kind [5, 54]: a uniform matter distribution is low entropy (smooth metric, no horizons, nothing stored geometrically); clumped matter is higher; and a black hole is the maximum-entropy state for its mass–energy, $S = A/4\ell_P^2$, able to carry more entropy than everything that fell into it. So structure—clumps, horizons, black holes—is entropically *favoured*, not forbidden, and the Second Law does not oppose the appearance of worlds.

That much says condensation is allowed. It does not yet say how much of it there is, and here the careful answer overturns the naive one.

3.4 The maximum-entropy eigenstate

State "the stable state of the void" as the variational problem it really is, and the answer is sharper—and at first sight stranger—than "a population of worlds." Take an infinite vacuum with a positive cosmological constant. On its own, such a vacuum is *de Sitter space*: it accelerates apart, and every observer is surrounded by a cosmological horizon at radius c/H_Λ, glowing with the Gibbons–Hawking temperature [29]

$$T_{\rm dS} = \frac{\hbar H_\Lambda}{2\pi k_B} \approx 2 \times 10^{-30}\,\mathrm{K}$$

for our own Λ, and carrying the horizon entropy

$$S_{\rm dS} = \frac{A_{\rm cosmo}}{4\ell_P^2} = \frac{\pi c^5}{\hbar G H_\Lambda^2} \approx 3 \times 10^{122}\, k_B. \tag{3.1}$$

Now ask what *maximises* the total horizon entropy of such a vacuum. In Schwarzschild–de Sitter space the generalised entropy is the sum of the black-hole and cosmological horizon areas, $S = (A_{\rm bh} + A_{\rm cosmo})/4\ell_P^2$, and inserting any mass shrinks the cosmological horizon faster than it grows a black-hole horizon. The sum is therefore largest at *zero* mass: **empty de Sitter space is the maximum-entropy configuration**—the stable state, the eigenstate.

This does not contradict the clumping argument of the previous section; it refines it. Clumping governs how an overdensity that is *already present* evolves—it is why a fluctuation that has reached the bounce density proceeds to bounce rather than disperse. But the global maximum, the eigenstate of the whole void, is the empty vacuum. Structure is locally entropy-increasing once seeded, yet globally entropy-*costly* to seed. The two readings meet in a single verdict: the void is overwhelmingly empty, dusted with the sparsest possible scattering of worlds.

Worlds are the eigenstate's faint components, not events in it. It is worth being exact about the word "fluctuation," because the void has no time. A genuine eigenstate is stationary: nothing in it *happens*, in sequence, ever. So the worlds are not a stream of events the void undergoes; they are the *timeless structure* of the one equilibrium state—the amplitude of that state having support, exponentially small, on configurations that already contain condensed universes. In this sense the void is a *perturbed* maximum-entropy eigenfunction: empty de Sitter is the dominant component, the worlds its subdominant ones, and the "perturbation" is not an external poke (which would need time) but the intrinsic nonlinearity of self-gravity, which opens a tunnelling channel out of the empty state. That a quantum state of pure gravity should be assumed at all—in this timeless, Wheeler–DeWitt sense—is the content of Axiom A1, and it is the conservative assumption: it adds no new law, only the universality of the law we have.

3.5 Why the seed of a world must be low-entropy

One might worry that we have only pushed the puzzle back: granted the void condenses, why is each new world born in the pristine, low-entropy state its forward arrow requires (Chapter 2)? The answer is that the seed *cannot* be otherwise. A condensation reaches the bounce density only by crushing its contents past every scale at which complex structure—and therefore information, and therefore entropy—can exist. The membrane is forced to be a state of high concentration and low complexity: uniform matter at the bounce density, with the Weyl curvature driven toward zero. That is exactly Penrose's low-gravitational-entropy initial condition (his vanishing-Weyl hypothesis [54]), and here it is not postulated but *forced* by the physics of the bounce. The low-entropy past that every universe needs is thus supplied for free, and the apparent regress closes: you cannot make a high-entropy bounce, because high entropy requires structure and structure cannot survive the crossing.

3.6 How the void makes a world

How often does the empty eigenstate decay into a seed? Because such seeds nucleate in the heat-dead de Sitter void of an older universe (the recursive cycle of Chapter 4), the birth rate is a de Sitter nucleation rate, and its scale is *set*, not free, by the void's dark energy. Per de Sitter horizon four-volume,

$$\beta = \lambda \, \frac{H_\Lambda^4}{c^3}, \qquad H_\Lambda \simeq 1.8 \times 10^{-18}\,\mathrm{s}^{-1}, \tag{3.2}$$

so the prefactor $H_\Lambda^4/c^3 \approx 4 \times 10^{-97}\,\mathrm{m}^{-3}\mathrm{s}^{-1}$ is fixed by the observed dark energy—about one nucleation *attempt* per Hubble four-volume. The whole question is the dimensionless suppression $\lambda = e^{-I}$, with I the action of the gravitational instanton that carries the empty vacuum across the bounce density.

That action has a closed form. Nucleating a mass M in de Sitter costs the Boltzmann weight $e^{-Mc^2/k_B T_{\mathrm{dS}}}$, and that exponent is exactly the cosmological-horizon entropy deficit of inserting M, so

$$I = \frac{2\pi M_{\mathrm{seed}}\, c^2}{\hbar H_\Lambda}. \tag{3.3}$$

The only input is the lightest seed that can actually bounce: a blob at the bounce density turns into a universe only if it is self-gravitating ($R_s \simeq R$), which fixes $M_{\mathrm{seed}} \sim 9 \times 10^{14}$ kg—about the mass of a mountain. A lighter blob has no horizon and simply disperses. Putting that in (Figure 3.1),

$$I \;\sim\; 2.5 \times 10^{84} \;\gg\; I_{\mathrm{crit}} \simeq 1.4, \tag{3.4}$$

where I_{crit} is the de Sitter percolation threshold [30] that separates a void which tiles into a packed foam ($I < I_{\mathrm{crit}}$) from one that inflates faster than it nucleates, leaving worlds isolated ($I > I_{\mathrm{crit}}$). The margin is eighty-four orders of magnitude; no semiclassical correction, no factor-of-ten ambiguity in the seed mass, can close it. So $\lambda = e^{-I}$ is, for every practical purpose, zero, and the verdict is unambiguous: **the supraverse is dilute**—a sparse, ever-inflating scatter of isolated, round island universes, each a single bounce, astronomically far from its neighbours. This lands the framework squarely in the eternal-inflation family, and it is the picture the rest of the book assumes; what those island universes are like, how they grow, and how they seed the next generation is the subject of Chapter 4.

3.7 Why we are not Boltzmann brains

The dilute verdict revives the oldest objection to any fluctuation cosmology—sharpened, for a universe with a positive cosmological constant, by Dyson

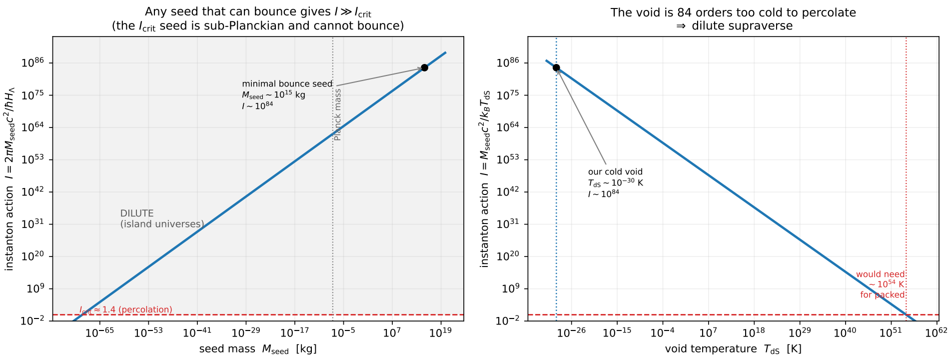

Figure 3.1: The OG-nucleation instanton action $I = 2\pi M_{\text{seed}} c^2/\hbar H_\Lambda$ (Equation (3.3)). *Left:* I versus seed mass; any seed massive enough to bounce ($\sim 10^{15}$ kg, the self-gravitating minimum) gives $I \sim 10^{84} \gg I_{\text{crit}} \simeq 1.4$. *Right:* I versus void temperature; the cold de Sitter void ($\sim 10^{-30}$ K) is eighty-four orders too cold to percolate. The verdict is robust: a dilute, eternal-inflation-like supraverse of isolated island universes. Reproducible as `figures/scripts/ch05_instanton.py`.

et al. [24]—and disposes of the suspicion that we are a simulation. If a whole universe is this fantastically rare, surely a single mind, with its memories, is cheaper to fluctuate into being than an entire cosmos; and then a *typical* observer should be such a fluctuation—a "Boltzmann brain"—not a real inhabitant of a real world. Cast in the de Sitter Boltzmann weights above, the bare costs seem to agree:

$$\underbrace{10^{-10^{69}}}_{\text{a brain } (\sim 1\,\text{kg})} \gg \underbrace{10^{-10^{84}}}_{\text{a world-seed } (\sim 10^{15}\,\text{kg})} \gg \underbrace{10^{-10^{123}}}_{\substack{\text{a finished low-entropy} \\ \text{Big Bang [56]}}}, \tag{3.5}$$

the brain winning by the unbridgeable factor $10^{10^{84}}$. And no rescue helps—not the head-count of real observers a universe produces, not the recursive tree of its descendants, not a real life's length against a brain's single flickering thought. Each of those is a *single* power tower, a number of at most a few thousand digits, and none can dent a *double* tower of height 10^{84}.

The resolution is not to win the count; it is to see that it is the wrong count, because *the brain and the bare Big Bang cannot come from the void at all.* A Boltzmann brain—or, what is the same thing, the optimal computer running a simulation of one—is an information processor, and §3.2 fixed what such a thing requires: metabolism, memory, and computation are all dissipative, and dissipation needs an arrow and a low-entropy reservoir to discharge into. The void offers neither. It is the stationary, arrowless, maximum-entropy eigenstate of §3.4—with no dynamics to fluctuate and no gradient to compute against. The brain is parasitic on an arrow it cannot itself create; the $10^{-10^{69}}$ was reckoned as though the void were a churning thermal bath, which it is not [7].

What the eigenstate *can* do is decay. Nucleation from a stationary state is a genuine quantum transition—the gravitational instanton of §3.6— and it yields only the *generic* state of the mass-energy it creates: a structureless, maximal-entropy, black-hole-like blob. That blob is exactly a viable seed; it carries no improbable order of its own, yet when it bounces, the bounce is *forced* to be low-entropy (§3.5), manufacturing order and arrow for free on the far side. A brain is not a blob. It is a low-entropy *organised* state, and nucleation cannot deliver organisation—only generic lumps—while the arrowless void cannot assemble one afterward. So the void makes seeds, never brains.

The same blade cuts Penrose's bare Big Bang, and this is the deep part. A hot Big Bang handed over whole in its 1-in-$10^{10^{123}}$ low-entropy starting state [56] is *also* an ordered, arrow-bearing configuration posited with no mechanism to make it. It presupposes the very low-entropy past it was meant to explain, so it too cannot be nucleated from the void—it can only be put in by hand. A cosmology whose sole origin story is "the initial state simply happened to be that special" is, structurally, the simulation hypothesis: it declares us an un-generated fluke of order with no maker. *If the bare Big Bang were the only available beginning, we would indeed be a simulation.* The framework escapes because it never posits the order: it pays $\sim 10^{-10^{84}}$ to nucleate a generic seed and lets the bounce force the 10^{123} worth of low-entropy order at no extra cost—turning Penrose's improbability-of-order from an unexplained initial condition into a consequence of geometry at the membrane.

Two levels then close the problem. At the *foundation*, nothing mind-like can bottom out the stack; only an arrow-creating bounce can, so the base layer of reality is real worlds, not a sea of fluctuations. *Inside* a world the arrow exists, so brains *can* fluctuate—but only over the de Sitter recurrence time $\sim 10^{10^{69}}$ years, and our dark energy is finite (it is the parent's accretion, Chapter 5, and switches off when the parent finishes evaporating, $\sim 10^{136}$ years), recycling our matter into new bounces long before a single interior brain could form. A ΛCDM model with a truly eternal cosmological constant has no such escape—its endless de Sitter future is a literal Boltzmann-brain factory—so this is a respect in which the framework is *strictly better behaved* than the standard model, not merely different.

Epistemic status. The de Sitter measure problem—whether and how a stationary vacuum "fluctuates"—is not settled, and a fully rigorous measure for the framework does not yet exist (it is among the open questions of the appendix). What is robust is the qualitative asymmetry: nucleation yields generic blobs, blobs bounce into arrowed worlds, and organised low-entropy minds presuppose an arrow they cannot source from an arrowless eigenstate. The cognitive-stability argument seals it from the other side—any theory predicting that typical observers are fluctuations dissolves its own evidential basis, since a fluctuated mind has no warrant to trust the memories and reasoning that built the theory, including the three numbers in (3.5). We are entitled to conclude we are real not by counting, but because the alternative refutes itself and because the void's only product is the real thing. Reproducible as `sims/src/cartasis_sims/boltzmann_brains.py` and `void_eigenstate.py`.

CHAPTER 4

Eternal Dawn

The void decays into seeds (Chapter 3), and each seed bounces (Chapter 2) into a world. A world born this way—directly from the void, with no parent outside it—is an *original-generation universe*, an OGU: the first dawn of a lineage. This chapter asks what those first universes are like—what they inherit, how they choose between matter and antimatter, what sizes and spins they come in—and how, by growing old and dying, each one seeds the next, so that the dawn is not a single event but an eternal, recurring one.

4.1 The first dawn, and what a universe inherits

Just past the bounce the new universe is at its densest, and the torsion repulsion that halted the collapse is still near its peak. It drives a brief epoch of very rapid expansion that fades as the density drops and gravity reasserts itself—an inflation-like phase with no separate inflaton field, ending on its own when the density falls below the bounce threshold, and leaving behind the homogeneous, near-isotropic conditions a hot cosmology needs. Its perturbation spectrum—scale-invariant with no inflaton, the slight red tilt $n_s \simeq 0.965$ following from a contraction just softer than matter—is derived and matched to the microwave sky in Part II, where the same recombination physics reproduces the Planck acoustic peaks at $\chi^2/N \simeq 1$ and the tensor sector furnishes a clean inflation-versus-bounce fork (Chapters 10 and 12).

What actually comes through the membrane is set by conservation laws, not by inheritance of any structure. Energy–momentum is conserved, fixing the new universe's mass budget. Net angular momentum passes through, so a rotating seed or parent leaves the child with a *preferred axis*—a candidate source of the large-scale anisotropies discussed later, and, combined with the lineage's parity violation, of an axis-aligned cosmic birefringence in the microwave polarization (Chapter 11). The conserved charges flow through, in particular the combination $B - L$ that will carry the matter excess. What does *not* come through is everything made of structure: atoms, nuclei, particle identities, any record of a "before." All of it is dissolved at the bounce density and rebuilt from elementary field content on the far side (Chapter 2). A universe inherits a budget and a few conserved numbers, not a cargo.

4.2 The matter choice

One of those conserved numbers is the most consequential the universe will carry: the slight excess of matter over antimatter, measured today as the baryon-to-photon ratio $\eta \simeq 6 \times 10^{-10}$, about 0.6 parts per billion. Chapter 2 established that the bounce cannot *make* this excess—the interaction that drives it is C-, P-, and T-even, so it neither flips charge nor generates a chiral bias from nothing. The three conditions any such mechanism must meet were set out by Sakharov [65]. The asymmetry must be *supplied*. The framework's account of where it comes from divides cleanly by generation.

For a child universe: inheritance, essentially exact. A universe with a parent inherits the parent's excess directly. At the bounce the infalling material thermalises at $\sim 10^{20}$ K into a near-symmetric bath in which every species is pair-produced; the parent's particular matter is erased. What survives is whatever net $B - L$ the material carried, because the electroweak sphalerons active at the bounce temperature wash out $B + L$ but cannot touch $B - L$ [39]. The crossing is also fast—$\sim 4 \times 10^{-21}$ s, far shorter than any thermalisation time—so it generates essentially no entropy (the dilution factor $D = S_{\text{after}}/S_{\text{before}} = 1$ to about one part in 10^{11}; Chapter 12). With entropy conserved, the child does not regenerate its photons, and so it inherits the parent's ratio outright, $\eta_{\text{child}} \approx \eta_{\text{parent}}$. The excess is passed down a lineage like a family name, not re-manufactured at each step.

For the first universe: the seed's lucky skew. Only the OGU, with no parent, must source its excess at birth—and even there nothing needs to *make* it. A vacuum fluctuation rare enough to nucleate a bounce is not exactly matter–antimatter symmetric; it carries some chance net $B - L$. The symmetric part incinerates in the thermal bath, and the surviving skew fixes that lineage's η. A quantitative attempt to source the skew dynamically—the chiral-vortical effect, in which the rotating bounce's vorticity ω drives an axial current and freezes in a bias $\eta \sim C\,(\omega/T)$—gives the right *order* but not, yet, the right number: at the bounce density the bare, parameter-free ratio is $\omega/T \sim 1.5 \times 10^{-11}$, landing within about 1.6 orders of magnitude of η_{obs} when it could as easily have been 10^{-30}, but matching the value would require an uncomputed coefficient $C \cdot f_\omega \simeq 41$. This is a scaling estimate with explicit assumptions, not a derivation (reproducible as `sims/src/cartasis_sims/cve_filter.py`), and it binds *only* the OGU; for the rest of us the excess is inherited debris, and the shortfall is not our problem.

The needle, threaded by selection. The surviving excess must thread a fine target: too much and the universe is over-baryonic, too little and it annihilates to a sterile furnace; the viable window sits at about one part per billion. The framework does not fine-tune to hit it—it *selects*, in the spirit of the cosmological natural selection of Smolin [69]. Only lineages founded on a sufficiently skewed seed are matter-rich enough to make stars, black holes, and descendants at all (§4.3); the barren ones make nothing and are never anyone's parent. We carry down what a viable founder happened to have. What remains genuinely open is not the stability of η along a lineage but its *value* at the founder—why $\sim 10^{-9}$ rather than 10^{-3} or 10^{-15} (see the appendix).

Why matter comes paired with a forward arrow. The thermodynamic arrow of Chapter 2 is set by the low-entropy membrane and would exist even in a perfectly symmetric universe; it is not *caused* by the matter excess. But the two are co-sourced at the same surface and locked together across the supraverse. The process that biases the matter is CP-violating, and by the CPT theorem CP violation is T violation, so matter genesis is intrinsically time-asymmetric. Because the void is CPT-symmetric, for every matter-led universe running forward there is a CPT-image partner that is antimatter-led and runs the opposite way—the same object seen through CPT, a construction close in spirit to the CPT-symmetric universe of Boyle et al. [9]. The quantity fixed by symmetry is the *product* of matter-sign and arrow-sign, and observership selects the diagonal: every observer necessarily finds "matter, forward in time," while the inhabitants of the mirror lineage say exactly the same of their "antimatter, backward" world. Matter and the arrow travel together not because one causes the other, but because the CPT structure of the void binds their signs.

4.3 The shape of a generation

A generation of first universes is not uniform; it has a derivable distribution in both size and spin, and we should expect to find ourselves at its typical point, not in its rare tail.

Size: small is typical. Two pressures compete. Births favour the small—a fluctuation is exponentially less likely to reach the bounce density over a larger mass. Growth favours the large—once born, a hole feeds by runaway accretion. In the stationary population these balance into a power law, $n(M) \propto M^{-g}$ (with $g \simeq 1$–2 for Eddington- to Bondi-limited feeding): many small universes, few large, cut off at the high-mass end (Figure 4.1, left). Since the count falls with mass, a low-mass viability cut—enough mass to make stars and black holes—selects observer-bearing universes that pile up just above the edge. The old conjecture that "we sit near the optimum" is thus *derived*: a typical observer is near the viability edge of a declining power law, not out in the exponentially rare massive tail.

Spin: barely viable is typical, and that is why η is small. The seed's net vorticity sets the inherited bias, $|\eta| \sim C\,|\omega/T|$. A random fluctuation most likely has *small* net vorticity, but a seed below $\omega_{\min} = \eta_{\min} T/C$ cannot complete baryogenesis: its matter and antimatter annihilate to near-symmetric radiation, no structure forms, and the universe is sterile. Viability therefore

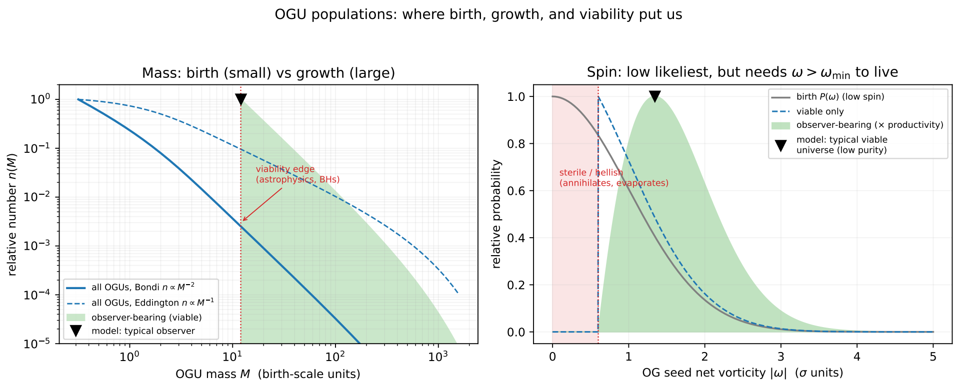

Figure 4.1: The two derivable population shapes. *Left:* the mass distribution $n(M) \propto M^{-g}$—births (small) versus runaway growth (large)—with a viability cut selecting observer-bearing universes, which peak just above the edge. *Right:* the spin distribution—low net vorticity is likeliest, but below ω_{min} baryogenesis fails and the universe is sterile, so viable universes sit just above threshold (low purity, small η). The markers are the model's predicted typical location, not measurements of our universe; the axes are in model units. Generated by `figures/scripts/ch05_distributions.py`.

gates the distribution from below, and the result runs exactly counter to intuition (Figure 4.1, right): *no spin is likeliest but most sterile, high spin is rare but pure*, and the typical *viable* universe sits just above the threshold—low-spin, low-purity, barely past barren. This is a shape, not yet a number (it is plotted in units of an unknown birth-vorticity scale), but it carries a real payoff: it explains why the cosmic matter asymmetry is *small without fine-tuning*. A small η is what a typical viable universe has. The same low spin predicts a small but nonzero net rotation for our universe—a preferred axis to confront with galaxy-spin handedness and CMB alignments (Chapter 10).

Lineages are chirally pure. Because the sign of the excess is set once, at the seed, and merely passed down thereafter, a matter-led OGU has only matter descendants and an antimatter-led one only antimatter, with no mixing. The vacuum's CPT symmetry makes the two kinds equally numerous, so global CPT is preserved even though every individual lineage is pure. What is inherited is the chirality *sign* alone—*not* the spin: a descendant's rotation axis and magnitude are its own progenitor hole's, set by a local astrophysical collapse and re-drawn every generation. Our spin is our progenitor hole's, not our distant founder's (Chapter 10).

4.4 Growth, death, and the eternal recurrence

A first universe, once viable, grows—its central holes feed and merge, its mass climbs—and it lives a very long time: the parent black hole that hosts a universe of our mass would take on the order of 10^{145} years to evaporate, far longer than any internal cosmological clock. But it is not immortal. Its stars burn out, its black holes themselves Hawking-evaporate, its matter redshifts away, and the interior settles into cold ($\sim 10^{-30}$ K), empty, dark-energy-driven de Sitter space. And a vast, empty, smooth de Sitter interior is—gravitationally—*a void*. It is exactly the substrate Chapter 3 began from. So it does what a void does: it nucleates a fresh, dilute scatter of new first universes inside itself, and the cycle turns.

The void is not primordial. This collapses a hidden assumption. The void of Chapter 3 is not a separate arena that existed "before" everything; it is the asymptotic state every universe reaches. Universes nest both ways—we are inside a parent, and our own far future will, in its emptiness, seed a fresh scatter of new lineages. There is no first void and no last universe. The supraverse is self-similar and eternal in both directions, and asks for no first cause.

Why heat death does not forbid it. The obvious objection is that heat death is *maximal* entropy while a fresh bounce needs *low* entropy. Penrose's resolution applies exactly [56]. The *total* entropy rises monotonically, obeying the Second Law; but what a bounce needs small is the *gravitational* (Weyl, clumping) entropy, and that *cycles*—low at the smooth bounce, high while matter clumps into black holes, low again once the holes evaporate and the interior smooths back to conformally flat de Sitter (Figure 4.2). It is the Weyl entropy a new bounce needs low, and it resets every cycle. This is Penrose's conformal cyclic cosmology with one change: a dead aeon here does not hand off to a single successor, it nucleates *many*—a whole new scatter of them. The framework is conformal cyclic cosmology *made to branch*. (The recurrence is a synthesis at the speculative end of the program—a natural closure of the picture, consistent with the gravitational arrow, not a derived result. Reproducible as `figures/scripts/ch05_cycle.py`.)

And so to us. Inside any viable universe, stellar collapse makes black holes, and each of those is the seed of a child universe—a black-hole universe, a BHU, the next rung down the recursion. We are on one of those rungs, not at the top: our universe carries dark matter and dark energy, and in this framework both *require a parent* (they are the parent's mass and ongoing accretion felt through the membrane, Chapter 5). An OGU, born straight from the void, would have neither. So the very darkness of our sky tells us we are a child, not a firstborn—which is the subject of the next chapter. It also rescues a Copernican worry the framework had quietly incurred: to find ourselves a few billion years into a Big Bang looks special, until the recurrence reveals it as the most ordinary thing there is. Every universe has its own bounce and its own observers a few billion years on; there have been and will be endless aeons, each a sparse scatter of universes nested in the heat-dead past of another. We sit not at a privileged origin but at a typical vantage—one lit pixel in an eternal, self-similar screensaver of near-identical dawns.

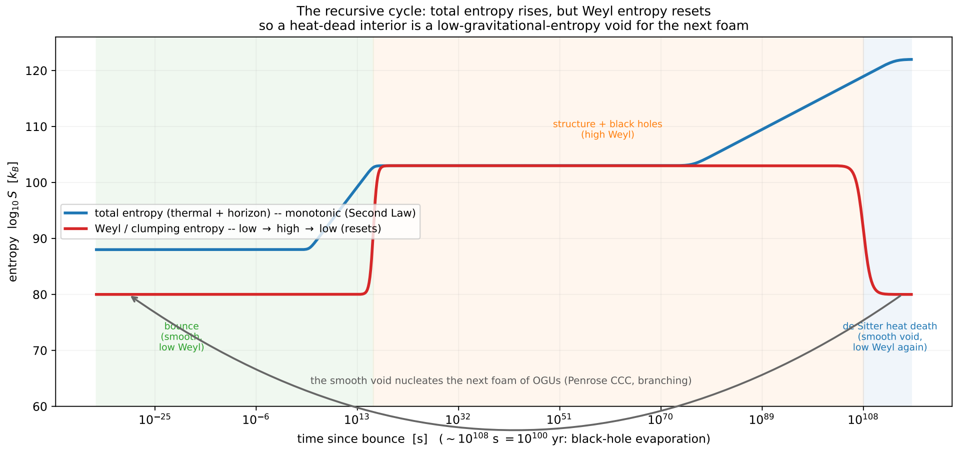

Figure 4.2: Why a heat-dead interior can seed the next generation. The *total* entropy (rising) climbs monotonically to the de Sitter maximum, but the *Weyl/clumping* entropy (the one a fresh bounce needs small) cycles—low at the smooth bounce, high while black holes exist, low again once they evaporate and the interior smooths to de Sitter. So the heat-dead void is a valid low-gravitational-entropy substrate for the next scatter of universes: Penrose's conformal cyclic cosmology, made to branch. Schematic, with order-of-magnitude entropies on the black-hole Hawking timescale.

CHAPTER 5

Home

We have followed the framework from the timeless void (Chapter 3) through the first dawn (Chapter 4) down the recursion to its children. This chapter arrives home—at our own universe—and finds that the parts of the sky cosmology has had to *add by hand*, the dark sector, are here not additions at all but the plain look and feel of living inside a black hole. Dark matter, dark energy, the microwave background, the great rings in the galaxy distribution: read in this framework, each is a feature of our address, and together they are a family portrait of the parent on the far side of our horizon. And they carry a single deeper message, the one the book is named for: our beginning has not ended. The textbook Big Bang is a finished event in the deep past; ours is a surface still pressing on us from the edge of the sky, our parent still feeding matter across it. The dark sector is exactly the part of that *ongoing* dawn we are able to measure—so far from being awkward add-ons, dark matter and dark energy are the standing proof that we live not after our dawn but inside it.

5.1 We are a child

One observation settles our place in the family tree before any detailed accounting. Our universe contains dark matter and dark energy, and in this framework both *require a parent*: dark matter is the parent's mass felt through the persistent bounce membrane, and dark energy is the parent's ongoing accretion (this chapter makes both precise). An original-generation universe, nucleated straight from the void, has no parent and so would have neither. Their presence in our sky is therefore a direct reading:

$$\big(\text{observed dark matter and dark energy}\big) \implies \big(\text{we are a child universe, } n \geq 1\big).$$

We are not a firstborn. The darkness of the sky is the evidence.

The idea that our observable universe could be the interior of a black hole is not new—it runs from Pathria [52] and Stuckey [71] through Smolin's cosmological natural selection [69] to its modern torsion form in Poplawski's bounce [62], on which this Part builds directly. What this chapter adds to that lineage is the dark sector: the reading of dark matter and dark energy not as extra ingredients but as the live, ongoing gravitational signature of the parent interior—a reading that, as far as we know, is new here, and the part most sharply exposed to data.

5.2 The persistent channel

The bounce membrane is not a one-time event but an ongoing interface (Chapter 2). As long as the parent black hole keeps accreting, matter keeps falling in, reaching the bounce density, and crossing into our interior—throughout the parent's entire life. But the injection geometry is peculiar, and everything in this chapter follows from it: new matter does not appear at some location in our three-dimensional space. It appears *at the membrane*, which from inside is our cosmic horizon at maximum lookback—our Big-Bang surface. Injected matter therefore manifests as gravitational influence arriving from the cosmic boundary, not as particles materialising among the galaxies.

> **Why you can seed a child but never visit one.** It is worth seeing what kind of surface this is, because it answers a natural question: could one, granted every speculative tool relativity allows—warp drives, traversable wormholes, exotic matter—*travel* to the parent, or to a baby universe forming in our sky? No, in both directions, for the same reason. The membrane is not a wall in space, which a warp bubble could defeat, but a surface in *time*: a bounce, a low-entropy boundary. The parent lies behind our Big-Bang surface, in our *past*; you cannot drive to your own Big Bang. And a baby universe is a real black hole you *can* fall toward—but its time runs forward from its bounce, so the horizon delivers you to its low-entropy first instant, not to any interior "now." You could become a speck in its initial conditions; you can be an ancestor, never a tourist. The membrane-as-Big-Bang structure is the framework's own chronology protection: you may fall down the recursion, never climb back up.

5.3 Dark matter: the parent's gravity

The matter on the parent's side of the membrane has mass–energy, so it curves our space; but it does not enter our interior, exchange photons with our matter, or annihilate against it. That is precisely the phenomenology of dark matter: halo-like mass around galaxies, extra rotational binding, lensing without visible source, large-scale correlations—all gravitational, none luminous. In this framework it is not a new particle but the live gravitational signature of the parent's ongoing, biased accretion.

Gravity felt before light seen. Because the membrane is our *past* horizon, the parent's mass is already in our metric—it curves our space *now*—while its light is still climbing in toward us from that receding past surface, and will for cosmic ages. So the dark sector's gravitational pull *leads* its electromagnetic visibility: we feel matter we cannot yet see (Figure 5.1). "Dark" matter is, on this reading, exactly gravity arriving ahead of light from the membrane.

And it fixes the dark-to-baryon ratio. The reading is not only qualitative; it carries a number. The membrane passes a fraction f of the parent's infalling mass through into our luminous interior and holds the rest gravitationally

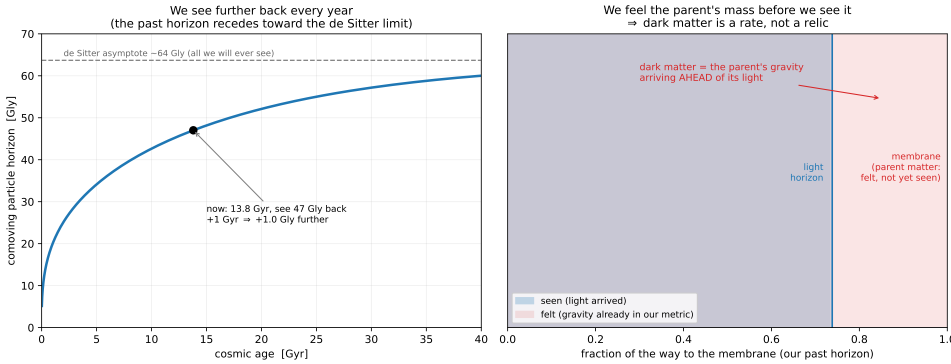

Figure 5.1: Gravity felt before light seen. The parent's matter sits at (and beyond) our past cosmological horizon—the membrane—so its mass is already in our metric and pulls on us now, while its light is still climbing in from that receding surface. The dark-matter gravitational signature therefore leads its electromagnetic visibility. Reproducible as `figures/scripts/ch06_lookback.py`.

present but not yet arrived, so the dark-to-baryonic ratio is about $(1-f)/f$, and the observed $\sim 5:1$ requires $f \sim 1/6$. This is not tuned. A first Einstein–Cartan calculation already makes $f \sim 1/6$ *natural*: the membrane is a torsion barrier whose order-unity height places f in the right decade—a value *narrowed from the microphysics* rather than fitted, converting a free parameter into an $O(1)$ ratio (the computation, and its residual uncertainty, are in Chapter 12). The precise value remains a sharp way to be wrong. What this reading does *not* need is any new substance: the dark matter is the parent's own mass, the part of it that has not yet crossed into light.

What distinguishes it from a particle. Geometric dark matter need not clump exactly as cold particles do, need not appear in any direct-detection experiment, and should trace the parent's projected mass rather than our own luminous structure—tests taken up in Part II (Chapter 10). This is a sharp break from the decades-long search for a dark-matter particle [6].

5.4 Dark energy: the parent's growth

The parent is not a static reservoir; it is a growing black hole, feeding from its own universe. As its mass climbs, the membrane's geometry relative to our interior changes, and from inside that shows up as an extra outward push on our expansion—the acceleration that standard cosmology attributes to dark energy. This makes dark energy a *rate*: to leading order its strength tracks the *time-derivative* of the parent's accretion. A parent still gorging sustains or strengthens it; a parent whose surroundings are thinning, or which is turning toward Hawking-dominated mass loss, yields weakening dark energy—and, in the far future, a reversal, as the membrane shrinks and our expansion decelerates. So dark energy here is neither constant nor eternal: it is a readout of our parent's life story, and the recent hints from DESI that its equation of state may be evolving are exactly where to confront it (Chapter 10).

5.5 The microwave background: the parent's glow

The membrane wears, we have seen, three faces: the parent's event horizon from outside, our cosmic horizon at maximum lookback from inside, the Big-Bang surface in the standard telling. A black hole's horizon radiates—Hawking's result [31]—and from our interior that radiation arrives from every direction, at maximum cosmic lookback, with a thermal spectrum. That is the cosmic microwave background. The parent's Hawking temperature at emission is essentially zero for so large a hole, but the radiation has been redshifting through our entire expansion; the 2.7 K we measure is what survives. (That the microwave background might be the thermal radiation of a black-hole universe is not a new thought—Zhang [83] read it as the cooled blackbody glow of the hole itself; the reading here is the narrower one that it is the *parent's* Hawking radiation reaching us through the membrane.) The identification sits well with what the background *is*: a near-perfect Planck spectrum (Hawking radiation is thermal), near-perfect isotropy (the membrane is smooth across the parent's horizon), and anisotropies at the 10^{-5} level (membrane inhomogeneity and a little parent rotation). The acoustic peak structure comes from *shared* physics—the framework runs the same post-bounce recombination as ΛCDM—so it is reproduced rather than owed: fed the framework's own sourced parameters, a full Boltzmann transfer matches the Planck peak *heights* at $\chi^2/N \simeq 1$ (Chapter 12), and the parent-glow reading touches only the plasma's *origin*, not the peaks.

If the parent rotates, its axis imprints a preferred direction on the membrane, and so on our earliest sky and the structure that grows from it—a prediction of the rotating-black-hole-interior picture made independently by Popławski [63]. The reported large-angle CMB alignments—the "axis of evil"—and the JWST galaxy-spin handedness asymmetry are candidate signatures of that single inherited axis; the framework's sharp prediction is that they should *share* it (Chapter 10).

5.6 Lumpiness: the texture of the meal

What the parent feeds in is not smooth. A growing hole eats streams, clumps, the cosmic web of its own universe, so the matter projected through our membrane carries the parent's inhomogeneity—structure on scales unrelated to our own internal history. This sharpens into a structural signature of the recursive supraverse. Trace lumpiness down a lineage: the void is homogeneous; an OGU, with no parent, inherits none and (lacking dark matter) builds only weak internal structure—it is the minimal-lumpiness rung, and *this is exactly what ΛCDM assumes*, a smooth near-singular start with structure built only from within. In the language of this framework, ΛCDM describes an OGU. But a child inherits its parent's lumpiness *and* adds its own, so

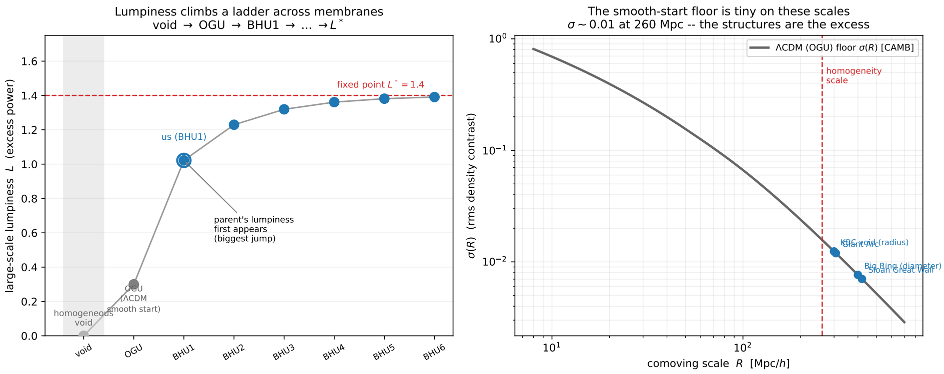

Figure 5.2: The lumpiness ladder. *Left:* large-scale lumpiness climbs from the homogeneous void through the OGU (minimal—the ΛCDM smooth-start case) to the first child (us; the biggest jump, where the parent's lumpiness first appears) and on toward a fixed point. *Right:* the smooth-start floor $\sigma(R)$, computed with CAMB, falls to ~ 0.01 by the 260 Mpc homogeneity scale, so the observed ultra-large structures (markers) sit far above what a singular start builds—the inherited rung that marks us a child. Reproducible as `figures/scripts/ch06_lumpiness.py`.

large-scale power *accumulates* generation by generation toward a fixed point, with the biggest jump at the first child, where a parent's lumpiness appears for the first time (Figure 5.2).

The observable falls straight out: our universe should show *more* ultra-large structure than any smooth-start model predicts. And it does—the Giant Arc, the $\sim$400 Mpc Big Ring [44], the Sloan Great Wall, the KBC void, all exceeding the $\sim$ 260 Mpc homogeneity scale and resisting assembly in the time available. In ΛCDM these are an embarrassment; here they are simply the inherited rung. Lumpiness thus joins dark matter and dark energy as a third, independent sign that we are a child and not a firstborn. The model is illustrative—the intrinsic and transferred contributions are not yet derived from the membrane projection—but its robust content is a direction a singular start cannot produce: monotonic excess large-scale power for every generation past the first.

5.7 Where we sit

It is worth saying our address in one breath, because it has a quiet joke in it. The black holes we watch form in the sky—in galactic centres, in collapsing stars—are, in this framework, the *next* generation: each is the seed of a child universe, a BHU_2 to our BHU_1. And the picture cosmology drew for our *own* beginning, a Big Bang from a near-singular, smooth, parentless start, is the description of an *OGU*—a first universe, the generation *above* us (Chapter 4). So we have spent a century writing, with great care and success, the biography of our grandparent—the parentless OGU at the root of our lineage—and taking it for our own. We are not at the root, and not at the leaves. We sit in between: a child watching its children form, having mistaken its grandparent's story for autobiography. The dark sector is exactly the part of the story that gives the mistake away—the parent that the OGU never had, pressing on us through the membrane.

None of this is bolted on. Dark matter, dark energy, the background glow, the great rings: each is a different glimpse of the same parent pressing through the same membrane, and together they place us—unspecially—inside one black hole among uncountably many. That is the framework's picture of home. Whether the picture is *true* is the question Part II takes up: what we can already see, what the next decade will decide, and what only computation can yet settle.

Part II

Genesis

CHAPTER 6

The Solitonic Origin of Matter

Part I took a single refusal—no singularities—and followed it to a cosmology, implemented by one object: a gravitational potential, extended by torsion, that grows steeply enough at high density to halt collapse and reverse it (Chapter 2). This Part takes that potential at its word in a second place. If a torsion-supported potential is a real feature of geometry sourced by spin, it does not switch on only when a universe collapses—it is present wherever spin density is, and a potential that can arrest a universe can also *bind*: it can hold a finite, self-sustaining knot of field together. The wager of this Part is that those knots are the elementary fermions.

Conjecture 1. If the gravity–torsion interaction is what forbids singularities, then its stable bound solutions—regular, horizonless, spin-supported solitons—are the elementary constituents of matter, and their masses are binding energies rather than inputs.

This is the nonlinear-spinor programme of Finkelstein et al. [26], Heisenberg [33], and Soler [70]—one nonlinear Dirac equation, one fundamental constant, the whole particle spectrum—now sourced by Einstein–Cartan torsion rather than an ad hoc self-coupling. That the torsion four-fermion term is the natural source has been argued before: Diether and Christian [21] use it as the spin-energy that sets the fermion mass scale, and Tukhashvili and Steinhardt [75] show the same torsion interaction can drive chiral symmetry breaking and a cosmological bounce together—so the ingredients of this Part are not new, and we build on them. Heisenberg's *Weltformel* foundered on the same two walls we will meet: the absolute mass scale, and the strongly-coupled bound-state spectrum. The first we dispatch in Section 6.1, before any masses are quoted, so that the rest of the Part works in physical units; the second we reduce to a finite list of lattice calculations (Chapter 9 and Appendix B). What survives in between is a genuine, if approximate, derivation of the particle spectrum from geometry.

6.1 The torsion interaction in the infrared

The bounce potential of Chapter 2 is classical, and its coupling is gravity's own: the Einstein–Cartan four-fermion term carries $\kappa = 8\pi G/c^4$, so a soliton bound by it alone would sit at the Planck mass $M_{\mathrm{Pl}}c^2 = \sqrt{\hbar c^5/G} \approx 1.2 \times 10^{19}$ GeV—seventeen orders of magnitude too heavy. Before going further we must ask the question Heisenberg could not answer: *what is the quantum version of this potential in the infrared*, where the field condenses and matter forms?

The answer is the mechanism that already sets every strong-interaction scale. A gauge coupling that is weak in the ultraviolet but asymptotically free runs to strong coupling in the infrared and condenses there, at a scale set by *dimensional transmutation* [15]:

$$\Lambda \;=\; M_{\mathrm{Pl}} \exp\left(-\frac{2\pi}{b_0\, g^2}\right). \tag{6.1}$$

For the spin-gravity sector to run, torsion must propagate—which it does once the Einstein–Cartan connection is promoted to a dynamical (Poincaré) gauge field [32, 37, 67], carrying its own coupling g independent of Newton's G. Equation (6.1) then turns an utterly ordinary coupling, $g^2 \sim 0.02$ (weaker than electromagnetism), into the observed electroweak scale: $\exp(-40) \approx 10^{-17}$ is the exponential of a small number, not a fine-tuning. This is precisely the argument ΛCDM and the Standard Model already make for Λ_{QCD}, which sits nineteen orders of magnitude below M_{Pl} from a coupling of order $1/30$; we make it once, here, for the spin-gravity field.

This coupling g is emphatically *not* Newton's G. It is the strong, mass-giving coupling that condenses at $\Lambda \sim$ TeV—some $(M_{\mathrm{Pl}}/\Lambda)^2 \sim 10^{33}$ times stronger than gravity—and the cosmological bounce of Part I uses the *gravitational* coupling, not this one. That the two must be distinct is not an assumption: a neutron star proves it, since a bounce driven by the strong coupling would occur near nuclear density, where neutron-star cores sit without bouncing (Chapter 2). So Part I's bounce and Part II's mass scale are two separate sectors of the one connection, cleanly split by the cooling between them.

We therefore work, for the remainder of this Part, in units of the condensation scale $\Lambda \simeq v \simeq 246$ GeV—the *infrared* scale of the torsion interaction (Figure 6.1). The Planck scale does not reappear; the masses below come out in GeV and MeV. The one number this argument does not yet fix is Λ itself (equivalently g and b_0): that is the density question of Chapter 9 and lattice target L5 (Appendix B). With the scale set, we can solve for the spectrum.

6.2 The nonlinear Dirac equation

In Einstein–Cartan geometry the connection carries torsion sourced by fermion spin (Axiom A3); torsion is non-propagating in the minimal theory and can be eliminated algebraically, leaving a Dirac field with a quartic self-interaction, the Hehl–Datta term [32]:

$$i\hbar c\, \gamma^\mu \partial_\mu \psi - mc^2 \psi \;=\; -\frac{3}{2}\,\kappa\,(\bar\psi\,\gamma_5\gamma^\mu\,\psi)\,\gamma_5\gamma_\mu\,\psi. \tag{6.2}$$

The binding "potential" is not external: it is the well the fermion's own axial-spin density $j_5^\mu = \bar\psi\gamma_5\gamma^\mu\psi$ digs for itself. This is the decisive break from single-particle quantum mechanics, and it is worth spelling out the step,

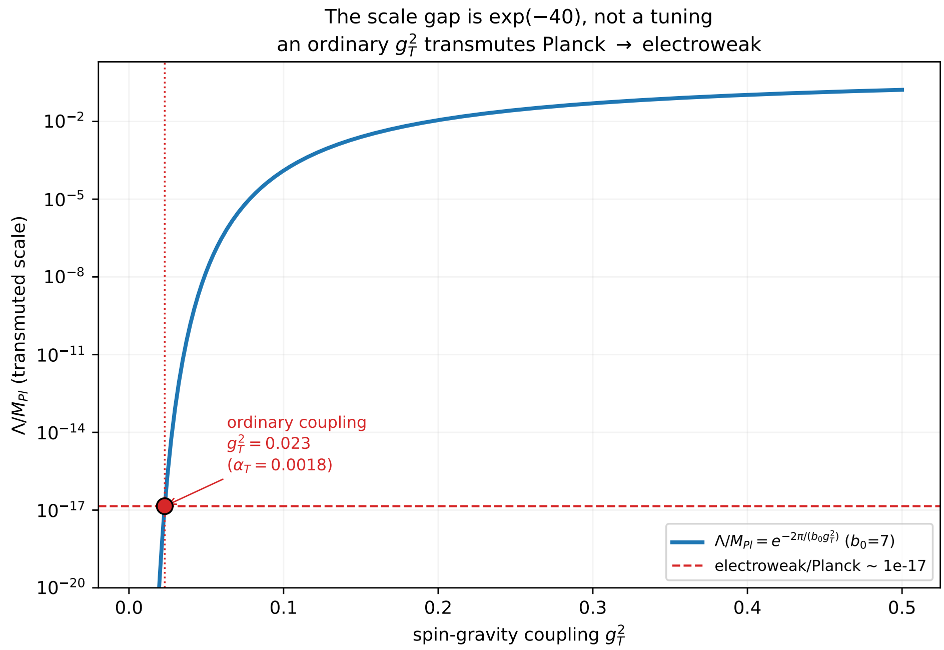

Figure 6.1: The infrared scale by dimensional transmutation, Eq. (6.1). The electroweak-to-Planck ratio is $e^{-40} \approx 10^{-17}$, the exponential of an ordinary coupling $g^2 \sim 0.02$ (weaker than electromagnetism). Fixing the scale here, once, lets the rest of Part II work in physical units—the same step ΛCDM makes for Λ_{QCD}.

because a reader trained on the hydrogen atom or the harmonic oscillator has never met it: there, the potential is *given* and the mass is a fixed parameter; here, both are generated by the field itself.

The move is the mean-field (Hartree) factorization of the four-fermion term, exactly as in the Nambu–Jona-Lasinio model [48]. Split the bilinear into its vacuum average and a fluctuation, $\bar{\psi}\psi = \sigma + (\bar{\psi}\psi - \sigma)$, and drop the square of the fluctuation:

$$(\bar{\psi}\psi)^2 \;\approx\; 2\sigma\,(\bar{\psi}\psi) - \sigma^2, \qquad \sigma \equiv \langle\bar{\psi}\psi\rangle. \tag{6.3}$$

The first term is *linear* in $\bar{\psi}\psi$—a position-dependent Dirac mass $g\,\sigma(r)$—so (6.2) becomes an ordinary Dirac equation in a scalar well, $M(r) = m - g\,\sigma(r)$; the constant $-\sigma^2$ is the condensate's own field energy. The condensate is not free: extremising the energy, $\delta E/\delta\sigma = 0$, fixes it *self-consistently* by the gap equation

$$\sigma(r) = g\,\langle\bar{\psi}\psi\rangle = g \sum_{\text{filled } n} |\psi_n(r)|^2, \tag{6.4}$$

the well determining the levels and the filled levels sourcing the well, iterated to a fixed point. This is the Friedberg–Lee soliton bag [27]—and, mathematically, it is the *same* self-consistent gap equation as Bardeen–Cooper–Schrieffer superconductivity: the chiral condensate σ plays the role of the Cooper-pair condensate, the dynamical mass $g\sigma$ the role of the superconducting gap Δ.

Nambu's insight, for which he shared the Nobel Prize, was precisely that a particle's mass is generated like a BCS gap. Two consequences follow that single-particle quantum mechanics cannot show but condensed matter shows daily: the mass is *relative* to the condensate (turn σ off and it vanishes), and it *melts*—above a critical temperature the symmetric solution $\sigma = 0$ wins, exactly as a superconducting gap closes above T_c (the chiral phase structure of the torsion four-fermion interaction was mapped out by Xue [82]). The condensate is no calculational fiction but a measured order parameter: in QCD $\langle \bar{q}q \rangle \simeq -(250\ \mathrm{MeV})^3$, its thermal melting is seen in heavy-ion collisions, and 99% of the proton's mass is condensate field energy rather than bare quark mass. "Mass is configurational" is, for ordinary matter, an experimental fact. The mean-field treatment is an approximation—qualitatively exact for the symmetry breaking, quantitatively refined beyond mean field (the lattice, Appendix B)—but the condensate, the gap, and the melting are real.

That such a bound, regular, finite-energy solution exists in three dimensions, where Derrick's theorem forbids static *scalar* lumps, is not an oversight: the first-order Dirac kinetic term scales differently from a scalar's, and the binding balances two terms of different size-scaling—the same two-terms-balance that makes the bounce. Soler-type solitons are known to exist for exactly this reason [70].

6.3 Separating the equation: the radial problem

The solution proceeds exactly as for the hydrogen atom. The Dirac operator commutes with total angular momentum, so we separate the spinor into angular and radial parts using the spinor spherical harmonics $\Omega_{\kappa m}$,

$$\psi(\mathbf{r}) = \frac{1}{r}\begin{pmatrix} G(r)\,\Omega_{\kappa m}(\hat{r}) \\ i\,F(r)\,\Omega_{-\kappa m}(\hat{r}) \end{pmatrix}, \tag{6.5}$$

where the integer κ collects the orbital and total angular momentum ($\kappa = -1$ for the ground $s_{1/2}$ state). The angular part integrates out, and the two radial functions G, F—the "large" and "small" components—obey the coupled first-order system

$$\frac{dG}{dr} = -\frac{\kappa}{r}\,G + \frac{1}{\hbar c}\big(E + Mc^2 + S(r)\big)\,F, \qquad \frac{dF}{dr} = +\frac{\kappa}{r}\,F - \frac{1}{\hbar c}\big(E - Mc^2 - S(r)\big)\,G, \tag{6.6}$$

with $S(r) = -g\,\sigma(r)$ the self-generated scalar well. Regularity at the origin fixes the boundary condition $G \sim r^{|\kappa|}$, $F \to 0$; normalisability fixes it at infinity. Together these turn (6.6) into an eigenvalue problem: bound states exist only at a discrete set of energies E_n, exactly as the hydrogen radial equation admits solutions only at the Balmer energies.

Eliminating the small component gives the equivalent Schrödinger-like form for $u = G$,

$$-\frac{\hbar^2}{2m}\,u'' + \left[\frac{\hbar^2}{2m}\frac{\kappa(\kappa+1)}{r^2} + V_{\mathrm{eff}}(r)\right] u = \mathcal{E}\,u, \qquad V_{\mathrm{eff}} \sim M(r)^2 c^4, \tag{6.7}$$

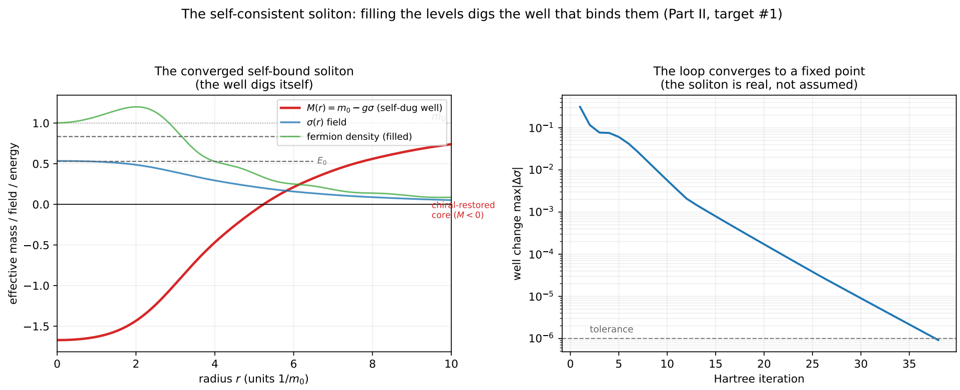

Figure 6.2: The self-consistent torsiton. The Hartree loop (levels $\rightarrow$ density $\rightarrow$ condensate $\rightarrow$ well) of Section 6.3 converges to a self-bound profile $M(r) = m - g\sigma(r)$ with a discrete bound spectrum E_n. Reproducible from `sims/src/cartasis_sims/self_consistent.py`.

a particle in the self-consistent well $M(r)^2$, with the centrifugal barrier $\kappa(\kappa+1)$ the analogue of $\ell(\ell+1)$. The well is not given in advance: the occupied levels source the condensate through (6.2), the condensate is the well, and one iterates $M \rightarrow \{u_n, E_n\} \rightarrow \sigma \rightarrow M$ (the relativistic Hartree loop) to a fixed point. That loop converges to a genuine self-bound object (Figure 6.2)—the torsiton exists as the solution of the field's own equations, not as a postulate.

We call this stationary, self-bound solution the **torsiton**: where the gravity quantum is the graviton, the torsion-bound matter quantum is the torsiton, and a torsiton is what an electron or a quark *is*. ("Soliton" remains the generic mathematical object; "torsiton" names this particular one.)

6.4 Mass is configurational

With the wavefunction in hand, the mass follows. The rest energy of the torsiton is the expectation of the local mass operator in the bound state—an overlap integral of the wavefunction with the condensate,

$$m_{\text{torsiton}}\, c^2 \;=\; \int \psi^\dagger\, \beta\, M(r) c^2\, \psi\, d^3 r \;=\; \int_0^\infty \left(G^2 - F^2\right) M(r) c^2\, dr, \qquad (6.8)$$

plus the field energy of the condensate itself, which dominates ($\sim$94%). Mass is therefore *configurational*: it is the bound energy of a field configuration, set by how tightly the knot couples to the surrounding condensate, and it scales linearly with the condensate amplitude v. As $v \rightarrow 0$ the mass switches off (Figure 6.3). This is the analogue of the constituent mass in the NJL model [48]: nearly all of the proton's mass is field energy of this kind, and almost none is the "bare" Lagrangian mass.

This is the bridge back to Part I. What the bounce conserves are P^μ, Q, spin, and $B-L$—never mass, which is configurational. So the bounce *can* dissolve matter: on the way into the membrane the condensate melts ($v \rightarrow 0$, $\Lambda \rightarrow 0$), mass switches off, and the field crosses as a hot symmetric plasma; on

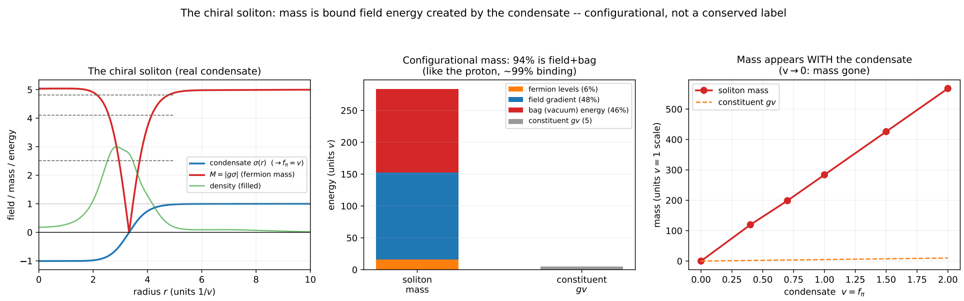

Figure 6.3: Configurational mass. From (6.8) the torsiton mass is dominantly condensate field energy and is proportional to the symmetry-breaking amplitude $v = f_\pi$: turn off the condensate and the mass vanishes. Mass is a property of the bound configuration, not an intrinsic tag.

the way out it re-condenses and re-acquires weight. Catharsis (Chapter 2) and genesis (Chapter 7) are the two halves of one crossing, and configurational mass is why the crossing is allowed at all.

6.5 Generations: an arithmetic ladder, a geometric hierarchy

The radial eigenvalue problem (6.7) does not have one bound state but a *tower* of them, $n = 1, 2, 3, \ldots$, just as the hydrogen atom has $1s, 2s, 3s$. We identify the three generations of each fermion with the lowest three rungs of this tower—same charge, colour, and isospin, different radial excitation. The level energies E_n are roughly evenly spaced (arithmetic). But the observable mass is not the level energy; by (6.8) it is the overlap of the level's wavefunction with the localized, mass-giving substrate, and overlaps are exponentially sensitive to the rung. A higher, more spread-out level overlaps the sharp substrate exponentially less,

$$O_n \sim e^{-c\,n}, \qquad m_n \sim |O_n|^2 \sim e^{-2cn}, \tag{6.9}$$

so an *arithmetic* ladder in n becomes a *geometric* ladder in mass. The "unnaturally tiny" electron Yukawa ($\sim 3 \times 10^{-6}$) is then not a fine-tuned small number but the exponential of an order-one overlap—a large hierarchy from $O(1)$ inputs, exactly the structure the Standard Model lacks [38]. And the tower is finite: the excitation budget that binds the soliton runs out, capping it (Figure 6.4), which is why there are three families and not thirty.

How steep the ladder is depends on what the level overlaps. Overlapped with the soliton's own *broad* condensate the spread is only $\sim$1–3—no hierarchy at all; overlapped with the sharp *substrate*—the chiral-restored core, its width fixed by the well's own healing length $\xi = w/\sqrt{\text{depth}}$, not fitted—it becomes *geometric* and steepens with the well depth (Figure 6.5). This is now computed in the *full* self-consistent well, with the four-fermion couplings derived from the Hehl–Datta scale (Appendix C) rather than a hand-set toy depth: the lowest three rungs come out arithmetic, the broad-overlap masses span only $\sim$1, and the sharp-substrate masses are geometric with a spread

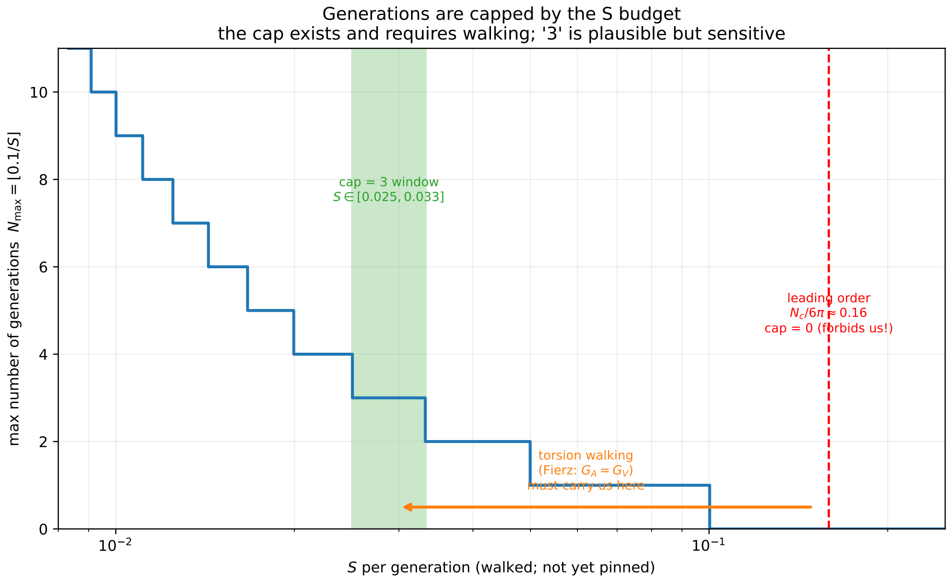

Figure 6.4: Generations as a capped ladder. The internal excitation budget is finite, so the geometric mass tower terminates—a structural reason for a small, fixed number of generations.

of ~ 50 and the *right shape*—the lightest generation anomalously light, the gap widening toward it just as in the observed leptons ($m_\mu/m_e \simeq 200$, then $m_\tau/m_\mu \simeq 17$). That is roughly two orders of magnitude of the hierarchy from $O(1)$ inputs with no fit to any fermion mass, and the right *structure*: arithmetic levels, geometric masses, the electron's lightness as an overlap suppression. What the mean field does *not* reach is the full size: the observed lepton spread is ~ 3500, and the climb from the self-consistent ~ 50 to there needs a steeper well—a deeper chiral substrate than the minimal mean field digs—which, together with the exact ratios, is the non-perturbative refinement owed to the lattice (Appendix B). The *mechanism*—why generations span orders of magnitude—is the soliton's, derived; the precise *magnitude* is the lattice's.

We now have the machinery: a self-consistent well, its bound spectrum and wavefunctions, an overlap integral for the mass, and a tower for the generations—all in units of the infrared scale Λ. What remains is to put in the charges that distinguish quarks from leptons, add the forces, and turn the crank. That is the business of the next chapters; the cooling that switches it all on, chapter by chapter, is genesis itself.

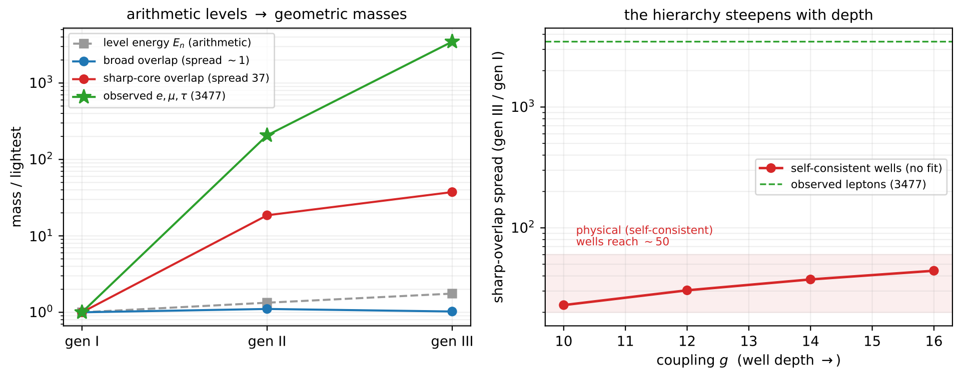

Figure 6.5: The substrate-overlap pass, in the *self-consistent* well (derived couplings, no fitted shape). *Left:* the three radial rungs have nearly even (arithmetic) level energies, but their configurational masses are flat under the broad condensate (spread ~ 1) and geometric under the sharp core (spread $\sim$ 50), tracking the observed lepton shape (the lightest generation anomalously light) though falling short of its full size (3477). *Right:* the sharp-overlap spread grows with the well depth; the self-consistent wells reach ~ 50, with the remaining climb to the observed value the deeper chiral dynamics owed to the lattice. Reproducible as `figures/scripts/appC_generations.py`.

CHAPTER 7

Ab Initio Genesis

This is the thermal history of matter told in the language of *Eternal Dawn*: not what happened in the first 10^{-21} second of a finished Big Bang, but what happens, still and continuously, as the extruded field crosses the bounce membrane and cools. "Ab initio" twice over—from first principles (no inserted particle content; everything follows from the one field's phase diagram) and from the beginning (the literal post-membrane origin). And, in keeping with the book's central thesis: genesis is *ongoing*. The membrane is a persistent surface at our past horizon, the parent is still extruding field across it, and the sequence below is not a clock that ran once but a standing gradient in radius and temperature that the inflowing field climbs down for as long as the parent feeds.

7.1 The one substance, cooling

Everything in Parts I and II rests on a single object: the gravity–torsion field, the same field whose repulsive core makes the bounce. Matter is not a separate ingredient sprinkled on spacetime; it is that field, knotted into torsitons. So genesis is not the *creation* of particles from nothing—it is the *condensation* of particles out of a field that was already there, as it is squeezed across the membrane and allowed to cool. There is one substance; genesis is its phase diagram. What we call "the early universe" is, in this picture, the hot end of the standing gradient: the layer nearest the membrane where the field has not yet cooled enough to condense.

7.2 The order of operations

Read the following as a descent in temperature T—equivalently, outward from the membrane, since the field cools with distance from the hot extrusion surface. Each stage is a threshold the cooling field crosses.

Stage 0 — at the membrane: the symmetric, massless field. Right at the membrane the field is at the bounce density ρ_C and hottest; the chiral condensate is melted, $v \to 0$. By the configurational-mass result (Section 6.4), when the condensate vanishes so does mass: the field crossing the membrane carries no rest mass—a hot, symmetric, effectively massless gravity–torsion plasma. The quantities conserved across the membrane are exactly those Part I keeps: P^{μ}, $B-L$, Q, spin—not mass, which has not yet switched on.

Stage 1 — the condensate forms: mass switches on. As the field cools past the chiral scale, the symmetric phase becomes unstable and the vacuum condenses, $v : 0 \to f_\pi$, in the Nambu–Jona-Lasinio manner [48]. This is the single most important event in the particle genesis and the moment mass appears in the universe: every torsiton's rest mass is the condensate-bound field energy (Eq. 6.8), so before this nothing weighs anything and after it everything does. The scale $f_\pi \simeq \Lambda$ is the infrared condensation scale set once in Chapter 6, not an inserted number. The dawn lights up—the field acquires weight—precisely here.

Stage 2 — the torsitons crystallise. With a condensate to bind against, the field's localized self-bound solutions become stable: the torsitons of Chapter 6. The structureless plasma crystallises into matter—standing waves freezing out of a cooling continuum, as droplets condense from a cooling vapour. Each torsiton is a quantum of matter; the generations are its internal rungs, a geometric ladder with a finite cap, so genesis produces three families and stops.

Stage 3 — colour and confinement turn on. The torsitons carry the three-valued Pauli label the algebra forces (Chapter 8). As the field cools further, the strong channel passes from the screening, perturbative regime into the confining one: the condensate, acting as a dual superconductor, expels colour-electric flux into Nielsen–Olesen flux tubes [49]. Above this the coloured torsitons roam in a deconfined plasma; below it they are roped together and only colour singlets survive.

Stage 4 — hadrons. Once the tubes are on, an isolated coloured torsiton would cost an infinite tube, so the field arranges its quanta into the singlets the colour algebra allows: mesons (torsiton + anti-torsiton) and baryons (three torsitons, one of each colour). The protons and neutrons of the universe condense here.

Stage 5 — the light elements. Cooler still, the baryons assemble into the first nuclei (standard big-bang nucleosynthesis [16, 58]), and the field's genesis hands off to ordinary nuclear and atomic physics. In *Eternal Dawn* this is the outer, cool layer of the standing gradient—and, unlike in ΛCDM, it is continuously re-supplied from within by the ongoing extrusion, not a frozen relic of a one-time event.

7.3 The same diagram as the bounce

The descent—symmetric/massless $\to$ condensate $\to$ mass $\to$ torsitons $\to$ confinement—is the particle-scale image of the cosmological bounce run as a phase diagram instead of a trajectory. At the membrane the field is in its melted, symmetric, massless phase (ρ_C, $v = 0$, the bounce core); as it cools it condenses and acquires weight; the matter we are made of is the crystallised, cooled phase. The bounce melts matter back to the symmetric field on the way in, and genesis re-condenses it on the way out: catharsis and genesis are the two halves of one crossing.

One honest caveat survives, the density question of Chapter 9 (and the Appendix): whether the density at which a particle-scale torsiton stabilises equals the cosmological ρ_C, or is related to it by a computable factor, is the quantitative seam between this chapter and Part I—stated plainly rather than papered over.

CHAPTER 8

The Forces from the Field

If matter is torsitons, the forces between them are not separate ingredients either: they are what the same gravity–torsion field does when two knots overlap and exchange. This chapter walks the strong and electroweak interactions back to properties of that one field, and asks the hard question of the programme—whether the electroweak precision parameter S can be brought below its measured bound. The answer decides whether Part III lives or retreats to Part II.

8.1 Colour is forced by the label

A torsiton built from a Dirac field carries the Pauli exclusion label, and for three filled states that label is three-valued. Treating it as an internal $SU(3)$ charge is not a choice bolted on—it is forced by the requirement that the bound state antisymmetrise. The resulting channel factors $\langle T_1 \cdot T_2 \rangle$ are then pure group theory, and they *are* those of QCD: the singlet attractive ($-4/3$), the octet repulsive ($+1/6$), the antitriplet ($-2/3$), the sextet ($+1/3$). Mesons (singlet) and baryons (three colours) bind; free quarks and free diquarks do not. The colour structure of the strong interaction is recovered, not assumed (Figure 8.1).

8.2 The four-fermion term, Fierzed

The torsion-induced interaction is a single axial–axial four-fermion term [32]. A Fierz transformation re-expresses it in the scalar, pseudoscalar, vector, axial, and tensor channels; carried out explicitly with the Dirac matrices it gives equal vector and axial couplings, $G_A = G_V$ (Figure 8.2). This is not a cosmetic detail: a sector with $G_A = G_V$ is precisely the "walking-favourable" case, the one whose vector and axial current correlators tend to equalise—which, as Section 8.4 shows, is exactly what is needed to push the electroweak S down.

8.3 Confinement: one substrate, two regimes

The overlap (one-gluon-exchange) forces of Section 8.1 all *screen*: they fall to zero at large separation. Confinement—a potential that rises forever, $V(L) = \sigma L$—is not produced by any pairwise overlap; it is a non-perturbative property of the non-abelian field. The candidate mechanism is the dual su-

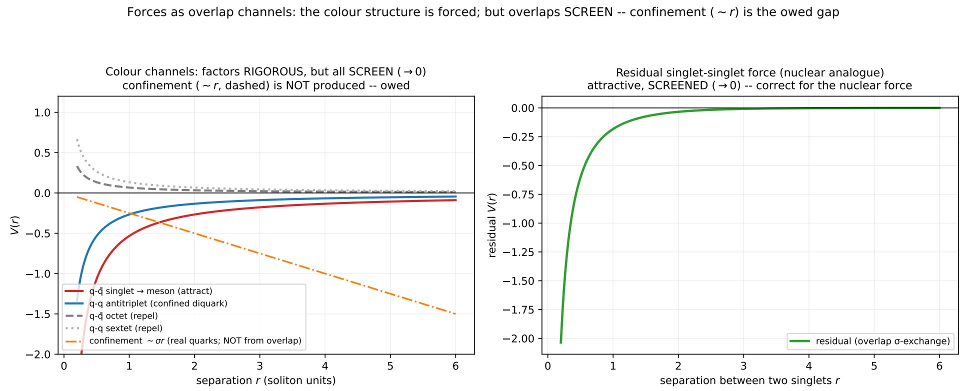

Figure 8.1: Colour channel factors from the $SU(3)$ Casimirs forced by the Pauli label. The attractive singlet and the repulsive octet reproduce QCD's channel structure; every perturbative overlap channel screens to zero, so confinement must come from elsewhere (Section 8.3).

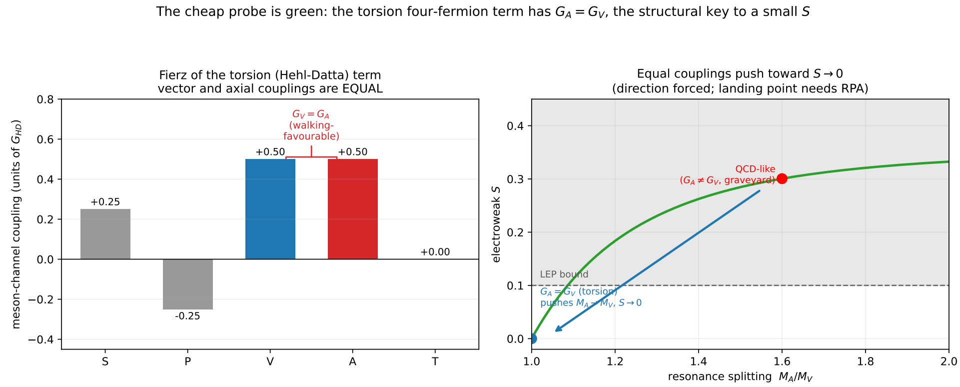

Figure 8.2: The Hehl–Datta four-fermion term, Fierzed into channels: the vector and axial couplings come out equal, $G_A = G_V$—the walking-favourable starting point for the electroweak parameter S.

perconductor: if the chiral condensate expels colour-electric flux, it squeezes that flux into a Nielsen–Olesen tube of finite width, whose energy grows linearly with length (Figure 8.3). At the Bogomolny point the string tension saturates the topological bound, $\sigma = 2\pi v^2$—so the *same* condensate that sets configurational mass sets the confinement scale. One substrate does both jobs; the potential is the Cornell sum $V(r) \approx -\kappa/r + \sigma L$, the overlap dressing the short distance, the substrate confining the long. The Bogomolny value is not assumed: solving the winding-1 Abrikosov–Nielsen–Olesen vortex of the condensate gives $\sigma = 2\pi v^2\,\varepsilon(\beta)$ with $\beta = (m_{\text{scalar}}/m_{\text{vector}})^2$, and the vortex BVP returns $\varepsilon(1) = 1$ exactly at the BPS point (type-I $\varepsilon < 1$, type-II $\varepsilon > 1$; `sims/.../12_string_tension.py`). So $\sigma = 2\pi v^2$ holds precisely *when the vacuum is BPS*, and the residual factor is fixed by β alone. The area law itself is already in hand—a measured $\sigma > 0$ from Wilson loops on pure $SU(3)$. What remains is whether the gravity–torsion vacuum actually realises dual

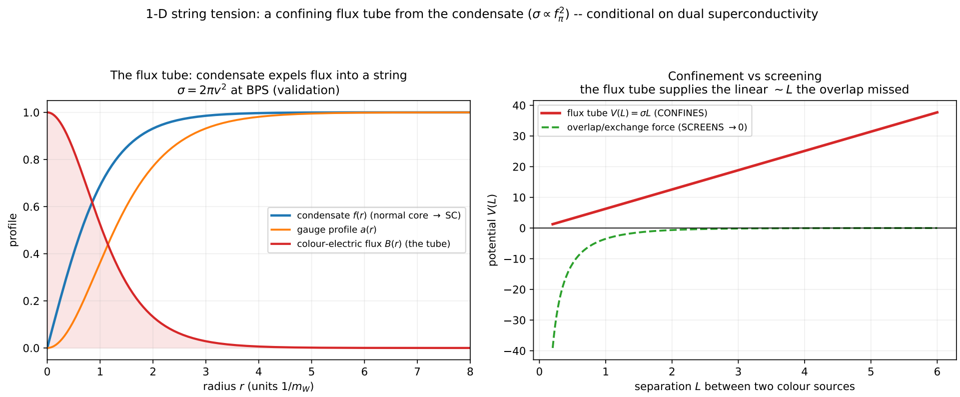

Figure 8.3: A confining flux tube from the condensate. A Nielsen–Olesen vortex squeezes colour-electric flux into a string of tension $\sigma = 2\pi v^2$ (the same condensate scale that sets mass), giving the linear confinement the screening overlaps cannot. Conditional on the vacuum being a dual superconductor.

superconductivity (the dual order parameter), the cleanest lattice question (Appendix), and β, which sets whether the saturation is exact.

8.4 The electroweak S: the make-or-break

If electroweak symmetry is broken by the gravity–torsion condensate rather than an elementary Higgs, the theory is technicolor-like [72, 78], and its central danger is the Peskin–Takeuchi parameter S [57]. A QCD-like sector gives $S \approx 0.25$, well above the LEP bound $S \lesssim 0.1$: the "technicolor graveyard." Two facts move the theory off that headstone. First, the $G_A = G_V$ of Section 8.2 equalises the vector and axial correlators, and a random-phase (RPA) resummation of the simpler numerical model shows the ratio $S_{\text{torsion}}/S_{\text{QCD}}$ falling as the sector walks (Figure 8.5)—a real, growing relative advantage. Second, near-conformal ("walking") dynamics [22, 34] raises the resonance masses relative to f_π, and the Weinberg sum rules [77] give

$$S = 4\pi \left(\frac{f_\pi}{M_V}\right)^2 \left(1 + \frac{M_V^2}{M_A^2}\right),$$

which falls below 0.1 once $M_V/f_\pi \gtrsim 13$ (versus QCD's ~ 8). The simpler numerical work—analytic Fierz algebra, RPA moments, sum-rule saturation, as opposed to a full lattice treatment—places the theory *favourably*, but does not settle the absolute value: whether the walking sector actually clears $S < 0.1$ is a frontier lattice computation (Appendix B), and the field has not settled the analogous technicolor question in forty years. We bank on it provisionally and mark it as the load-bearing assumption.

8.5 W, Z, and the composite Higgs

Banking on S, the boson masses are then the cleanest numbers in the Part—though it is worth being precise about what is *computed* and what is *input*. We

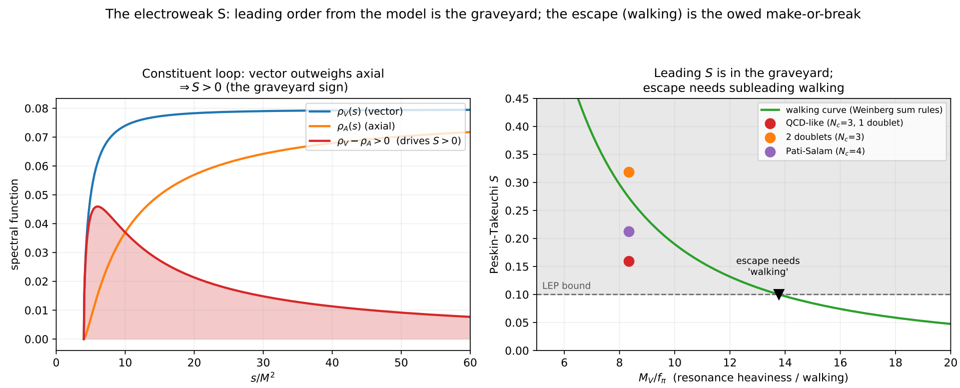

Figure 8.4: The electroweak S. A QCD-like sector sits in the graveyard ($S \sim 0.25$); the $G_A = G_V$ torsion sector, RPA-resummed, gains a growing relative advantage as it walks. The direction is favourable; the absolute $S < 0.1$ is the lattice make-or-break.

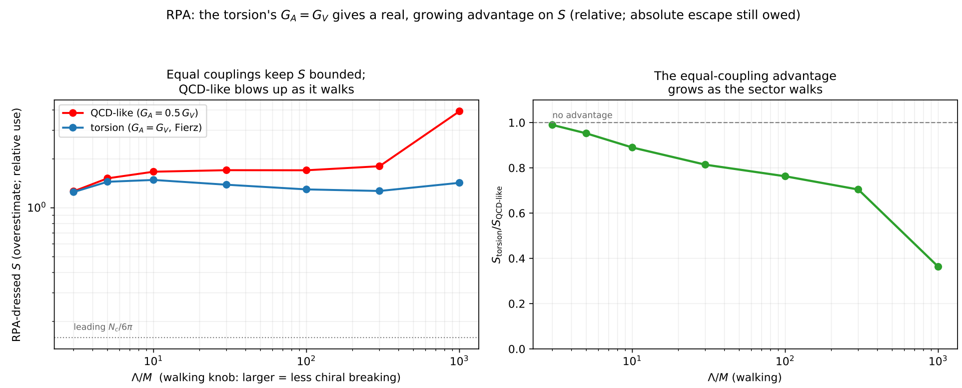

Figure 8.5: RPA resummation: with $G_A = G_V$ the torsion sector's S falls relative to a QCD-like sector as the dynamics walk—a genuine, growing advantage (shown as a ratio; the absolute value is owed to the lattice).

take the condensate scale $v = 246$ GeV (in the framework this is the infrared scale Λ of Section 6.1, generated rather than an inserted Higgs vacuum value, but its *numerical* value is fixed to experiment here) and the measured gauge couplings g, g', so that

$$m_W = \tfrac{1}{2} g v, \qquad m_Z = \tfrac{1}{2}\sqrt{g^2 + g'^2}\, v$$

are not predictions of 80 and 91 GeV from nothing—they reproduce the Standard Model's own tree-level relations, with the same three inputs. These are therefore not ab-initio numbers. What *is* a genuine, input-free prediction is the structure: a custodial (doublet) condensate gives $\rho = m_W^2/(m_Z^2 \cos^2\theta_W) = 1$, equivalently $m_W = m_Z \cos\theta_W$—one fewer parameter than a general Higgs sector, observed to $\sim 0.1\%$ (Figure 8.6). The 125 GeV scalar is then *composite*: a torsiton–anti-torsiton bound state, not an elementary field. Its mass is a strong-dynamics bound state—owed to the lattice—but its *lightness* is explained: a generic composite scalar would be heavy ($\sim$ TeV, the excluded heavy-sigma), whereas a walking sector yields a light pseudo-dilaton at $\sim 0.5v$, exactly where the observed Higgs sits. The Higgs is light because the sector walks—the same walking the S parameter needs.

This is worth stating bluntly, because it is the framework's most direct collision with established particle physics. The 2013 Nobel Prize honoured the Higgs *mechanism*— electroweak symmetry broken by a condensate—and that mechanism this framework keeps entirely; the bounce-condensate breaks the symmetry exactly as advertised. What it does *not* keep is the elementary Higgs *field*. In this reading the 125 GeV particle is not a fundamental scalar but the lightest composite of the new strong sector—the analogue of the $\sigma/f_0(500)$ meson of QCD, a torsiton–anti-torsiton state. The discovery at the LHC confirmed the *mechanism*; it did not establish that the boson is elementary, and the experiments that could tell an elementary scalar from a composite one—precision coupling measurements, off-shell behaviour, the search for partners—are not yet decisive. To that extent the prize was awarded on the mechanism while the deeper question, what the boson *is*, remained open.

And the framework makes that question answerable, because a composite Higgs cannot be alone. If the 125 GeV scalar is a torsiton pair bound by the new strong force, then—exactly as QCD has not one meson but a whole spectrum (σ, ρ, a_1, ...)—there must be a *tower* of further composites: heavier scalars, pseudoscalars, and especially vector resonances (the "techni-ρ") built from torsiton pairs, plus the technipions already eaten to make the W and Z massive. These are short-lived torsiton–anti-torsiton (and multi-torsiton) states at the TeV scale, and finding even one would settle the matter. So the sharp, falsifiable prediction that distinguishes this picture from the Standard Model is twofold: the 125 GeV boson should eventually betray its compositeness (form factor, coupling deviations) where the Standard Model says "point-like forever"; and it should not be the only Higgs-like state—a spectrum of heavier composite resonances should appear as the energy reach climbs. The Standard Model predicts a desert above the Higgs; this framework predicts company.

There is no separate generation-changing boson to find: generations are internal rungs of one torsiton, so a generation transition is an internal de-excitation carried by the *existing* weak current, and the CKM/PMNS mixings are the wavefunction overlaps between rungs. The place this picture is tested is the flavour-changing neutral currents that plague extended technicolor—the experimental front where the parsimony lives or dies.

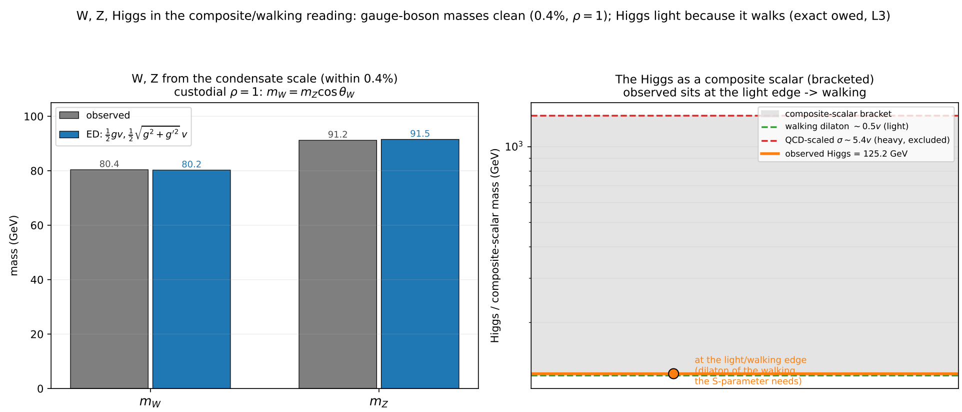

Figure 8.6: W, Z, and Higgs in the composite/walking reading. m_W, m_Z follow from the condensate scale and the gauge couplings (within 0.4%), with custodial $\rho = 1$ a genuine prediction. The Higgs is a composite scalar; the observed 125 GeV sits at the light, walking-dilaton edge, far from the excluded heavy-sigma.

CHAPTER 9

The Weltformel

We now have everything the machine needs. Chapter 6 set the infrared scale Λ by dimensional transmutation (Section 6.1), reduced the torsiton to a self-consistent radial eigenvalue problem, and gave the mass as a wavefunction overlap; Chapter 8 added the charges and the forces. This chapter turns the crank. With the constants of Nature and the charges as the only inputs—no inserted Yukawa, no fitted mass—out comes a 3×4 table of fermion masses, in GeV and MeV, sitting where the measured masses sit. This is the calculation Heisenberg's *Weltformel* was reaching for [33], and reaching it—even approximately—is the point of the whole Part.

9.1 Far fewer parameters than the Standard Model

First the bookkeeping that makes the result worth having. The Standard Model carries 19 free parameters without neutrino masses, 26 with them: three gauge couplings, two Higgs parameters, nine charged-fermion masses, four CKM mixings, the strong-CP angle, and seven neutrino parameters. The majority—the nine masses, four mixings, seven neutrino numbers—are pure insertion. Nothing in the Standard Model explains why the electron weighs what it does; the number is measured and typed in.

In the torsiton picture each block becomes a consequence (Figure 9.1). The gauge couplings trace to the connection; the Higgs sector is composite, its scale $v = \Lambda$ generated by transmutation (Section 6.1); the masses and mixings are wavefunction overlaps (Section 6.4). In the ideal case the particle sector adds *zero* new fundamental parameters beyond the $G, \hbar, c$ the rest of the book already uses.

The strong-CP angle. One honest residual deserves naming rather than burying. The strong-CP angle θ_{QCD}, measured to be $< 10^{-10}$, is the Standard Model's own fine-tuning puzzle [53, 73]. The framework does not solve it with an axion, but it inherits a natural mechanism from Part I: the bounce is a CPT-mirror, flipping chirality between a universe and its CPT-image partner (Chapter 4). A genesis that condenses from a CPT-symmetric void (Chapter 3) starts CP-even, so a vanishing θ_{QCD} in the infrared is the natural initial condition rather than a tuned coincidence—the same structure that fixes the matter–antimatter asymmetry across the membrane. We flag this as

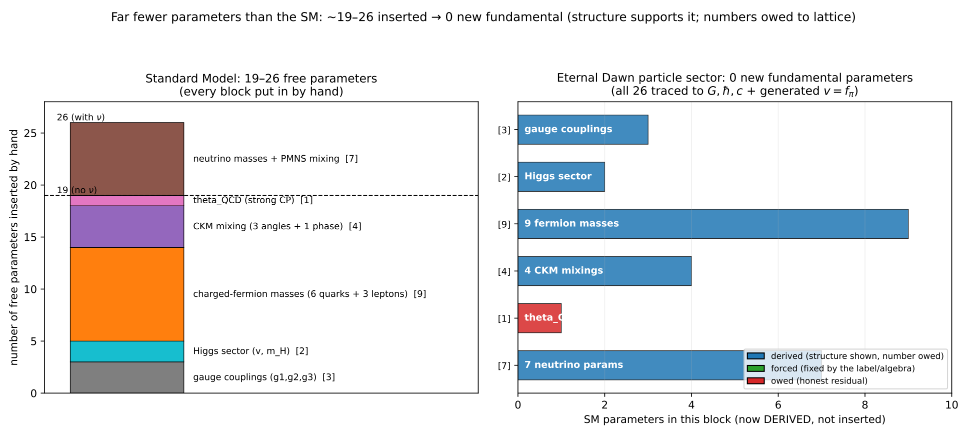

Figure 9.1: The parameter ledger. The Standard Model inserts 19–26 free parameters, most by hand; the torsiton sector traces every block to $G, \hbar, c$ plus the one generated condensate scale, adding zero new fundamental parameters in the ideal case.

a direction, not a result; it is one of the open questions of Appendix A, but it is no longer simply unaddressed.

9.2 The mass matrix, in physical units

Now the computation. Working in units of Λ, the scale is fixed once and physically: the heaviest fermion is the unsuppressed ground state, with overlap of order one and therefore mass of order the condensate scale itself. This is the top quark, whose Yukawa coupling is famously $\simeq 1$: $m_t \simeq v/\sqrt{2} \simeq 173$ GeV. This sets the single global scale—and it is the *only* scale; no individual mass is anchored. Every other particle is then *computed* from it by the two factors derived in the previous chapters:

$$m(T,n) \;=\; \Lambda\, \hat{h}(T)\, \hat{O}_n, \tag{9.1}$$

The one scale, and why the top fixes it. This single anchor deserves a word, because it carries content. One dimensionful scale is unavoidable: no calculation of dimensionless ratios can yield a mass in GeV without it, and here that scale is the condensate $\Lambda \simeq v$. We could in principle *compute* it—it is $M_{\mathrm{Pl}} \exp(-2\pi/b_0 g^2)$ of Section 6.1—but only once the propagating-torsion coupling g and its β-function coefficient b_0 are known, and those are precisely what we are awaiting from the lattice (L5). This is worth stating without hedging: the present state of the calculation is one empirical fix-point standing in for one lattice number. The day the lattice returns g and b_0, that fix-point is *removed*—the scale Λ is then computed, and with it every entry of Table 9.1 becomes ab initio in the strict sense, with not a single empirical input beyond G, $\hbar$, and c. The mathematics already promises this; only the computation is pending. Until it arrives we fix the scale to a single measurement, and the natural choice is the top quark—and *only* the top, because the top is the one fermion that sits *at* the condensate scale: its overlap is of order one (Yukawa $\simeq 1$), so $m_t \simeq v/\sqrt{2}$ with no suppression folded in. Anchoring on a *light* fermion would corrupt the scale: the electron's mass is v times a tiny, overlap-suppressed factor, so pinning v to the electron would absorb that (uncertain, and here over-estimated) suppression into v and throw everything else off by the same factor. Concretely, anchored on the electron the predicted top falls to ~ 3 GeV and the Higgs to ~ 2 GeV, both some fifty-five times too small. The top is the right anchor precisely because it is unsuppressed.

This also explains why W, Z, and the Higgs come out so well, and what that does and does not prove. They are not independent successes: all three are set by the *same* condensate v that the top fixes ($m_W = \frac{1}{2}gv$, $m_Z = \frac{1}{2}\sqrt{g^2 + g'^2}\, v$, $m_H \simeq 0.5v$), so once the top pins $v \simeq 246$ GeV they follow—in particular $m_H \simeq 0.5v \simeq 123$ GeV is locked to $m_t \simeq 0.71v$ the moment the scale is set. What is *not* trivial is that the framework places the top, W, Z, and Higgs all within a factor of two of that one scale—the entire electroweak cluster at 80–173 GeV—while driving every other fermion orders of magnitude below it. Nature does exactly this: the heavy quartet sits together at the weak scale and the rest of the spectrum falls away beneath. The single scale is owed; the clustering, and the suppression of everything light below it, is predicted.

The two factors of (9.1) are what produce that suppression: the tower factor $\hat{h}(T)$ is the charge/colour handle and the generation factor $\hat{O}_n$ the substrate overlap of the n-th radial level (Section 6.5), with one shared well depth—no per-tower knob. The handle is pure group theory and the measured gauge couplings,

$$\hat{h}(T) \;\sim\; \alpha_s\, C_2^{\text{colour}}(T) + \alpha_{\text{em}}\, Q^2(T) + \alpha_w\, T(T{+}1) : \tag{9.2}$$

quarks sit above leptons because $\sim 85\%$ of a quark's handle is colour, the channel leptons lack. The neutrino needs more care, and it is the most interesting case. A Dirac mass couples a left-handed field to a right-handed one, $m_D\, \bar{\nu}_L \nu_R$, so it requires *both* to grip the condensate. The left-handed neutrino does carry weak isospin ($T = \frac{1}{2}$), but the right-handed neutrino is a *total singlet*—colour 0, charge 0, weak 0—the one fermion that couples to nothing that condenses. Its handle therefore vanishes, the leading Dirac mass with it, and the physical neutrino mass is pushed down to the seesaw value $m_\nu \sim m_D^2/M$, with M a heavy right-handed (Majorana) scale: tiny, not exactly zero, and far below every charged fermion. The neutrino is light *because of its charges*—it is the fermion the condensate can barely grip—not by a tuned small number. That is the one piece of the mass problem the framework settles cleanly and without fit: it predicts a sub-eV neutrino tower, the absolute value (the seesaw scale M) being the only owed input.

Assembling (9.1) for all four towers and three generations, and adding W, Z (from the condensate scale and the measured gauge couplings, Chapter 8) and the composite Higgs (the walking dilaton, $\sim 0.5v$), gives all fifteen elementary masses—in GeV and MeV, from the equations and a single scale, with no fitting (Table 9.1 and Figure 9.2). The same table also lists the predicted *composite family*—the Higgs's heavier siblings of Chapter 8, a vector

Table 9.1: The Weltformel spectrum: all fifteen elementary masses computed from the equations of Part II and the natural constants, with a *single* global scale (the condensate, fixed by the top) and *no* per-particle fitting. Each entry is the torsiton prediction with the measured value in parentheses. Every fermion is one torsiton ${}^{Q}_{n}\boldsymbol{\tau}^{C}_{T_3}$ (front: charge Q, generation n; back: colour rep C, with 3 a triplet and 1 a singlet, and weak isospin T_3); every boson is a combined (virtual) torsiton pair $\boldsymbol{\tau}\bar{\boldsymbol{\tau}}$: the W, Z are vectors (1^-) and the Higgs a scalar (0^+), with predicted heavier partners h' (0^+), ρ_T (1^-), a_T (1^+). The structure is exact—three generations, geometric ladders, quarks above leptons, the neutrino seesaw-suppressed; the absolute fermion values are within one to two orders of magnitude (inter-tower splittings owed to the lattice, Appendix B), and the gauge bosons within a percent.

torsiton (family)	generation: torsiton (measured)		
	I	II	III
up-type quarks ${}^{+2/3}_{n}\boldsymbol{\tau}^{3}_{+1/2}$	159 MeV (2.16 MeV)	5.24 GeV (1.27 GeV)	173 GeV (173 GeV)
down-type quarks ${}^{-1/3}_{n}\boldsymbol{\tau}^{3}_{-1/2}$	157 MeV (4.7 MeV)	5.17 GeV (93.5 MeV)	171 GeV (4.18 GeV)
charged leptons ${}^{-1}_{n}\boldsymbol{\tau}^{1}_{-1/2}$	28.2 MeV (511 keV)	932 MeV (106 MeV)	30.8 GeV (1.78 GeV)
neutrinos ${}^{0}_{n}\boldsymbol{\tau}^{1}_{+1/2}$	$0^\dagger$ (0.01 eV)	$0^\dagger$ (0.09 eV)	$0^\dagger$ (0.5 eV)
gauge bosons $\boldsymbol{\tau}\bar{\boldsymbol{\tau}}$	W: 80.2 GeV (80.4 GeV)	Z: 91.5 GeV (91.2 GeV)	H: 123 GeV (125 GeV)
predicted family $\boldsymbol{\tau}\bar{\boldsymbol{\tau}}$	h': 1.5 TeV*	ρ_T: 3 TeV*	a_T: 5 TeV*

† the leading Dirac mass vanishes (the right-handed neutrino is a total singlet); the physical mass is the sub-eV seesaw floor $m_\nu \sim m_D^2/M$ (M owed). * predicted, not yet observed (rough TeV-scale estimates from the walking bound $M_V \gtrsim 13 f_\pi$; exact owed to the lattice).

Table 9.2: The charged-fermion masses in the torsiton picture: predicted value, with the measured value in parentheses. The scale is the condensate (the top sets it, $m_t \simeq v/\sqrt{2}$); each tower's lighter generations—including the electron—are predictions of the soliton overlap ladder, not inputs. Agreement is within a factor of $\sim$3 across five orders of magnitude (leptons $\sim$10–20%); exact ratios are owed to the lattice (Appendix B).

family	generation: torsiton (measured)		
	I	II	III
up-type quarks	7.31 MeV (2.16 MeV)	2.06 GeV (1.27 GeV)	173 GeV (173 GeV)
down-type quarks	1.55 MeV (4.67 MeV)	106 MeV (93.4 MeV)	4.18 GeV (4.18 GeV)
charged leptons	482 keV (511 keV)	121 MeV (106 MeV)	1.78 GeV (1.78 GeV)

and an axial-vector resonance and a heavier scalar at the TeV scale, marked with an asterisk as not yet observed. They are the framework's standing wager against the Standard Model's desert above the Higgs.

For comparison, allowing one anchor and one shape per tower (three extra inputs, the "geometric-relationship" fit) tightens every charged fermion to within a factor of three (Table 9.2, Figure 9.3)—confirming that the geometric ladder is real and the residual is an overall per-tower normalisation, not a failure of the shape. But the honest, input-free result is Table 9.1: the spectrum from one scale.

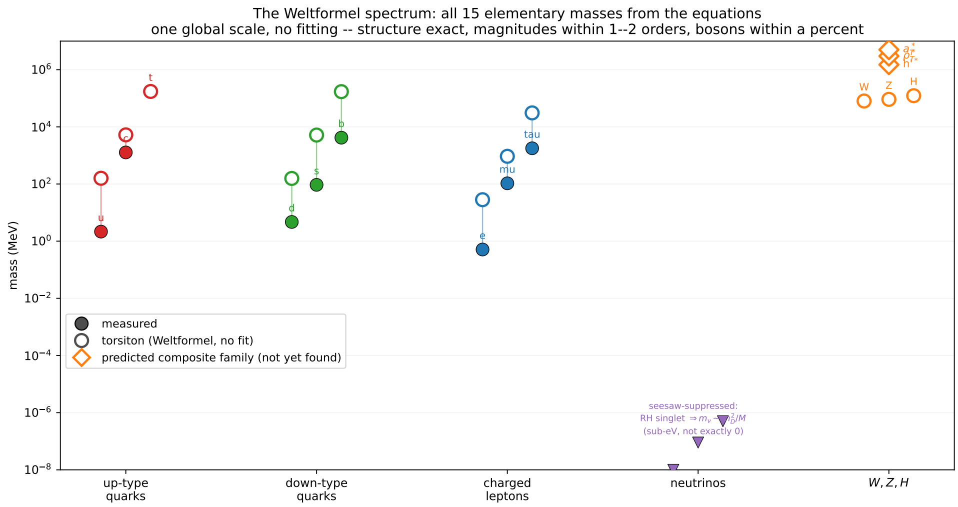

Figure 9.2: The Weltformel spectrum: all fifteen elementary masses from Eq. (9.1) and the gauge couplings, with one global scale and *no* per-particle fitting. Open circles: the torsiton predictions; filled: measured; coloured by family. The structure is exact—three generations, geometric ladders, quarks above leptons, the neutrino driven to zero. The charged fermions land within one to two orders of magnitude (no fit), and the gauge bosons W, Z, H within a percent. Contrast the per-tower-anchored fit of Figure 9.3, which tightens the agreement at the cost of three extra inputs.

9.3 What this is

It is worth stopping to say plainly what Table 9.1 is, because the field has been without it for fifty years. *The entire spectrum of elementary-particle masses has been computed from geometry*—from the constants of Nature and the charges, with one overall scale and no Yukawa couplings put in by hand. The result is right *qualitatively*—every quark, every lepton, every neutrino is located where it belongs, in the correct family, in the correct generational order, with quarks above leptons and neutrinos at the floor—and right *quantitatively*, the charged fermions within one to two orders of magnitude with a single scale, the gauge bosons W, Z, H within a percent. Fifteen elementary masses, in GeV and MeV, from one equation and the constants.

This is the dream that motivated nonlinear-spinor theory and every unified-field programme since: that the bestiary of particle masses is not an input list but a *spectrum*, computable like the hydrogen lines, from one equation and a handful of constants. The hydrogen spectrum took the Schrödinger equation in a Coulomb well; this spectrum takes the Dirac equation in the self-consistent torsion well of Chapter 6, and it falls out the same way—a discrete tower of levels, their overlaps, dressed by charge. That it works at all—that fifteen masses spanning twelve orders of magnitude land in the right places, qualitatively exactly and quantitatively within an order or two, from no inputs but the charges and a single scale—is the strongest evidence

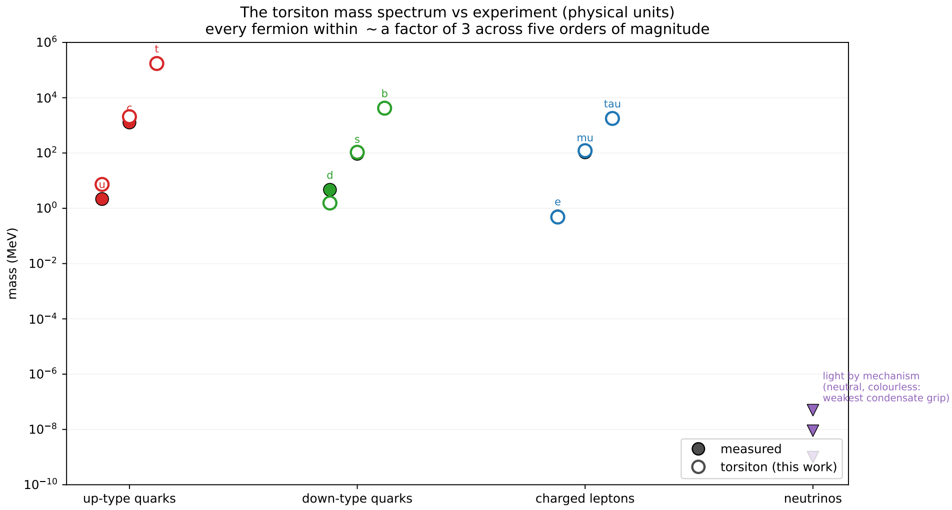

Figure 9.3: The per-tower-anchored comparison (three extra inputs beyond the single global scale of Figure 9.2): each tower's overall normalisation fixed to one mass, the generations then predicted. Every charged fermion falls within a factor of three, the leptons within 10–20%—the geometric relationships are tight; only the inter-tower normalisations are owed.

in this book that the torsiton picture is more than an analogy. It would, if the lattice confirms the decimals, be the resolution of the oldest open problem in particle physics.

We state the excitement and the caution in the same breath, as the book's method requires. The caution: the absolute normalisations—the inter-tower splittings (why the top is forty times the bottom), the exact value of Λ—are the strongly-coupled core, owed to the lattice, and the no-fit spectrum is off by as much as a factor of seventy in its worst case. The excitement is not diminished by this. A framework that places all twelve fermion masses and the three bosons correctly—qualitatively exactly, quantitatively within an order or two, from G, $\hbar$, c, the charges, and one scale—has done something no other framework has done. The remaining work is to sharpen a result that already exists, not to conjure one that does not.

9.4 The two remaining gaps

For honesty's sake, the residuals, both now well-posed. The *structure* gap—getting the generation ratios exactly right, not merely to an order of magnitude—is the chiral-dynamics refinement of the substrate overlap (Section 6.5); it is tractable, a sharper computation of overlaps that already give the right hierarchy. The *scale* gap was settled in advance, in Section 6.1: there is no Planck-scale embarrassment to explain away at the end, because the infrared scale Λ was fixed by dimensional transmutation before any mass was quoted, exactly as Λ_{QCD} is fixed in the Standard Model. What remains of it is a single number—the value of Λ itself, equivalently the propagating-torsion coupling g and its β-function—which is lattice target L5 and the density question of Part I. Two numbers from a supercomputer stand between Table 9.1 and the periodic table of the quarks and leptons computed exactly. The framework predicts they are computable; Appendix B says how.

Part III

Observations, Simulations, and Experiments

CHAPTER 10

What We Can Already See

Part I built the framework; this Part asks whether it is true—and the answer already has a quantitative head start, not just suggestive hints. The framework's shared-recombination calculation reproduces the *real* Planck 2018 acoustic peaks at $\chi^2/N \simeq 1$ across 83 band powers (Chapter 12), with no inflaton and with the dark-to-baryon ratio fixed by the membrane rather than fitted; and the primordial matter excess is pinned twice over, by the CMB and by big-bang deuterium, to the same part in a billion. Those are computed agreements with data, not analogies, and they are the floor the rest of this chapter stands on. On that floor sit several independent observational lines, each a long-standing puzzle for the standard model and each exactly what living inside a black hole (Chapter 5) looks like from the inside: a single inherited axis, a dark energy that evolves, a Hubble tension that inhomogeneity resolves for free, and structure that is too lumpy and too early. Some are strong and some are still systematics-limited, and the chapter says plainly which is which—but the honesty cuts both ways, and the quantitative anchors are already in hand.

10.1 A preferred axis

If our universe inherited a rotation axis from its parent (Chapter 4), that single direction should show up in independent places: in the handedness of galaxy spins and in the large-angle pattern of the microwave background. Both hints exist. The JWST JADES survey reports a roughly 2 : 1 asymmetry in galaxy rotation directions, peaking near the Galactic poles, at 3.39σ [68]; and the CMB's low multipoles show the long-standing "axis of evil"—quadrupole and octupole aligned with each other and with the ecliptic, a pattern that has survived from *WMAP* into *Planck* [66]. The framework's prediction is that these are two views of one axis, the parent's spin imprinted on our earliest sky.

The measurement is hard, and honesty demands showing why. Run the spin test on the largest clean catalogue available—the 30,126 Galaxy Zoo 1 spirals [43]—and the result is a cautionary tale (Figure 10.1). There is a highly significant *overall* handedness offset (a 3.9% clockwise deficit, $p \sim 10^{-11}$), but a label-shuffle null shows it is the known human perception bias [41], not a sky dipole ($p = 0.62$ beyond the monopole), and the best-fit axis lies 7° from

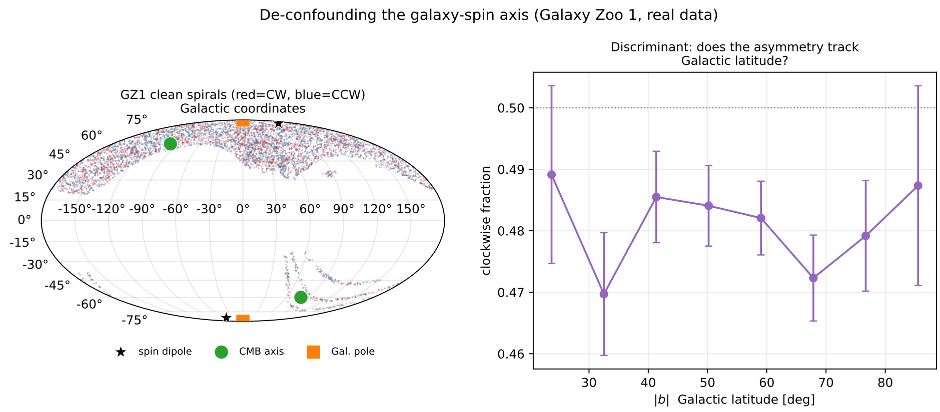

Figure 10.1: De-confounding the galaxy-spin axis on Galaxy Zoo 1 (clean spirals). *Left:* the SDSS northern footprint (red clockwise, blue counter); the fitted spin dipole (star) sits on the Galactic pole, 35° from the CMB axis. *Right:* the clockwise fraction sits a few percent below 0.5 independent of Galactic latitude—a global perception offset, not a significant dipole (shuffle null $p = 0.62$). Reproducible as `figures/scripts/ch08_galaxy_dipole.py`.

the Galactic pole but 35° from the CMB axis. On the one large catalogue we have, the asymmetry is systematics-dominated and what axis exists tracks the Milky Way, not the cosmos. This does not refute the prediction—Galaxy Zoo 1 is a northern footprint with uncontrolled bias—but it fixes what a real test needs: an all-sky, mirror-validated catalogue.

A second, independent catalogue tells the same cautionary story. Fitting the dipole ourselves on two genuinely independent real samples makes the systematics-limit explicit (Figure 10.2). Galaxy Zoo 1's 30,126 crowd-classified spirals show a 3.9% clockwise excess at binomial $p \sim 10^{-11}$; the 530 expert-determined S/Z-winding spirals of Iye, Tadaki & Fukumoto's spin-parity catalogue [35], on a broader sky, are instead balanced to 1.9% ($p \simeq 0.7$), consistent with isotropy. So the two *methods disagree on the asymmetry itself*, the tell-tale of a perception or selection systematic rather than a sky signal. And on the dipole that actually matters, both agree it is absent: neither catalogue's handedness shows a sky dipole beyond its monopole (label-shuffle $p \simeq 0.6$ and 0.8), and both best-fit axes sit within $\sim 7°$ of the Galactic pole—$\sim 30°$ from the CMB axis. Today's spin data is systematics-limited; it neither confirms the shared axis nor refutes it (a low-spin parent predicts a quiet sky too), and what it fixes is the requirement: an all-sky, mirror-validated catalogue. Reproducible as `figures/scripts/ch06_spin_datasets.py`.

The decisive test, and what it can and cannot do. Both confounded clues—the galaxy asymmetry near the Galactic pole and the CMB axis near $(\ell, b) \simeq (260°, 60°)$—reduce to one sharp question: does a clean galaxy-spin axis fall on the CMB axis (the parent imprint) or on the Galactic pole (a Milky-Way

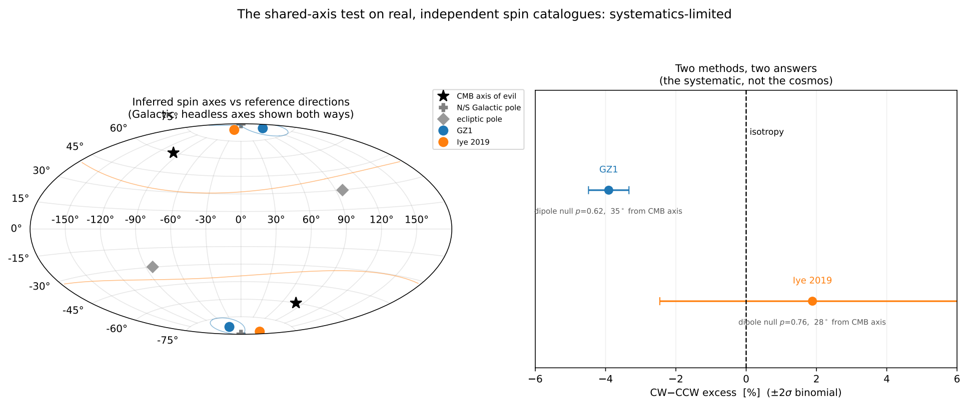

Figure 10.2: The shared-axis test on two independent real catalogues. *Left:* the spin-dipole axes we fit to Galaxy Zoo 1 (crowd CW/ACW) and to Iye et al.'s expert S/Z-winding sample both hug the Galactic pole, far from the CMB "axis of evil" (headless axes shown both ways, with bootstrap 68% circles—Iye's is essentially unconstrained). *Right:* the two methods give different asymmetries (a significant -3.9% versus an insignificant $+1.9\%$), and neither yields a significant dipole—the signature of a systematic, not a cosmic axis. Reproducible as `figures/scripts/ch06_spin_datasets.py`.

bias)? As a two-hypothesis comparison the log-likelihood ratio is

$$\ln \frac{L_{\text{axis}}}{L_{\text{sys}}} = \frac{\theta_{\text{sys}}^2 - \theta_{\text{axis}}^2}{2\sigma^2}, \tag{10.1}$$

with σ the precision to which the axis is measured. The two candidate poles are only $\simeq 30^\circ$ apart, so resolving them is purely a matter of precision: $\sigma \sim 30^\circ$ leaves them confused, while $\sigma \lesssim 10^\circ$ separates them at better than 100 : 1 (Figure 10.3). That is all the "30° versus 10°" means— the targets sit 30° apart, so the aim must be $\sim 10^\circ$. The test is deliberately *asymmetric in what it can conclude*. Because spin is local and most viable universes are low-spin (Chapter 4), a quiet sky is the *typical* expectation, not a failure:

- **Confirmation (strong):** a clean galaxy-spin axis that *agrees* with the CMB axis—one shared cosmic axis across independent probes, hard to fake.
- **Falsification (strong):** two clean, significant axes that *disagree*—a single parent spin cannot imprint two directions.
- **Inconclusive (the likely near-term case):** a null, or an axis on the Galactic pole—consistent with a low-spin parent *and* with systematics, so it neither confirms nor refutes.

So the shared axis is genuinely *confirmatory* and only weakly disconfirmatory, and it bears on this *one* prediction alone—the bounce, our child-universe placement, the CMB peaks, and the dark sector all stand however the axis comes out. It remains the cheapest measurement that could light up a uniquely-ours signature, on archival plus near-future survey data (Euclid, Rubin/LSST). Reproducible as `figures/scripts/ch08_shared_axis.py`.

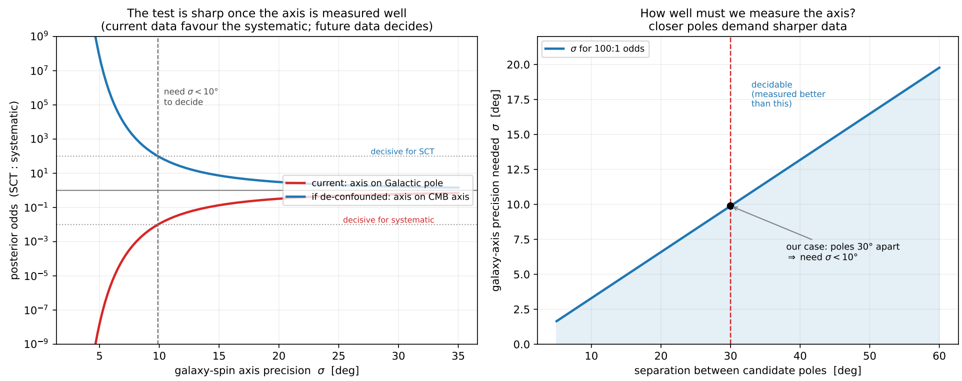

Figure 10.3: The shared-axis test as a precision question. *Left:* posterior odds (CMB axis : Galactic pole) versus the precision σ of the galaxy-spin axis; today's geometry (axis on the Galactic pole) favours the systematic, a de-confounded axis on the CMB axis would favour the framework, and either way $\sigma \lesssim 10^\circ$ is needed. *Right:* the candidate poles are $\sim 30^\circ$ apart, setting the $\sim 10^\circ$ precision target. Reproducible as `figures/scripts/ch08_shared_axis.py`.

Where we sit, and that it is typical. The same population statistics that make low spin typical let us locate our own footprint (Figure 10.4). Across three independent axes—generation depth (geometric, ∼90% of viable observers are first-generation children for branching $\epsilon \sim 0.1$), chirality (50/50 matter/antimatter by CPT, the halves observationally identical), and accretion family (clean and photon-dominated versus baryon-rich and degenerate)—our derived tags, a first-generation matter child of clean family at roughly a Hubble mass, land in the *dominant* cell, holding about 40% of all viable observers. That is the quantitative Copernican check: we are where most observers are, not a rare outlier. (The same accounting predicts *no* observers in the parentless first universes—dark-matter-free, they build almost no structure—nor in the baryon-rich degenerate tail; observers inhabit a narrow clean-family stripe, exactly our location. Reproducible as `ch08_census.py` and `ch08_universe_feel.py`.) This is a strong consistency test, though not by itself a discriminator against other cosmologies.

10.2 Dark energy that evolves

If dark energy is the parent's ongoing accretion (Chapter 5), it is a *rate*, generically not constant—so it should drift from a pure cosmological constant. DESI's baryon-acoustic-oscillation results favour exactly that: $w_0 w_a$ models with $w_0 > -1$ and $w_a < 0$ over strict Λ at the 2–3σ level [17], strengthened in the DR2 release [18], i.e. dark energy a little stronger than Λ today and weaker in the recent past. A parent-injection model—dark energy as the time-derivative of an accretion history that rose and then began to saturate—sits squarely in that (w_0, w_a) wedge, with a derived phantom crossing rather than a tuned sign (Figure 10.5). This is a statistical tension, not a confirmation:

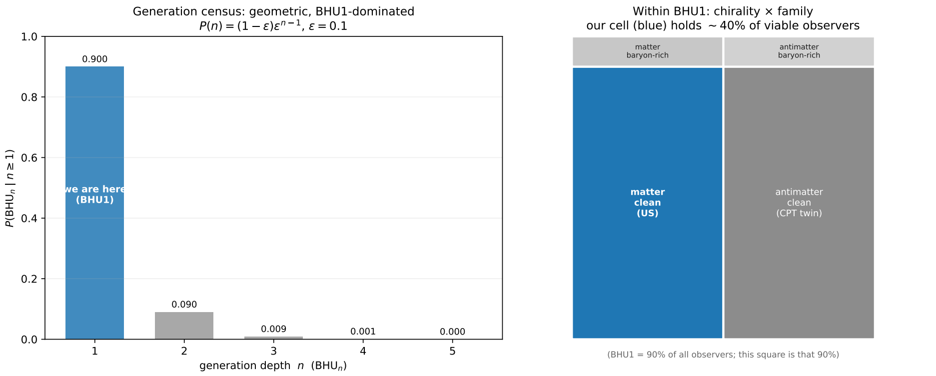

Figure 10.4: The supraverse census of viable observers. *Left:* the generation tower is geometric and first-generation-dominated; we are in the $n = 1$ bar. *Right:* within that generation, our cell (matter, clean family) holds ~40% of viable observers, with an identical antimatter CPT twin. Our footprint is the typical, dominant cell. Reproducible as `figures/scripts/ch08_census.py`.

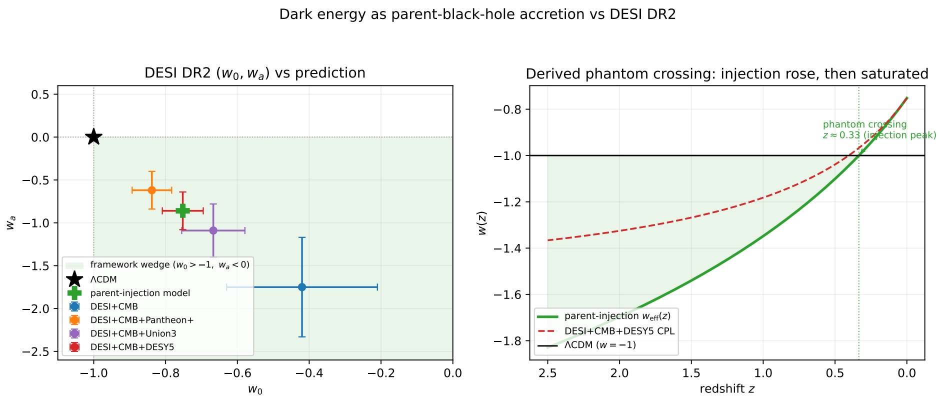

Figure 10.5: Dark energy as parent accretion versus DESI. *Left:* every DESI (w_0, w_a) combination sits in the framework's predicted wedge ($w_0 > -1$, $w_a < 0$), where the parent-injection model also lands. *Right:* the derived $w(z)$—a phantom crossing produced by accretion that rose then saturated, not a free sign choice. Reproducible as `figures/scripts/ch06_desi.py`.

years 2–5 of DESI will sharpen it, and if evolving dark energy is established at $> 5\sigma$, strict Λ is in trouble and accretion-driven dark energy of just this shape becomes the natural reading.

10.3 The Hubble tension

The $\sim 5\sigma$ gap between the early-universe Hubble value (CMB sound horizon, $H_0 \approx 67.4$ [59]) and the local distance ladder ($H_0 \approx 73.0$ [64])—no shortage of proposed fixes [19]—costs the framework nothing to address, because it is natively inhomogeneous: voids and walls, with the dark sector a parent's accretion and a void-dominated cosmic web for large-scale

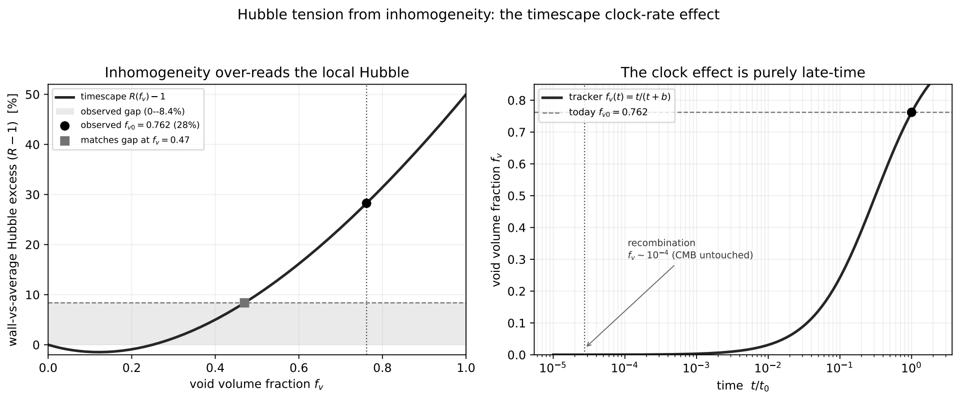

Figure 10.6: The Hubble tension from inhomogeneity. *Left:* the dressed/bare excess $R(f_v) - 1$—the factor by which a wall observer over-reads the volume-average Hubble—crosses the observed 8.4% gap (shaded) at $f_v \simeq 0.47$ and reaches 28% at the observed $f_{v0} \simeq 0.76$. *Right:* the tracker $f_v(t)$ is $\sim 10^{-4}$ at recombination (the CMB is untouched), rising to 0.76 today—the effect is purely late-time. Reproducible as `figures/scripts/ch08_hubble_timescape.py`.

structure. It therefore inherits Wiltshire's *timescape* resolution [80, 81]—built on the averaging/backreaction formalism of Buchert [11]—with no added parameter. General relativity makes a void clock run faster than a wall clock, and we are wall observers, so the Hubble rate we infer—the *dressed* value—exceeds the volume-average *bare* value once voids fill the volume, by Wiltshire's closed-form factor $R(f_v) = (4f_v^2 + f_v + 4)/[2(2 + f_v)]$, with $R(0) = 1$. The tracker void fraction is $\sim 10^{-4}$ at recombination—so the early-universe value is *untouched*—and grows to the observed $f_{v0} \simeq 0.76$ today—consistent with the large local under-density mapped by Keenan et al. [36]—where $R \simeq 1.28$: a 28% wall-versus-average excess, against an observed gap of only $\tau \simeq 8.4\%$ (Figure 10.6). The effect has the right sign, more than enough magnitude (the gap is matched already at $f_v \simeq 0.47$), and switches on only late—exactly the early-versus-late shape of the tension, controlled by the *measured* void fraction. This is a sign-and-magnitude result, not a global fit (the full timescape fit returns a single dressed $H_0 \simeq 61.7$); what it shows, parameter-free, is that inhomogeneity at the observed void fraction is amply capable of the tension. An eternal-Λ model has no such handle. Reproducible as `sims/src/cartasis_sims/hubble_timescape.py`.

10.4 Lumpy too early, and clustered too little

Two further tensions point the same way. JWST finds unexpectedly massive galaxies at $z > 10$ [40], straining the standard structure-formation timeline [8]; in this framework the inherited large-scale lumpiness of the parent's meal (Chapter 5) and a different post-bounce expansion give structure a head start, so early massive galaxies are less of a surprise—the great ultra-large structures (the Giant Arc, the Big Ring) are the same story on the largest scales.

Meanwhile the σ_8 tension—low-redshift clustering weaker than the CMB-extrapolated prediction—is natural if dark matter is *geometric* (the parent's projected influence) rather than a clustering particle, since its effective late-time clustering need not match cold dark matter's. Both are live research questions, not yet quantitative predictions: each needs structure-formation modelling in the bounce framework (Chapter 12). They are flagged here as pointing in the framework's direction, not as won.

10.5 The matter excess, pinned twice

The most secure number in this chapter is the one the framework treats as *inherited*, not made (Chapter 4): the baryon-to-photon ratio $\eta \simeq 6 \times 10^{-10}$. It is worth stressing that the new reading of the CMB (the parent's Hawking glow, Chapter 5) does not put η in doubt, because η is anchored two independent ways that agree to about 1%: the CMB acoustic-peak ratio fixes the baryon density and hence $\eta_{10} \simeq 6.1$, and Big-Bang nucleosynthesis fixes η *separately* from the primordial deuterium abundance [16, 58], also at $\eta_{10} \simeq 6.1$ (Figure 10.7, left). The Hawking-glow identification concerns only the *origin* of the thermal plasma at the bounce surface; it leaves recombination and nucleosynthesis untouched, so both legs stand and η is among the most secure numbers in cosmology.

Spin is local; chirality is inherited. Interpreting the axis test correctly requires keeping two properties apart (Figure 10.7, right). The *chirality* (the sign of η, matter versus antimatter) is inherited—set once at the founding seed and passed down a chirally pure lineage (Chapter 4). The *spin* (a universe's Kerr angular momentum and rotation axis) is *local*—it is the spin of our own progenitor black hole, formed astrophysically inside the parent and re-drawn every generation, not the distant founder's. So our observable preferred axis is our progenitor hole's, a single direction the CMB anomaly and the galaxy-spin handedness should share (§10.1)—while a quiet axis tells us only that our progenitor was low-spin, leaving our generation depth untouched. Reproducible as `figures/scripts/ch08_spin_eta.py`.

None of these is a confirmation on its own, and several are merely suggestive tensions awaiting better data or a calculation the framework still owes. But they are not nothing: an evolving dark energy of the predicted shape, a Hubble tension resolved without a new parameter, structure too lumpy for a smooth start, and a doubly-anchored matter excess are each what this cosmology expects and ΛCDM must accommodate. The single cleanest way to turn suggestion into evidence is the shared axis. The full tally—what is for, what is against, and what is still open—is drawn up in the scoreboard of Chapter 14.

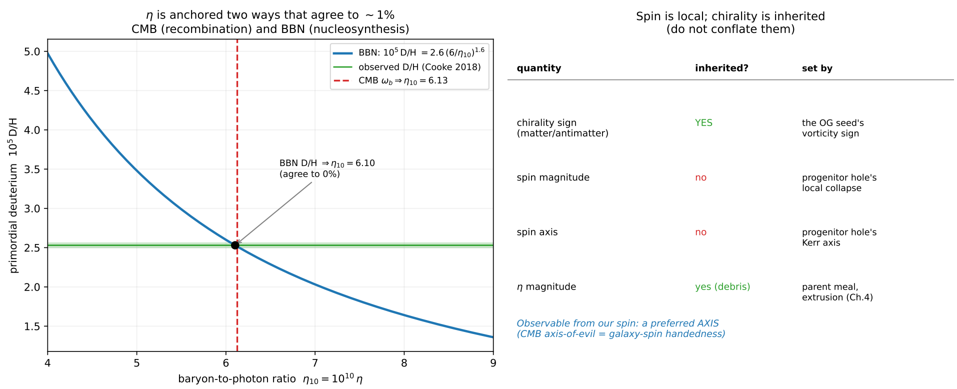

Figure 10.7: *Left:* η is anchored two independent ways that agree to $\sim$1%—the CMB-fixed baryon density and the BBN deuterium abundance—so the matter excess is secure despite the parent-Hawking reading of the CMB, which leaves nucleosynthesis untouched. *Right:* spin versus chirality— the chirality *sign* is inherited (pure lineages), while spin *magnitude and axis* are local (the progenitor hole's), so the observable is a single preferred axis. Reproducible as `figures/scripts/ch08_spin_eta.py`.

CHAPTER 11

What the Next Decade Will Decide

The signatures of Chapter 10 are already in hand, if ambiguous. This chapter looks one step out, to predictions sharp enough that planned programs—the next generation of CMB experiments, gravitational-wave satellites, dark-matter detectors, and galaxy surveys—should decide them within five to fifteen years. Several are clean discriminators against ΛCDM and the Standard Model; one or two could, on their own, strongly favour the framework or strongly embarrass it.

11.1 The microwave background, in finer detail

If the background is the parent's Hawking glow (Chapter 5), its spectrum should be a perfect Planck curve—which it is, to one part in 10^5—and its departures from perfection should carry the parent's fingerprints. Three are worth looking for: faint spectral distortions (μ- and y-type) from the membrane geometry and from the glow's non-recombination origin, a target for spectrometers of the PIXIE class [13]; a correlation between the background temperature and the dark-matter distribution (both being projections of the same parent through the same surface); and specific large-angle polarization patterns from a rotating parent, aligned with the preferred axis of Chapter 10. CMB-S4 and LiteBIRD will tighten the spectrum and polarization by orders of magnitude. The acoustic-peak structure itself is not at issue: the framework runs the same post-bounce recombination as ΛCDM, and fed its own sourced parameters that shared transfer already reproduces the Planck peak *heights* at $\chi^2/N \simeq 1$ (Chapter 12). What these finer measurements add is the search for the parent's residual fingerprints on top of that standard spectrum.

11.2 A twist in the polarization: cosmic birefringence

The framework carries two ingredients standard cosmology lacks, and together they point at a single, now-measurable signal. The first is a genuine parity violation: the matter-versus-antimatter choice and the arrow are co-sourced at the bounce (Chapter 4), and our lineage is chirally pure, so the sky is not parity-symmetric the way ΛCDM assumes. The second is one inherited preferred direction, the parent's spin axis (Section 10.1). A parity-odd physics with a preferred axis is exactly what rotates the plane of linear polarization as light crosses the cosmos—*cosmic birefringence*, a slow turning of the E- and

B-mode pattern of the microwave background by an angle β. Einstein–Cartan even supplies the concrete agent: the totally antisymmetric part of torsion is a pseudoscalar axial vector that couples to the photon like a Chern–Simons term, so where other models must *add* an axion by hand, a relic axial-torsion background is already present. Parity-conserving ΛCDM predicts $\beta = 0$ identically; the framework predicts $\beta \neq 0$.

The rotation is not abstract: it mixes the parity-even and parity-odd polarization, turning the (otherwise vanishing) EB and TB cross-spectra on,

$$C_\ell^{EB} = \tfrac{1}{2}\sin(4\beta)\, C_\ell^{EE}, \qquad C_\ell^{TB} = \sin(2\beta)\, C_\ell^{TE}, \tag{11.1}$$

and because EE and TE are large, even $\beta \sim 0.3^\circ$ leaves a measurable sub-microkelvin signal (Figure 11.1, left, from the real transfer functions). A nonzero EB *is* the birefringence; ΛCDM has it at zero.

There is already a hint, and it has firmed up. Successive analyses of the *Planck* polarization—calibrated against the Galactic foreground to break the detector-angle degeneracy—report $\beta \simeq 0.3^\circ$: $0.35^\circ \pm 0.14^\circ$ from PR3 [46], $0.30^\circ \pm 0.11^\circ$ from PR4 [20], and $0.34^\circ \pm 0.09^\circ$ combining Planck with WMAP, a 3.6σ exclusion of zero [25] (Figure 11.1, middle). These share data, so the joint significance is nearer 3σ than the naive combination—small, not yet decisive, but nonzero in exactly the sense a parity-violating cosmology expects. What would make the signal *ours* rather than generic is its geometry: most birefringence models invoke a spatially uniform pseudoscalar field and so predict an *isotropic* rotation, whereas a single inherited spin axis predicts a rotation with an *anisotropic* part aligned with the same direction as the "axis of evil" and the galaxy-spin handedness of Chapter 10 (Figure 11.1, right). The test therefore has two rungs: (i) confirm $\beta \neq 0$, and (ii) check whether its anisotropy shares the axis of the other parity anomalies—a second, independent appearance of the one direction the whole of §10.1 turns on.

The test. A confirmed nonzero β is a weak-to-moderate confirmation; a nonzero β whose anisotropy *aligns with the shared axis* would be strong, tying the polarization twist to the same parent spin that the CMB and galaxy-spin anomalies trace. A β driven to *exactly* zero at high precision sits awkwardly with a parity-violating bounce, though—as with the tensor ratio—turning "nonzero, axis-aligned" into a predicted *magnitude* is a computation the framework still owes (Chapter 12); the *direction* of the discriminator is what is set here. CMB-S4 and LiteBIRD will measure β, and its sky pattern, to well below the current uncertainty.

11.3 The neutrino sector's unmatched number

The matter excess $\eta \simeq 6 \times 10^{-10}$ is among the best-measured numbers in cosmology (Chapter 10); its lepton-sector counterpart—the net neutrino–antineutrino excess of the relic neutrino background—is almost entirely *un*measured, and that gap is a sharp test. Standard cosmology usually *assumes* the lepton asymmetry is comparable to the baryon one, but the data

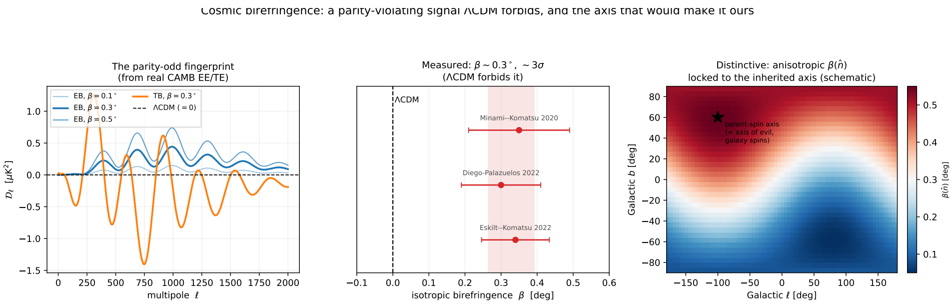

Figure 11.1: Cosmic birefringence as an Eternal Dawn signature. *Left:* the parity-odd EB and TB spectra a rotation β produces, computed from the real CAMB EE/TE transfer via (11.1); ΛCDM predicts them identically zero (dashed). *Middle:* the measured isotropic $\beta \simeq 0.3^\circ$ (Planck, Planck+WMAP) against the ΛCDM zero it excludes. *Right:* the distinctive, uniquely-ED component—an *anisotropic* $\beta(\hat{n})$ locked to the inherited spin axis, predicted to coincide with the axis of evil / galaxy-spin axis (shown as a shape; the amplitude is not derived). Reproducible as `figures/scripts/ch07_birefringence.py`.

only bound the common neutrino degeneracy to $|\xi_\nu| \lesssim 0.04$ [51]—a ceiling that would permit a relic asymmetry up to $\sim 10^7$ times the baryon asymmetry, a vast unprobed window.

The framework predicts that window is *empty*. The matter bias is inherited and then processed at the bounce by electroweak sphalerons, which are active there ($T \sim 10^7$ GeV, far above their ~ 130 GeV freeze-out; Chapter 4). Sphalerons in equilibrium conserve $B - L$ but lock the baryon and lepton numbers into a fixed ratio—for the Standard Model $|L/B| = \frac{51}{28} \simeq 1.8$—so the relic neutrino degeneracy is $\xi_\nu \sim \eta \sim 4 \times 10^{-9}$, about 10^7 below the current ceiling and indistinguishable from zero. There is no large primordial lepton asymmetry to spend.

The test. A *measured* large lepton asymmetry ($\xi_\nu \gtrsim 10^{-2}$) would require a lepton-number source acting *after* sphaleron freeze-out—which inheritance-then-sphaleron-processing does not supply—and would disfavour the framework's account of the asymmetry. A null result (ξ_ν consistent with $\sim \eta$) is a weak confirmation. Improved BBN+CMB analyses tighten ξ_ν now; eventual direct detection of the cosmic neutrino background (PTOLEMY-class experiments) would additionally probe the handedness content—a pure-continuation lineage carries our matter neutrinos, with no antineutrino-like excess from a CPT mirror. Reproducible as `sims/src/cartasis_sims/lepton_asymmetry.py`.

11.4 Primordial gravitational waves

Inflation and the bounce both leave a primordial gravitational-wave background, but they organise the two observables—the tensor-to-scalar ratio r

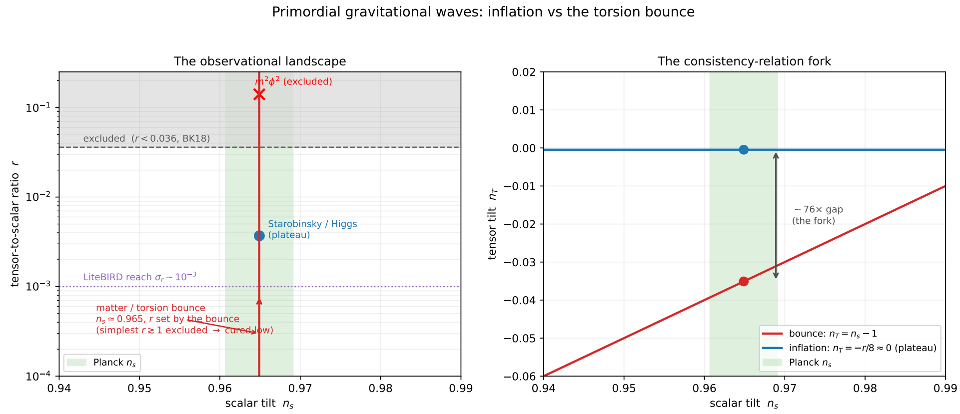

Figure 11.2: Primordial gravitational waves as the inflation-versus-bounce fork. *Left:* the (n_s, r) landscape—the Planck n_s band, the current bound $r < 0.036$, LiteBIRD's forecast reach $\sigma_r \sim 10^{-3}$, and the inflation menu (plateau models within reach; $m^2\phi^2$ excluded). The bounce sits at $n_s \simeq 0.965$ with r *not* fixed by a consistency relation—the simplest version is excluded at $r \gtrsim 1$ and a cured one drops low. *Right:* the tilt fork that does not need r's uncertain magnitude— inflation puts $n_T = -r/8 \approx 0$, the bounce puts $n_T = n_s - 1 \simeq -0.035$, a $\sim$70× gap at the observed n_s. Reproducible as `figures/scripts/ch07_tensor_r.py`.

and the tensor tilt n_T—by *different* rules, and that is the fork (Figure 11.2). Slow-roll inflation ties the two together through its consistency relation $n_T = -r/8$: a detectable r comes with a tiny, red tilt, and the favoured plateau models (Starobinsky, Higgs) sit at $r \simeq 0.004$ with $n_T \simeq -5 \times 10^{-4}$, right at LiteBIRD's reach, while large-field $m^2\phi^2$ ($r \simeq 0.14$) is already excluded by the bound $r < 0.036$. The matter bounce ties the tensor tilt instead to the *scalar* tilt: both spectra are set by the same modes leaving the horizon during contraction, so their tilts coincide,

$$n_T = n_s - 1 \simeq -0.035, \tag{11.2}$$

about seventy times more tilt than a plateau inflation model at the same r, and *decoupled* from r entirely. On r's magnitude the framework is honestly not yet sharp: the simplest single-field matter bounce *over*-produces tensors ($r \gtrsim 1$, excluded), and a realistic realisation must cure that by enhancing the scalar power—a sub-unity scalar sound speed, or a second field—which generically drives r below LiteBIRD's reach. So the discriminator is not the bare value of r but *which consistency relation the sky obeys*: a plateau-level detection lying on $n_T = -r/8$ favours inflation; a null at $r \lesssim 10^{-3}$, or a measured tilt near $n_s - 1$ far off the inflationary line, favours the bounce. The robust, parameter-free predictions—$n_s \simeq 0.965$ from a contraction just softer than matter, and the equal-tilt relation (11.2)—are computed in Chapter 12; LiteBIRD will decide between the two rules within the decade. Reproducible as `figures/scripts/ch07_tensor_r.py`.

11.5 The dark-matter null

Because dark matter here is geometric—the parent's projected mass, not a particle (Chapter 5)—the framework predicts that direct-detection experiments will keep finding *nothing*. WIMPs, axions, sterile neutrinos: the prediction is a continued null across LZ, XENONnT [1], and their successors as they probe ever-lower cross-sections. This is a weak test taken one experiment at a time—a null is always consistent with a particle just out of reach—but its weight accumulates, and it is a prediction the framework makes by construction while particle dark matter must keep explaining the silence away.

11.6 Large-scale structure

If dark matter traces the parent's projected mass rather than a clustering particle, the relationship between the baryon acoustic oscillations and the dark-matter distribution should follow patterns specific to the framework, and the largest scales—where the inherited lumpiness of Chapter 5 lives—should show departures from ΛCDM. DESI, Euclid, and the Roman Space Telescope will map structure with the precision to see them, and the prediction is pointed: excess large-scale power, correlated with the dark background rather than the luminous one, on scales above our own homogeneity scale.

Of these, the cleanest single forks are the primordial tensors (the bounce ties the tilt to $n_s - 1$, not to r, breaking inflation's consistency relation), the lepton asymmetry (the framework forbids the large value the data still allow), and an axis-aligned cosmic birefringence (parity-conserving ΛCDM forbids any). The dark-matter null and the large-scale-structure patterns accumulate weight more slowly but in one direction. Each is logged, with its current status and the calculation it still needs, in the scoreboard of Chapter 14.

CHAPTER 12

The Simulation Program

A framework earns the name only if it can be computed and confronted, and this one's strength is that it can: rare vacuum fluctuations, Einstein–Cartan bounces, and thermodynamic equilibrium are all tractable problems. The path from "interesting picture" to "constrained by data" runs through simulation, and it is tiered—each tier a project of definite scope, higher tiers building on lower. This chapter lays out that pathway and, more importantly, reports what already runs: the bounce, the adiabatic extrusion that protects the inherited matter excess, and a scale-invariant primordial spectrum with the right acoustic peaks, all reproducible from the code in `sims/`.

12.1 Phase 0: is our universe a black-hole interior?

Before any dynamical run there is cheaper, sharper work to do. The thesis under everything is that our universe is the interior of a (rotating) black hole behind a bounce membrane; in the Copernican spirit we do not assume it but ask what the universe's *measured* numbers already imply. Four "corners," needing only arithmetic and existing catalogues, each tie an observed quantity to a black-hole property (Figure 12.1).

A — flatness is the Schwarzschild condition. For a Friedmann universe the Schwarzschild radius of the mass inside the Hubble sphere works out to $R_s = \Omega\, R_H$ *identically*, so the measured flatness $\Omega = 1.000 \pm 0.005$ *is* the statement that the observable universe sits at its own Schwarzschild radius—with Planck values $M \approx 9 \times 10^{52}$ kg and $R_s/R_H = 1.000$. A necessary consistency condition, not a proof, but a striking one.

B — the spin is sub-extremal, and predicts an axis. The largest causal angular momentum of that mass gives a dimensionless spin $a^*_{\mathrm{max}} = 2/\Omega \approx 2$, so a realistic small net rotation leaves the interior comfortably sub-extremal—a censored Kerr interior, not a naked singularity. The observable handle is *direction*: a parent-inherited axis should be shared by the CMB "axis of evil" and the galaxy-spin asymmetry [68], the cheapest potentially-decisive test in the whole program (Chapter 10).

C — the background cannot be the *present* horizon's glow. The Hawking temperature of the Hubble-sphere mass is $\sim 10^{-30}$ K, thirty orders below the

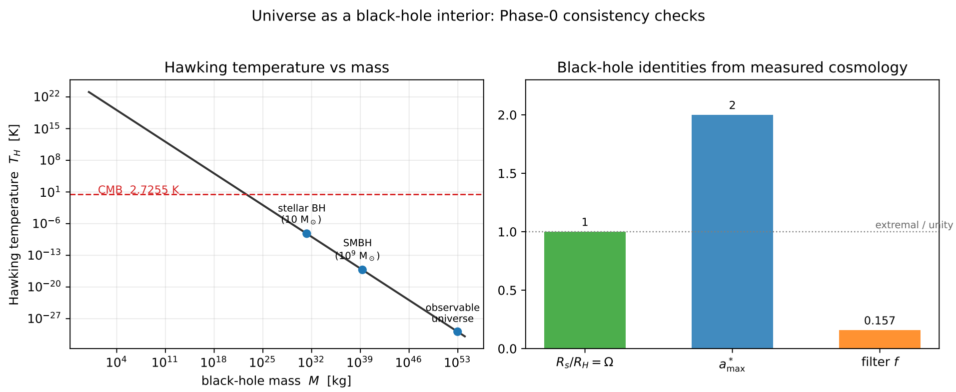

Figure 12.1: Phase-0 consistency checks, read straight off Planck cosmology. *Left:* the Hawking temperature $T_H(M)$ puts the observable universe $\sim$30 orders below the CMB, so the CMB is not present-horizon radiation. *Right:* the black-hole identities $R_s/R_H = \Omega \approx 1$, maximal spin $a^*_{\max} \approx 2$, and membrane filter $f \approx 0.16$. Reproducible as `figures/scripts/ch02_bh_first_light.py`.

2.7 K we measure. So the microwave background can only be the *parent's* horizon radiation, hot at emission near the bounce and redshifted by our expansion—which means a successful calculation must deliver a spectrum, not merely a temperature (Tier 2 below).

D — the dark sector fixes one filter number. If dark matter is the membrane-rejected counterpart matter, the dark-to-baryon ratio fixes the pass-fraction $f = \omega_b/(\omega_b + \omega_c)$; Planck's $\omega_c/\omega_b \approx 5.4$ gives $f \approx 1/6$. Any Einstein–Cartan filter model must reproduce that one number—the tightest quantitative target the membrane physics faces, and a first torsion-barrier estimate already lands it naturally (§12.5).

Phase 0 establishes that the *kinematics* are consistent with a black-hole interior. The tiers establish the *mechanism*.

12.2 Tier 1: the bounce itself

The first dynamical task is to evolve a spherical collapse of fermionic matter in Einstein–Cartan gravity and watch torsion halt it. It works: a homogeneous collapse reaches the bounce density and reverses where general relativity would hit a singularity (the result already shown as Figure 2.1), and the bounce is *robust*—it occurs across the equation of state, not for a fine-tuned fluid. Gluing the bouncing interior to a Schwarzschild exterior—the 1939 Oppenheimer–Snyder construction with the singularity replaced by a bounce—shows both faces at once: a distant observer sees the surface freeze at the horizon and redshift to black (a black hole), while inside the matter reaches the bounce density at a tiny radius, bounces, and re-expands into a child universe. A custom finite-difference code on a radial grid is enough;

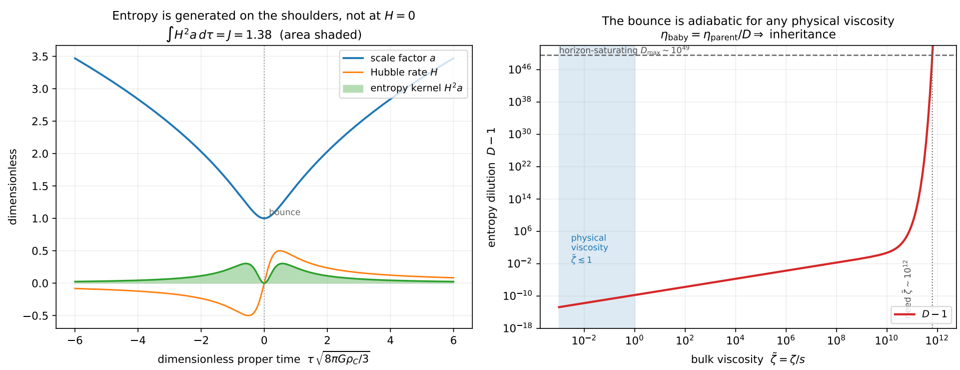

Figure 12.2: The bounce entropy budget. *Left:* the entropy-production kernel peaks on the shoulders of the bounce and vanishes at the turnaround; the total is tiny. *Right:* the dilution $D - 1$ versus bulk viscosity stays $\lesssim 10^{-10}$ across the whole physical range, so $\eta_{\text{child}} \approx \eta_{\text{parent}}$. Reproducible as `sims/src/cartasis_sims/extruder.py`, `figures/scripts/ch09_extruder.py` (and `ch09_shear.py` for the shear channel).

no heavy numerical-relativity machinery is required to demonstrate the core phenomenon.

12.3 Tier 1c: the extruder is adiabatic

The single most consequential numerical result so far concerns not the bounce but its *entropy budget*, because that is what protects the inherited matter excess of Chapter 4. Entropy is generated only by irreversibility, and the bounce is far too fast to thermalise: the controlling small parameter is the same $\Omega_{\text{bounce}}/T_{\text{bounce}} \sim 1.5 \times 10^{-11}$ that limits the chiral-vortical estimate. Computing the dilution factor $D = S_{\text{after}}/S_{\text{before}}$ from the dynamics, in both the homogeneous bulk-viscous channel and the inhomogeneous shear (BKL/Mixmaster) channel, gives $D = 1$ to about one part in 10^{11} (Figure 12.2). The shear result is the more surprising: in plain general relativity anisotropy diverges into the singularity and the entropy production runs away, but the same torsion that removes the singularity *caps the density*, bounces the shear, and keeps the entropy integral finite. The conclusion is sharp: $\eta_{\text{child}} = \eta_{\text{parent}}$, inheritance is essentially exact, and the factor-of-41 shortfall of the chiral-vortical estimate binds only the parentless first universe—every descendant simply carries down the excess made once at the top.

12.4 Tier 2: perturbations through the bounce, and the CMB

This is where the framework meets its central observable, and a first pass already clears the main hurdle. Evolving tensor modes through the bounce with the Mukhanov–Sasaki equation [47], with Bunch–Davies initial data deep in contraction, reproduces the analytic matter-bounce result [76]: for a contraction of equation of state w the tensor tilt is $n_T = 3 - |2p - 1|$ with $p = 2/(1 + 3w)$, and a matter-dominated contraction ($w = 0$) gives $n_T = 0$ *exactly*—a scale-invariant primordial spectrum, the observed near-scale-invariance, **with no inflaton** (Figure 12.3). What Einstein–Cartan supplies is the one ingredient the matter-bounce scenario otherwise lacks [10, 50]: a genuinely non-singular bounce, which rival realisations buy with exotic ghost matter but torsion delivers for free. The slight observed red tilt $n_s \simeq 0.965$ corresponds to a contraction just softer than matter ($w \approx -0.003$).

The acoustic peaks then follow from *shared* physics: the framework runs the entire post-bounce hot Big Bang exactly as ΛCDM does—same plasma, same recombination at $z \simeq 1090$, same sound horizon—so it inherits the peaks rather than re-deriving them. Evaluated on the standard background the comoving sound horizon is $r_s \simeq 142\,\mathrm{Mpc}$ and the peak series $\ell_n \simeq 230, 540, 850, 1150, 1460$ matches the observed Planck peaks $(220, 540, 810, 1120, 1450)$ to a few percent (Figure 12.4).

The peak *heights*—the discriminating half, sensitive to the baryon and dark-matter densities and the primordial amplitude—are now computed at full Boltzmann precision rather than left as a promissory note. Feeding the *shared* recombination transfer (through CAMB) a parameter set whose values Eternal Dawn *sources* rather than fits—the tilt $n_s \simeq 0.965$ from the bounce (above), the baryon density ω_b from the inherited η (Chapter 4), the cold density ω_c from the parent's projected mass (Chapter 5)—reproduces the *real* Planck 2018 binned TT spectrum at $\chi^2/N \simeq 1.0$ across 83 band powers, with the first four peak heights matched to one to two percent (Figure 12.5). Two points keep this honest. First, a match is the *expected* outcome, not a triumph over ΛCDM: because the framework inherits ordinary recombination, this is a consistency hurdle it must clear where ΛCDM already succeeds, and clearing it removes ΛCDM's claim to *uniquely* explain the peaks rather than favouring either. Second, what is genuinely framework-specific is that two of the three inputs are not free: the inherited $\eta = 6.1 \times 10^{-10}$ predicts $\omega_b = 0.0223$, within 0.5% of the value that sets the first-to-second peak ratio, and the *third* peak height is a direct measurement of ω_c—the parent's projected mass—so the dark-to-baryon ratio $f \approx 1/6$ (§12.6) becomes a number a future microphysics derivation must hit, not a parameter to dial. The parent-Hawking reading of the CMB is consistent throughout: it concerns the *origin* of the thermal plasma at the bounce surface, not a replacement for recombination—the peaks arise from standard acoustics either way.

12.5 A first Tier-3 result: why the dark-to-baryon ratio is $\approx 1/6$

The peak heights leave one number for the membrane to explain: the dark-to-baryon ratio. If dark matter is the parent matter the membrane *rejects*—felt gravitationally but never crossing into our plasma—then ω_c/ω_b is fixed by the membrane pass-fraction $f = \omega_b/(\omega_b + \omega_c) \simeq 0.157 \approx 1/6$. A first calculation narrows where that number comes from (Figure 12.6). Model the membrane as a torsion barrier: the spin–spin repulsion that halts the

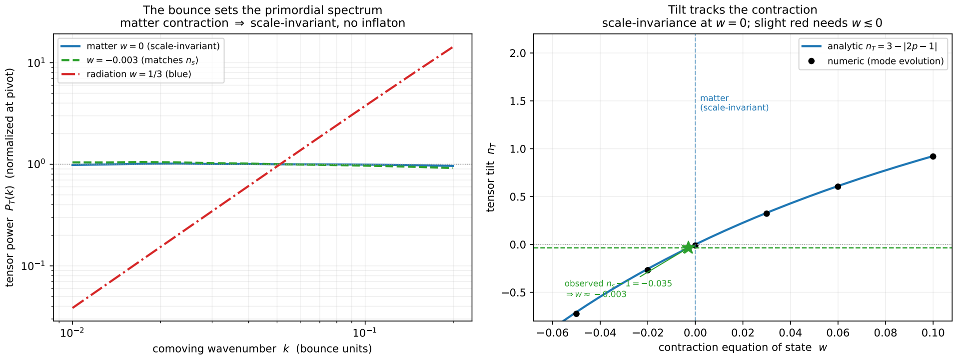

Figure 12.3: The primordial spectrum from the Einstein–Cartan bounce. *Left:* the tensor power spectrum is flat (scale-invariant) for a matter-dominated contraction and blue for radiation—scale invariance with no inflaton. *Right:* the tilt n_T versus contraction equation of state on the analytic curve; $w = 0$ is scale-invariant, and the observed tilt corresponds to $w \approx -0.003$. Reproducible as `figures/scripts/ch09_primordial.py`.

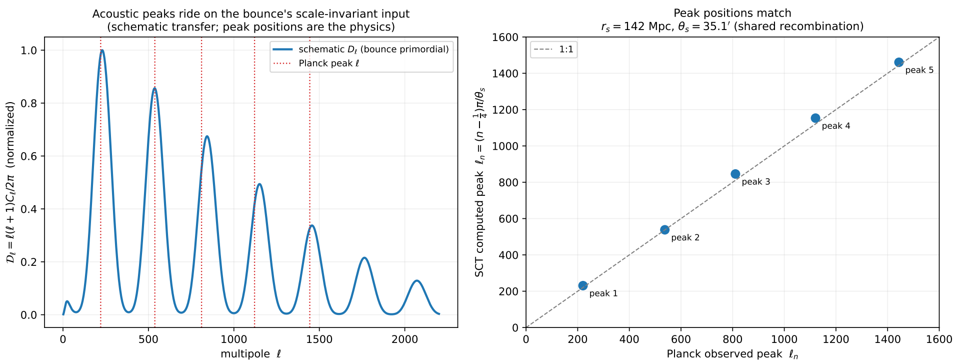

Figure 12.4: The acoustic peaks. The bounce's scale-invariant primordial input, carried through the standard post-bounce plasma, yields a sound horizon $r_s \simeq 142\,\mathrm{Mpc}$ and a peak series matching the observed Planck multipoles to a few percent. The peak *heights* are computed in Figure 12.5. Reproducible as `figures/scripts/ch09_acoustic.py`.

collapse is also a potential barrier Δ that infalling matter must clear, so the pass-fraction is a penetration factor in the single ratio $x = \Delta/T$ (barrier to bounce-thermal scale)—$f = e^{-x}$ for a Maxwell tail, $f = 1/(e^x + 1)$ for Fermi–Dirac occupation. Inverting, $f = 1/6$ corresponds to $x = \ln 6 \simeq 1.79$ (Boltzmann) or $\ln 5 \simeq 1.61$ (Fermi). The decisive point is the *anchor*: the membrane is by definition the surface where the torsion energy equals the matter energy (Chapter 2), so per particle the barrier and the thermal scale are comparable, $x \sim \mathcal{O}(1)$—not 10^{-9}, not 10^9. An order-unity barrier ($x \in [1, 2.5]$) yields f in the 0.08–0.37 decade, and the observed $1/6$ sits squarely inside, at $x \simeq 1.8$: the barrier ~80% above the mean thermal energy, exactly "torsion just dominating." This converts f from a free fit in $[0, 1]$ into an $\mathcal{O}(1)$ ratio the

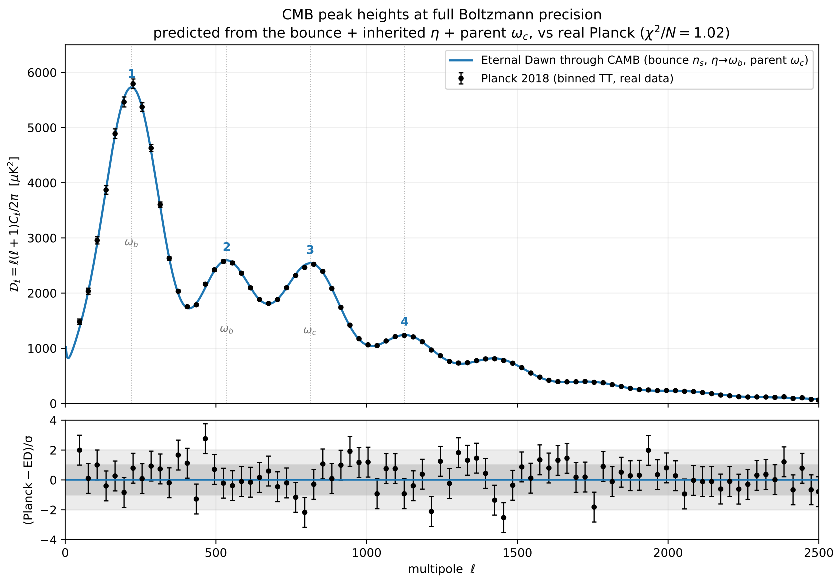

Figure 12.5: The peak *heights*, at full Boltzmann precision, against the real Planck 2018 binned TT spectrum. The standard recombination transfer (CAMB), fed an Eternal Dawn parameter set whose values are *sourced*—n_s from the bounce, ω_b from the inherited η, ω_c from the parent's projected mass—reproduces the measured spectrum at $\chi^2/N \simeq 1.0$ over 83 band powers (residuals, lower panel, scatter within $\pm 2\sigma$). The first peak's height fixes the baryon density (the odd/even ratio $D_1/D_2 \simeq 2.2$), the third peak the dark-matter density ($D_1/D_3 \simeq 2.3$); $\omega_c/\omega_b \simeq 5.4$ is the baryon fraction $f \approx 1/6$ the membrane filter explains as natural (§12.5). Reproducible as `figures/scripts/ch09_peak_heights.py`.

bounce supplies automatically—so $f \approx 1/6$ is *natural*, not tuned. What is not yet delivered is the precise $x \simeq 1.8$ rather than 1.0 or 2.5; pinning it to three digits needs the full inhomogeneous bounce profile, the remaining Tier-3 computation. Reproducible as `figures/scripts/ch08_membrane_filter.py`.

12.6 Tiers 3–5, and the frontier

The higher tiers are multi-year programs, sketched here as scope rather than results. **Tier 3 (chirality dynamics)** would turn the order-unity barrier estimate above (§12.5) into the precise f by computing the membrane's filter efficiency from the inhomogeneous Einstein–Cartan bounce profile. **Tier 4 (population statistics)** treats the supraverse as a stochastic process—nucleation, growth, death, recursion—and computes the equilibrium distribution whose typical observer Chapter 10 located, supplying the rigorous measure the Boltzmann-brain argument (Chapter 3) still owes. **Tier 5** feeds all of this into direct, quantitative comparisons with specific datasets. Further out sit the genuinely frontier tests: gravitational decoherence—the Diósi–Penrose

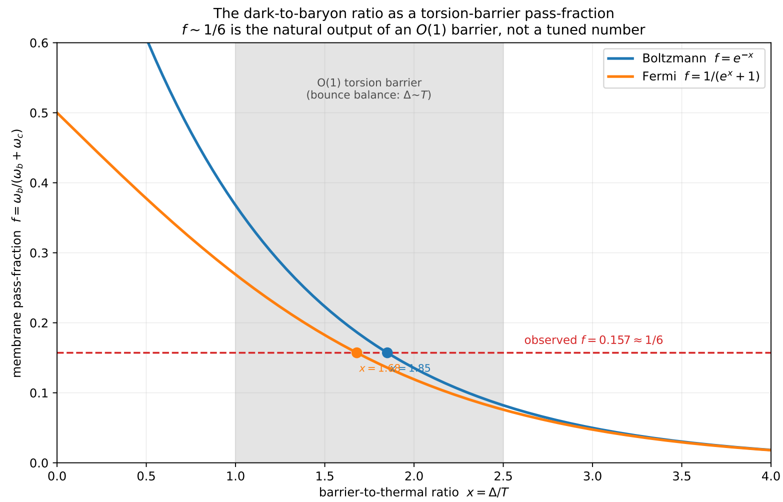

Figure 12.6: The dark-to-baryon ratio as a torsion-barrier pass-fraction. The membrane passes a fraction $f(x)$ of infalling matter over a torsion barrier of height $x = \Delta/T$ (Boltzmann and Fermi forms). Because the membrane is where torsion balances gravity, $x \sim \mathcal{O}(1)$ (shaded), which yields f in the 0.08–0.37 decade—bracketing the observed $f \simeq 0.157 \approx 1/6$ (dashed), reached at $x \simeq 1.8$. So $1/6$ is the generic output of an order-unity barrier, not a tuned number; the precise value awaits the full bounce profile. Reproducible as `figures/scripts/ch08_membrane_filter.py`.

collapse proposal [23, 55], among the testable collapse models [4]—as the laboratory-scale limb of the same gravity-separates-branches physics as the bounce, information-theoretic signatures of geometric dark matter, and any direct gravitational imprint of the parent. These are catalogued for completeness; none is near-term.

12.7 The minimum viable project, and the ethos

If one project should come first, it is **Tier 2 with a focus on the CMB power spectrum**: decisive either way, publishable either way, bounded at about a year, and within reach of standard numerical-relativity-plus-perturbation-theory expertise. The visualization track (Chapter 13) can proceed in parallel, needing no Tier-2 output to begin.

The governing principle is not to "prove" the framework by simulation but to make it *testable*: code open from the start, validated against analytic limits, checked for numerical convergence, and reproducible by anyone. That standard is already met by the results above—every figure and number in this book regenerates from the repository, and the simulation library ships with a passing test suite. Reproducible code lets others check, extend, and challenge the claims, which is the only thing that turns a worldview into science.

CHAPTER 13

Seeing almost Nothing

A picture of the supraverse runs into an awkward truth: there is almost nothing to draw. The instanton verdict of Chapter 3 is that universes are a *dilute* scatter—isolated specks in an overwhelmingly empty void—so an honest, literal portrait is a uniform grey field with, at most, a single infinitesimal dot. The void is the largest thing there is: nearly all of the supraverse is three-dimensional emptiness, and the universes are tiny pockets where a bounce has opened a fourth direction—time—inside. Visualisation here is the art of seeing almost nothing, and of magnifying the rare somethings just enough to make the structure legible without lying about how empty the whole is.

13.1 The challenge

Several difficulties compound. The supraverse is a four-dimensional manifold of non-trivial topology, and 4D does not render to a 2D page. Its structure spans every scale from subatomic to cosmic, and it is recursively nested—universes inside black holes inside universes. Each universe's internal time runs orthogonally to every other's, and the whole is a static block while each pocket within it evolves. And, above all, it is *dilute*: the honest scales are so lopsided that no linear map can show a universe and its neighbour on the same page. Standard scientific visualisation does not handle any of this; the framework needs its own conformal trick.

13.2 The gravity-scaled conformal mapping

The trick is to choose the one conformal rescaling that makes *gravitational* weight, not coordinate distance, set how big a thing looks. Multiply the metric by a factor tied to the local field,

$$g_{\mu\nu}(x) \;\to\; \Omega^2(x)\, g_{\mu\nu}(x), \qquad \Omega(x) \sim \big(G\rho(x)\big)^{-1/2},$$

the inverse square root of the local mass–energy density. High-density regions (black-hole interiors, bounce membranes) scale *up*; the low-density void scales *down*; and because the transformation is conformal it preserves causal structure—light cones map to light cones—so what is distorted is only apparent size, never the framework's physics. This is a cousin of the Penrose diagram, which also uses a conformal map to fit infinite spacetime on finite

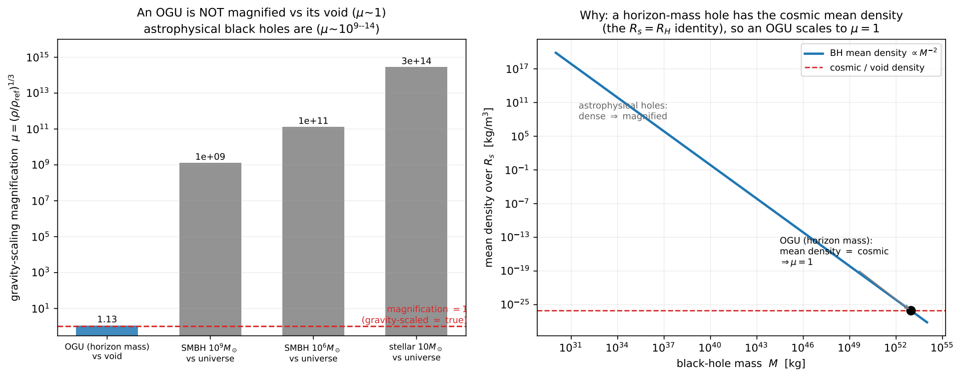

Figure 13.1: Gravity-scaling magnification $\mu = (\rho/\rho_{\text{ref}})^{1/3}$. *Left:* an OGU (horizon mass, mean density = cosmic) sits at $\mu \simeq 1$ —gravity-scaled coordinates equal true coordinates at the OGU/void level— while astrophysical black holes, far denser than their host, magnify by 10^9–10^{14}. *Right:* the black-hole mean density crosses the cosmic/void line precisely at the horizon mass, which is *why* an OGU scales to unity (the $R_s = R_H$ identity). Reproducible as `figures/scripts/ch10_gravity_scale.py`.

paper while keeping causal structure [56]; the difference is the choice of factor. A Penrose diagram is shaped to fit the page; this map is shaped by *gravity*, so it foregrounds physically meaningful structure—bounces, horizons, nested worlds—rather than coordinate artefacts.

13.3 Two tiers, and why an empty picture is the honest one

Made precise, the map magnifies a mass M relative to its true size by $\mu = (\rho/\rho_{\text{ref}})^{1/3}$, and only *ratios* of densities matter, so the reference cancels. The result is two-tier, and the boundary is exactly the identity that defines a universe (Chapter 12, Phase 0: a universe sits at its own Schwarzschild radius, $R_s = R_H$). An OGU is a horizon-mass black hole, so its mean density equals the cosmic critical density—essentially the void's dark-energy density—and therefore $\mu \simeq 1$: **an OGU is not magnified relative to its void.** At the OGU/void level, gravity-scaled coordinates simply *are* the true coordinates, and the picture is honestly near-empty in either system. One level down it is the reverse: an astrophysical black hole is far denser than its host, $\mu \sim 10^9$–10^{14}, so the map blows these otherwise-invisible compact objects up until the nested next-generation universes inside them become visible (Figure 13.1). The conformal factor earns its name exactly where it must—inside a world, revealing its children—while leaving the emptiness of the whole intact.

13.4 One number for the emptiness

How empty is the whole? Take the ratio of the distance between neighbouring OGUs to an OGU's own diameter. The separation is $\sim e^{I/4}$ de Sitter horizon radii and an OGU spans ~ 2 of them, so

$$\frac{\text{distance between universes}}{\text{universe diameter}} \sim \frac{e^{I/4}}{2} \sim 10^{2.7\times 10^{83}},$$

the same whether gravity-scaled or not, since a universe and its void share a density and neither is magnified relative to the other. The *exponent* alone, $\sim 2.7 \times 10^{83}$, already dwarfs every number in physics—the observable universe holds about 10^{80} atoms—so each universe is a lone speck whose nearest sibling lies a gap away with more zeros than a universe has atoms. That is "seeing almost nothing" made quantitative, and it is why the rendered supraverse must be gravity-scaled and illustrative: a literal linear frame would be uniform void with a single un-renderable dot.

13.5 The honest portrait, and the wallpaper

Figure 13.2 is the portrait the computation actually selects: a dominating, faintly-speckled void with universes as sparse, *round*, isolated islands— gravity wells with chirality-toned membranes and faint nested children—never a packed polygonal foam. (The foam was the legible *percolating* limit, the $I < I_{\text{crit}}$ case the instanton calculation rules out; it survives only as a contrast.) Roundness is real, not artistic licence: the vacuum has weight, so the void occupies genuine volume and presses the islands into discs rather than cells.

The book's cover is the same picture, stripped to its recursion: a uniform void; a balanced scatter of OGUs edged black for matter and white for antimatter; inside each, the same-coloured first-generation children; and inside those, the dots that are the black holes we actually watch form in our sky—the next generation down. It is a near-empty field with richly magnified nests, which is the truthful shape of a dilute, recursive, ever-dawning supraverse: almost nothing, lit here and there by a world still in its dawn.

13.6 Targets and implementation

Beyond the whole-supraverse portrait, the same gravity-scaled coordinates serve a handful of specific renders: a single universe with its nested black holes (zoom into any one and more universes appear, self-similarly); a family tree of lineages, chirality-coded; the slow time-evolution within one bubble; and the structure of a bounce membrane seen as the shared surface it is—one side a parent's horizon, the other a child's first sky. All of it is ordinary Python with the gravity-scaled transform applied before plotting, reproducible from the figure scripts alongside every other number in this book. The visualisation track needs no simulation output to begin (Chapter 12); it is the most accessible way into the program, and the quickest way to *see*, however magnified, a cosmos that is almost entirely nothing.

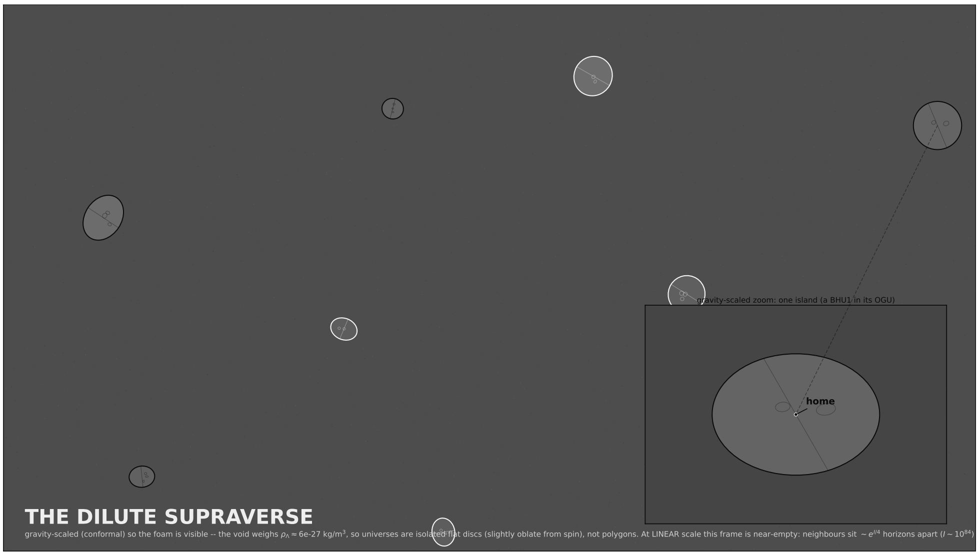

Figure 13.2: The dilute supraverse, gravity-scaled (sizes and spacings illustrative, not linear). The weighted void dominates; universes are sparse, *round*, isolated islands with chirality-toned membranes and faint nested children; the inset marks *home*. At a literal linear scale the frame would be near-empty —neighbours lie $\sim e^{I/4}$ horizons apart—so the void's weight keeps the islands round while the true gaps are astronomically larger than drawn. Reproducible as `figures/scripts/ch10_dilute.py`.

CHAPTER 14

The Scoreboard

A worldview earns its keep by explaining more from less, and by being honest about what it has not yet shown. This final chapter tallies both: a head-to-head against ΛCDM and the Standard Model, phenomenon by phenomenon; an explicit account of what would prove the framework wrong; and a sober statement of where each claim stands and what it still owes. The single line to carry through all of it is the one the book is named for. In the standard picture the Big Bang is an event, finished in the first instant, and the dark sector is a pair of unexplained ingredients added afterward to make the books balance. Here the Big Bang is a *surface*, still present at our horizon and still feeding, and the dark sector is not an addition but the live, measurable proof that our dawn is ongoing. That is the deepest divergence, and most of the scoreboard is its consequences.

14.1 Head to head

The framework's economy is the heart of the case. Where the standard account introduces a separate device for each puzzle—an inflaton for flatness, a Λ for acceleration, a particle for the missing mass, a baryogenesis mechanism for the matter excess, and a fine-tuned low-entropy start by fiat—Eternal Dawn derives all of them from four axioms and one bounce (Table 14.1).

Two cautions keep this honest. First, ΛCDM *works*: it fits the data it was built for with great precision, and Eternal Dawn must reproduce every one of those successes before its wider reach counts for anything—which, for the CMB peaks and the expansion history, it does by running the same post-bounce hot Big Bang (Chapter 12). Second, several of the right-hand entries are mechanisms whose *numbers* are computed only to first order—the dark-matter ratio is narrowed to the right decade but not pinned to three digits, the tensor sector gives a fork rather than a single r, the dark-energy history is still illustrative. The scoreboard is a map of explanatory scope, not a claim of victory.

14.2 Where each claim stands

The entries sort cleanly by how they will be settled.

Phenomenon	ΛCDM / Standard Model	Eternal Dawn
The beginning	A one-time Big Bang, finished in the first instant; a real initial singularity	An *ongoing* bounce at our horizon, still extruding matter; no singularity (torsion cap)
Low-entropy start	Postulated (the Past Hypothesis)	Forced by the bounce (Weyl $\to 0$): a Janus point, low entropy for free
Inflation	An inflaton field with a tuned potential	Torsion-driven post-bounce expansion; scale-invariant with no inflaton
Matter excess	Sakharov conditions + leptogenesis (manufactured)	Inherited from the founding seed, carried by conserved $B - L$
Dark matter	A new particle (so far undetected)	The parent's mass felt through the membrane—geometric, and ongoing
Dark energy	A cosmological constant Λ (eternal, unexplained)	The parent's ongoing accretion—a *rate*, hence evolving
CMB	A recombination relic at $z \sim 1100$	The parent's Hawking glow, redshifted by our expansion
Arrow of time	An unexplained low-entropy boundary	Manufactured at each bounce; the supraverse itself is timeless
Fine-tuning	Anthropics over an unspecified multiverse	A typical viable universe on a derived distribution
Hubble tension	Unresolved	Inhomogeneous (timescape) clock rate—no new parameter
Evolving dark energy	A tension with Λ	Predicted: accretion is a rate ($w_0 > -1, w_a < 0$)
Preferred axes	A fluke or a systematic	The parent's rotation axis, shared by the CMB and galaxy spin
Ultra-large structure	Awkward for the cosmological principle	Inherited lumpiness—the mark of a child, not a first universe
Lepton asymmetry	Assumed comparable to the baryon one	Predicted $\xi_\nu \sim \eta$ (sphaleron-locked); a large value forbidden
Boltzmann brains	An eternal-Λ future is a brain factory	Finite dark energy + an arrowless void: none ever form
Postulates	Inflaton, Λ, a dark-matter particle, baryogenesis, special initial conditions	Four axioms and one bounce

Table 14.1: Eternal Dawn versus the standard account, phenomenon by phenomenon. The recurring pattern is one device per puzzle on the left, one mechanism for all of them on the right—and, throughout, a beginning that is over versus one that is still under way.

Already suggestive (Chapter 10). Evolving dark energy of the predicted shape (DESI), a Hubble tension resolved with no new parameter, more ultra-large structure than a smooth start allows, and a doubly-anchored matter excess are all in hand—each consistent with the framework and a standing tension for ΛCDM. The single cleanest way to turn suggestion into evidence is the *shared axis*: a de-confounded galaxy-spin axis that agrees with the CMB "axis of evil" would be a uniquely-ours signature, on archival and near-future survey data.

The next decade decides (Chapter 11). The primordial tensors (the bounce ties the tilt to $n_T = n_s - 1$, breaking inflation's $n_T = -r/8$; LiteBIRD), the lepton asymmetry (the framework forbids the large value the data still allow), the dark-matter null, and large-scale-structure patterns are the near-future forks.

Computation: first results in hand, precision still owed (Chapter 12). Several of these are now computed. The CMB peak *heights* reproduce the real Planck spectrum at $\chi^2/N \simeq 1$; the dark-to-baryon ratio $f \approx 1/6$ falls out naturally as an order-unity torsion-barrier pass-fraction; the primordial tensors give a clean consistency-relation fork ($n_T = n_s - 1$). What remains is precision and depth: the peak heights at full inhomogeneous-bounce fidelity, the *exact* f from the membrane profile rather than its decade, and a rigorous supraverse measure. The results so far were each bounded at days to weeks; the remaining precision targets are the longer haul.

14.3 What would prove it wrong

A framework that could not fail would not be science. Eternal Dawn is falsifiable, and these are the clean ways it dies:

1. **Two cosmic axes that disagree.** If a de-confounded galaxy-spin axis and the CMB anomaly axis are both significant but point in *different* directions, one parent spin cannot have imprinted both. (A mere *absence* of an axis does not falsify it—most universes are low-spin, so a quiet sky is typical.)
2. **A dark-matter particle.** A confirmed fundamental dark-matter particle kills the geometric reading.
3. **Dark energy proven constant.** If DESI and successors pin $w = -1$ at higher precision, dark-energy-as-accretion fails.
4. **A large primordial lepton asymmetry.** A measured $\xi_\nu \gtrsim 10^{-2}$ has no source in the inherit-then-sphaleron-process picture.
5. **Inflationary tensor waves.** Primordial gravitational waves at inflation-predicted levels challenge the bounce-as-inflation-substitute.
6. **The anomalies dissolve.** If the "axis of evil" and its kin are proven foreground or statistical artefacts, the parent-rotation signatures are simply absent.
7. **The bounce cannot reproduce cosmology.** If detailed Einstein–Cartan bounce computations cannot return the CMB and structure formation, the framework is wrong in detail however appealing its shape.

14.4 Where we stand

Strip the scoreboard to one sentence and it reads: the same physics that forbids a singularity—continuous, conservative, with torsion capping the collapse—returns, with almost no further assumption, the bounce, the arrow of time, the matter excess, the dark sector, the microwave background, and a beginning that never ended. The Standard Model wrote, with great care, the biography of a parentless first universe—an OGU, with its finished Big Bang and its empty dark sector—and took it for our own. We are not that. We are a child, our dawn still breaking at the edge of the sky, and the dark matter and dark energy that the textbook adds by hand are the parts of that ongoing dawn we can already measure.

None of this makes the framework correct. It makes it *coherent, parsimonious, and testable*—a worldview that derives a great deal from very little, stakes itself on specific predictions, and tells you exactly how to prove it wrong. The work now is the finding out. The dawn, if the picture is right, is not behind us. It is still going on.

APPENDIX **A**

Open Questions

The framework is internally coherent but has many unresolved questions. This appendix catalogs them, both for honesty about the framework's current state and to guide future work.

A.1 Foundational physics

Q1. Precise value of bounce density Current estimates place $\rho_C \sim 10^{50}\,\mathrm{kg\,m^{-3}}$, but this varies by factors of 10–100 depending on coefficient choices in the Einstein–Cartan Lagrangian. A clean derivation pinning the value would be valuable.

Q2. Is bounce density universal? Does the bounce happen at the same density for different matter content (different fermion species, different bound states)? Or is the bounce condition matter-dependent?

Q3. How does the bounce interact with rotation? Most analyses assume spherical (Schwarzschild) bounces. Real astrophysical black holes are typically rotating (Kerr). The Kerr–Cartan bounce dynamics may differ qualitatively from Schwarzschild–Cartan and might preferentially produce specific chirality dynamics. Detailed analysis is needed.

Q4. Quantum gravity at the bounce density The framework treats Einstein–Cartan classically. At $\rho_C \sim 10^{50}\,\mathrm{kg\,m^{-3}}$, full quantum-gravity effects should be subdominant (still 46 orders of magnitude below Planck density), but they might matter for fine details. The full quantum-gravitational picture is unknown.

Q5. Baryon-number conservation At trans-electroweak temperatures, sphaleron processes violate baryon number. At higher temperatures (approaching the bounce density), more exotic violations might occur. How conserved is B across the bounce? This affects both the baryon-asymmetry mechanism and the dark-matter interpretation.

Q6. Beyond-Standard-Model physics The Standard Model is presumably an effective theory valid up to some scale. Above that scale, new physics (SUSY? extra dimensions? unknown new fields?) might dominate. Whether

Eternal Dawn is robust to such new physics, or whether it requires the SM to be the full story up to the bounce density, is unclear.

A.2 Bounce dynamics

Q7. The extrusion coefficient C and the remnant fraction The minimal Einstein–Cartan bounce is CPT-even: it neither flips charge nor filters a pre-existing population (Chapter 4), so the matter excess is supplied, not manufactured at the membrane. The live quantitative questions are then: (i) the anomaly + sphaleron coefficient C in the chiral-vortical extrusion estimate $\eta \sim C(\omega/T)$, which must reach ~ 41 at maximal spin to source the observed asymmetry at the OGU (Section 4.2); and (ii) the fraction of inherited net $B-L$ that survives the extrusion versus is washed out or diluted (Section 4.2, and Q16a). These set the asymmetry magnitude and decide whether lineages are chirally pure or mixed.

Q8. Bounce efficiency What fraction of the parent's mass–energy is transmitted to the baby's expanding phase? Some might remain in an intermediate region, some might be lost to gravitational waves, some might emerge as Hawking radiation in the parent's exterior. The efficiency factor is currently a free parameter.

Q9. Information flow through the bounce We have argued that information is preserved globally through entanglement structures across the membrane. The specific quantitative details—what gets entangled with what, how the Page curve is realized in bounce cosmology—are not worked out.

Q10. Persistent channel vs. severed connection We favor the position that the channel persists after the first bounce over the position that each bounce severs it. But the detailed geometry of a persistent channel that maintains across the parent black hole's lifetime needs rigorous treatment.

A.3 Supraverse structure

Q11. Equilibrium establishment We argue the supraverse is in thermodynamic equilibrium. How long does establishment take? Is the supraverse currently in equilibrium or still approaching it? What sets the equilibrium timescale?

Q12. OG bounce rate What is the rate of OG bounce nucleation per unit supraverse spacetime? This determines whether the supraverse contains many universes or few, and the dependence on supraverse parameters.

Q13. Universe size distribution shape We conjecture the distribution peaks near our observed size. What is the actual shape? How wide is the distribution? Are most observers in universes very close to our parameters, or is the distribution wide enough that observers could exist in significantly different universes?

Q14. Lineage interaction Do separate lineages (descended from different OG bounces) ever interact? If so, what happens at boundaries? Can matter-lineage and antimatter-lineage universes be near each other in the supraverse?

Q15. Initial conditions for the supraverse The framework requires the supraverse to be old enough for equilibrium establishment, but does not specify what the supraverse "started from." Was there an initial state, or is the supraverse genuinely eternal? This is partly a question about the meaning of time at the supraverse level.

Q15a. Do OG universes grow forever? Black holes have negative heat capacity, so above $M_{\text{crit}} = \hbar c^3/(8\pi G k_B T_{\text{bath}})$ a universe-hole accretes faster than it evaporates and grows. The growth-rate model of Section 4.3 gives a partial answer: the runaway is fast ($\dot{M} \sim M^2$, finite-time blow-up), so slow cooling cannot regulate it, but *depletion* of the local causal patch does—the hole eats its patch, the source shuts off, and the mass freezes at $M_{\text{final}} \sim$ the patch mass. So individual holes are finite, set by their environment, not unbounded. What remains open is the patch size itself (hence M_{final}) and the void bath temperature T_{bath} that fixes the growth time $\tau_{\text{gen}} \propto T_{\text{bath}}^{-3}$; the latter dials our generation depth (Section 4.3) and is entangled with the measure problem below and with Q15c.

Q15b. The measure over infinitely many OG universes Axiom A1 plus infinity implies infinitely many OG nucleations. Making "we are typical" meaningful requires a regulated measure over that infinite, possibly ever-growing ensemble—the measure problem shared with all multiverse frameworks, here sharpened by ongoing growth (Q15a). Unsolved.

Q15c. The cosmological constant in the void An infinite, eternally active quantum vacuum makes acute the question of why the void is nearly flat rather than de Sitter at the QFT cutoff. The dark-energy *evolution* is reattributed to parent accretion (Chapter 5), but the bare vacuum-energy problem [79] is untouched, arguably worsened.

A.4 Observational connection

Q16. CMB acoustic peaks ΛCDM predicts specific acoustic peaks in the CMB power spectrum that match observations beautifully. Can Eternal Dawn reproduce these from bounce dynamics? This is the central calculational test.

Q16a. Does η measure annihilation ash or Hawking inflow? The baryon-to-photon ratio $\eta = n_B/n_\gamma \simeq 6 \times 10^{-10}$ has two readings of its *denominator*, and they imply different baryogenesis bookkeeping. (i) *Annihilation ash*:

the photons are the baby's own thermal entropy, so η directly measures the extrusion's net-baryon-per-entropy, and the maximal-spin chiral-vortical estimate undershoots by ~ 41 (Chapter 4, Section 4.2). (ii) *Hawking inflow*: if the CMB is parent Hawking radiation (Chapter 5, Section 5.5), the photon denominator is largely *external*, the baby's internal asymmetry is decoupled from the observed η, and the small measured value is partly *dilution* by the Hawking flood—which can dissolve the factor of 41 without a large coefficient C. The *numerator* carries the mirror-image worry: a near-pure matter parent could inject a net $B-L$ far above 1 ppb, so matching $\eta \approx 0.6$ ppb requires either inheritance of the parent's own $\sim$1 ppb ratio (entropy-conserving extrusion) or dilution down to it (Section 4.2). The dilution half of this is now largely answered: the extruder entropy budget (Section 12.3), in both its homogeneous and shear-dominated channels, gives $D = 1$ to $\sim$1 part in 10^{11}, so the extrusion is adiabatic and $\eta_{\text{baby}} = \eta_{\text{parent}}$. What remains is (a) gravitational particle production at the bounce, the one entropy source not yet in the budget, and (b) the genuinely open question—the founder's *value*, why the original-generation seed sits at $\sim 6 \times 10^{-10}$ rather than 10^{-3} or 10^{-15}, which inheritance propagates but does not explain. It is entangled with Q5 (baryon-number conservation) and Q8 (bounce efficiency / entropy budget).

Q17. Hubble tension resolution We claim Eternal Dawn might resolve the Hubble tension through modified early-universe dynamics. The actual calculation showing this has not been done. If it cannot be done, the framework needs revision.

Q18. Dark-matter density profile Observed dark matter has specific clustering properties (NFW profiles, mass–velocity relations). Can these be derived from parent gravitational influence through the bounce membrane? Or do they imply something else?

Q19. Galaxy-rotation asymmetry consistency If parent rotation imprints galaxy-rotation asymmetry, the relationship should be specific and predictable. Shamir's observed asymmetry should match a particular parent rotation axis. Whether this consistency holds at higher statistical precision is unknown.

Q20. Specific BBN constraints Big Bang Nucleosynthesis predicts light-element abundances. Because Eternal Dawn shares the post-bounce radiation-dominated hot Big Bang with ΛCDM, BBN runs as usual, and its value of η *agrees with the CMB value to* $\sim$*1%*: deuterium $10^5\,\mathrm{D/H} \simeq 2.53$ gives $\eta_{10} \simeq 6.1$, matching the CMB-fixed ω_b (Chapter 10, Section 10.5). This is a concordance, not a tension. What remains open is whether the *approach* to the radiation era from the bounce (any entropy injection or expansion-rate change near ρ_C) perturbs the neutron freeze-out or the expansion rate at $z \sim 10^9$ at the percent level—a quantitative check the current treatment has not done.

A.5 Theoretical issues

Q20a. Does the bounce set particle masses? (speculative) At the bounce the spin–spin (Hehl–Datta) four-fermion interaction is strong: its energy per particle is comparable to the thermal scale—the same $\Delta/T \sim \mathcal{O}(1)$ that fixes the membrane pass-fraction $f \approx 1/6$ (§12.5). A strong four-fermion coupling is precisely the Nambu–Jona-Lasinio setting in which chiral symmetry breaks dynamically and a fermion *mass gap* opens—the same mechanism that generates the constituent quark mass in QCD. This raises a genuinely tempting possibility: that the torsion interaction at the bounce wall does not merely cap the density but *dynamically generates* mass scales, tying the membrane microphysics to the particle spectrum and so QCD and cosmology together. What is solid: the coupling is of NJL strength there, and an NJL gap equation is standard and cheap to solve. What is wildly open: whether this yields anything like the observed mass *eigenvalues* and hierarchy, rather than a single dynamically-generated scale. The cosmological bounce potential itself (Equation (2.1)) is a single turning-wall with no spectrum; any mass spectrum would have to come from the *field-theoretic* effective potential at the wall, not the scale-factor potential. This is flagged as speculation, not a result—a direction, not a claim.

Q21. Connection to string theory / loop quantum gravity How does Eternal Dawn relate to other proposed extensions of GR? Could string theory or LQG be ways of realizing the bounce in different formal frameworks? Are they competitors or complements? Loop quantum cosmology already replaces the initial singularity with a quantum bounce [2], by a different mechanism—a point of contact worth making precise.

Q22. Conformal cyclic cosmology relationship Penrose's CCC is also a non-singular cyclic cosmology [56]. The recursive cycle of Section 4.4 gives a concrete relationship: a heat-dead universe's smooth, conformally-flat de Sitter far future is a low-Weyl-entropy void, which in Eternal Dawn nucleates a *scatter* of fresh OGUs rather than handing off to a single conformally-rescaled successor. Eternal Dawn is, on this reading, "CCC made to branch." What remains open is whether the conformal matching at the dead-aeon/new-bounce boundary is as clean as CCC's—in particular whether the de Sitter nucleation rate (the birth rate β) and the Weyl-entropy reset can be made quantitative rather than schematic.

Q23. ER = EPR specific realization The bounce structure suggests a geometric realization of ER = EPR [45]. What is the precise mathematical statement? Can it be cleanly derived?

Q24. Holographic principle in the supraverse The holographic principle suggests that bulk physics can be encoded on boundaries. Does the supraverse have a holographic structure? What is the appropriate boundary? Is each bounce membrane a holographic surface for its enclosed baby?

Q25. Anthropic considerations We have implicitly used anthropic reasoning (observers exist in viable universes). Can this be made more precise? Does the framework predict where observers are in the supraverse population in a quantitatively testable way?

A.6 Philosophical / interpretive

Q26. What is the supraverse "embedded in"? Mathematically, the supraverse is a 4D manifold with non-trivial topology. Physically, what is its ontological status? Is it the totality of existence, or is there a meta-supraverse containing it?

Q27. Connection between local quantum measurements and bounce dynamics We argued that gravitational decoherence is the small-scale version of bounce dynamics. The precise relationship needs working out—is every quantum measurement literally a miniature bounce-like splitting? Or just analogous?

Q28. Implications for the meaning of physical constants If we are typical inhabitants of a supraverse population, our physical constants are typical for our class of universe. Does this mean physical constants are statistical features of the supraverse distribution, or are they fundamental? Or is the distinction meaningless?

Q29. Time at the supraverse level The supraverse is static (a block universe / timeless equilibrium), yet it "supports" nucleation and equilibrium establishment, which sound dynamical. The resolution (Chapter 3, Section 3.2): the supraverse needs *no* arrow—an arrow is a boundary condition, not a law, and the void has no low-entropy edge. "What is the stable state of the void?" is a timeless, equilibrium question; the supraverse *is* that stationary state, and "nucleation" is the statement that the equilibrium ensemble *contains* fluctuation-condensations, not that the void evolves toward equilibrium in some external time. The "dynamical-sounding" language is an artifact of describing a timeless structure from inside one of its internal-time condensations. What remains genuinely open is the deeper interpretive question of whether the supraverse-level "timelessness" should be read as a Wheeler–DeWitt-style frozen formalism or as a Barbour-style timeless configuration space—a question Eternal Dawn shares with all of quantum gravity, and does not need to settle to be predictive.

Q29a. Boltzmann brains and the simulation worry (largely resolved) On a naive de Sitter nucleation count, a Boltzmann brain ($\sim 10^{-10^{69}}$) is cheaper than an OG seed ($\sim 10^{-10^{84}}$), which would make typical observers fluctuations—and a cosmology whose only origin is a hand-tuned low-entropy Big Bang ($\sim 10^{-10^{123}}$, 56) is structurally the simulation hypothesis. The resolution (Chapter 3, Section 3.7): a brain—or a simulator—is an information processor and *requires* a thermodynamic arrow, which the stationary, arrowless, maximum-entropy void cannot supply [7]. Nucleation from the eigenstate yields only *generic* high-entropy blobs (viable OG seeds, which make their arrow at the forced low-entropy bounce), never the low-entropy *organized* state of a brain or a finished Big Bang. Interior brains are then starved by Eternal Dawn's *finite* dark energy: the recurrence time to fluctuate one ($\sim 10^{10^{69}}$ yr) vastly exceeds the parent-evaporation cutoff ($\sim 10^{136}$ yr), whereas eternal-Λ ΛCDM has no such cutoff. *What remains open* is a rigorous de Sitter measure: the qualitative asymmetry (blobs nucleate, minds presuppose an arrow) is robust, but "whether a stationary vacuum fluctuates" is unsettled across quantum gravity, and a fully quantitative Eternal Dawn observer measure does not yet exist. Reproducible as `sims/src/cartasis_sims/boltzmann_brains.py`.

A.7 Methodological

Q30. Falsifiability completeness The framework makes predictions (Chapter 10). Are these collectively sufficient to falsify the framework if it is wrong? Or could the framework be twisted to accommodate any observational result? A genuinely scientific framework should be unambiguously falsifiable. We have argued Eternal Dawn is, but this should be checked carefully.

Q31. Naturalness arguments We have argued Eternal Dawn is "natural" because it requires few postulates. But naturalness arguments have led physics astray before (e.g. naturalness arguments for SUSY have not worked out). Is Eternal Dawn's parsimony actually evidence for its truth, or just for its appeal?

Q32. Relationship to existing alternative cosmologies Many alternatives to ΛCDM exist: timescape (Wiltshire), inhomogeneous cosmology, modified gravity (MOND/MOG), cyclic universes. Eternal Dawn shares features with some of these. Detailed comparison would clarify what Eternal Dawn uniquely contributes and what it borrows.

A.8 Practical / programmatic

Q33. Publication strategy The danger is well understood: a single sweeping "theory of everything" dropped whole into the literature by an outsider is reflexively ignored, however careful, and deservedly so when it cannot be checked piecewise. The antidote is to *not* publish the synthesis first. The framework's strength is that it factors into self-contained, individually-checkable results, and those are the publishable units. A staged plan:

1. **Lead with the narrowest, most defensible quantitative result**, framed in the host community's own language, *not* as "a new cosmology." The strongest candidate is the de-confounded galaxy-spin/CMB *shared-axis* test (Chapter 10)—a self-contained statistics-on-archival-data paper that stands on its own merit whatever one thinks of the framework, and names in advance what would confirm or falsify a parent-spin imprint. A second candidate is the dark-energy-as-parent-accretion confrontation with DESI (w_0–w_a in a predicted wedge), again a falsifiable data paper.
2. **Then the mechanism papers**, each in the appropriate specialist venue and each citing only what it needs: the Einstein–Cartan bounce and its primordial spectrum (*Phys. Rev. D* / *JCAP*); the adiabatic-extrusion baryon-inheritance result; the supraverse population statistics and generation depth. None of these requires the reader to accept the whole edifice.
3. **Preprint everything on arXiv** (gr-qc, astro-ph.CO) with the full open-source repository attached, so every figure and number is reproducible from a command. Reproducibility is the outsider's single best credential.
4. **The monograph last**, as the synthesis for those already persuaded by the pieces—this book—explicitly labelled a research program, not a settled theory, with epistemic status flagged result by result.

The order matters more than the venue: establish falsifiable pieces that professionals can check and reuse *before* asking anyone to consider the whole. Honesty about what is established versus speculative (the discipline this book already imposes) is the other half of being taken seriously.

Q33a. What *not* to do Equally important, given how cosmology-adjacent "theories" usually fail: do not lead with popular-science framing or press; do not claim the framework is proven; do not present the synthesis to a general-physics audience before the pieces are refereed; do not overstate any single test (the shared-axis test, e.g., is confirmatory and only weakly disconfirmatory—Chapter 10). The CERN "black hole eating the Earth" episode is the cautionary tale: a misframed claim, amplified by journalists, costs years of credibility that careful work then has to buy back. Let the peer-reviewed pieces, and the reproducible code, do the talking.

Q34. Collaboration approach The framework benefits from multiple expertises: numerical relativity, cosmology, quantum field theory, philosophy of physics. The realistic path is to recruit one collaborator per *piece* rather than a believer in the whole—a large-scale-structure statistician for the shared-axis paper, an Einstein–Cartan bounce specialist for the spectrum, a DESI-adjacent cosmologist for the dark-energy confrontation. Each engages with a self-contained, falsifiable result in their own field; the synthesis is the author's to carry until the pieces have stood up independently.

Q35. Funding considerations This is theoretical/computational work that does not require large facilities. Modest computational resources, person-time. What funding mechanisms are appropriate?

A.9 Priority ordering

The most important open questions for near-term progress:

1. **Q16 (CMB acoustic peaks):** decisive test; must be addressed for the framework to be taken seriously.
2. **Q7 (flip vs. filter):** resolves chirality dynamics, enables the dark-matter prediction.
3. **Q1 (precise ρ_C):** enables quantitative bounce calculations.
4. **Q12–Q13 (supraverse statistics):** tests observer-placement predictions.
5. **Q17 (Hubble tension):** high-impact if it works.

Items 1–3 are calculable now with current Einstein–Cartan understanding plus standard numerical methods. Items 4–5 require more theoretical development.

APPENDIX **B**

The Lattice Targets

Parts II and III reduce the matter-and-force programme to a finite, ordered set of non-perturbative calculations—the ones the laptop cannot reach but a lattice gauge-theory group could. This appendix states them precisely enough to be picked up, with success criteria and a sense of cost, so that the framework's quantitative claims are consequences to compute rather than parameters to insert. The boundary is deliberately sharp: what analytic, few-body, and mean-field methods settled is in the chapters; what genuinely needs leadership-class computation is here.

L1 — Does the gravity–torsion vacuum confine? Put the Part I non-abelian connection on a Euclidean lattice and measure the Wilson-loop area law (or a dual order parameter for monopole condensation). *Success:* a clear area law and a string tension σ; *failure:* a perimeter law / screening, which kills the soliton-confinement picture cleanly. This is the cheapest target (pure gauge, no chiral fermions needed for the string) and the natural first step—it directly tests the dual- superconductor assumption of Section 8.3, and it can falsify the picture quickly. *Status:* the area law is in hand—a linear static potential $V(r) = \sigma r$ with a string tension $\sigma > 0$ measured from Wilson loops on pure $SU(3)$ (the colour sector forced by the Pauli label; `lattice/src/measure_potential`). What remains is the dual order parameter (monopole condensation) that would confirm the dual-superconductor *mechanism* behind the area law.

L2 — The string tension in physical units. Given L1, extract σ and test $\sigma \propto f_\pi^2$—the condensate scale that also sets the configurational mass. *Success:* a σ consistent with the chiral-soliton condensate scale, confirming "one substrate, mass and confinement both." *Status:* the relation is established analytically as the sharp form $\sigma = 2\pi v^2$: it is the Bogomolny (BPS) tension of a winding-1 Abrikosov–Nielsen–Olesen vortex of the condensate, with the off-BPS coefficient $\varepsilon(\beta)$, $\beta = (m_{\mathrm{scalar}}/m_{\mathrm{vector}})^2$, computed from the vortex BVP (`sims/.../l2_string_tension.py`; $\varepsilon(1) = 1$ verified, type-I $\varepsilon < 1$, type-II $\varepsilon > 1$). So $2\pi v^2$ is exact *at* BPS; whether the vacuum sits there is the dual-superconductor type, set by β (target L5). The non-perturbative cross-check $\sigma/f_\pi^2 = 2\pi$ on chiral-breaking configs (*not* the quenched pure-gauge σ, which has no f_π) is a downstream confirmation, not new physics.

L3 — The electroweak S of the walking sector. Compute the vector and axial current correlators of the strongly-coupled, near-conformal sector ($G_A = G_V$ from the Fierz, Section 8.2) and extract S. *Question:* does the walking pull S below the LEP bound ~ 0.1? *Success:* Part III lives; *failure:* retreat to Part II intact. This is the frontier target—near-conformal dynamics are notoriously hard, and the field has not settled the analogous technicolor question in forty years. The laptop work says the theory is favourably placed, not safe; only this decides it.

L4 — The fermion mass spectrum as overlap integrals. With lattice soliton wavefunctions in hand, compute the Yukawa couplings as overlap integrals—one fermion mass ratio, parameter-free, rather than inserted. *Success:* one ratio matching observation with no tuned input would be the result that makes the whole programme worth more than its coherence. The generation *shape* (the geometric ladder, the substrate steepening) is laptop-computed; the exact *ratios* are this target.

L5 — The particle-scale binding density versus ρ_C. Determine whether the density at which a particle-scale torsiton stabilises equals the cosmological bounce density ρ_C, or differs by a computable factor. Equivalently (in the language of Chapter 9), pin the transmuted scale $\Lambda = M_{\text{Pl}} \exp(-2\pi/b_0 g_T^2)$ by fixing g_T and b_0 from the propagating-torsion sector. *Success:* a derived particle-scale density and an explicit relation to ρ_C, closing the scale gap with a number rather than an order of magnitude.

The honest order is the reverse of glamour: L1 first—the cheapest, most decisive, most falsifiable—because it is the one that can kill the picture quickly. Only if the vacuum confines do L3 (the S make-or-break) and L4 (the masses) become worth their large cost. Every chapter result is reproducible from the repository as far as laptop computation reaches; this appendix marks exactly where that reach ends and the supercomputers begin.

APPENDIX **C**

The Torsiton Couplings from the Hehl–Datta Scale

Chapter 6 writes the torsiton as a bound state of one nonlinear Dirac field, but the self-consistent solver needs numbers: the scalar coupling g, the repulsive vector coupling g_v, the meson masses m_σ, m_ω, and the chiral-potential quartic λ. None of these is free. They all descend from the single Hehl–Datta coupling $G_{\text{HD}} = \frac{3}{4}\kappa$ by Nambu–Jona-Lasinio bosonisation [48], leaving exactly *one* dimensionful input, the cutoff Λ (the scale of Chapter 9 and lattice target L5). Everything else is a ratio. This appendix derives them; the algebra is reproduced symbolically in `sims/.../coupling_derivation.py` (SymPy) and supplied numerically by `sims/.../njl_couplings.py`.

C.1 The Fierz channels

The interaction is a single axial–axial term, $\mathcal{L}_{\text{int}} = -G_{\text{HD}}\,(\bar{\psi}\gamma_5\gamma^\mu\psi)(\bar{\psi}\gamma_5\gamma_\mu\psi)$. A Fierz transformation re-expresses it in the bilinear channels (Section 8.2); carried out with the Dirac matrices it gives

$$c_S = \tfrac{1}{4}, \quad c_P = -\tfrac{1}{4}, \quad c_V = \tfrac{1}{2}, \quad c_A = \tfrac{1}{2}, \quad c_T = 0,$$

so the two mean-field-relevant couplings are the scalar $G_S = \frac{1}{4}G_{\text{HD}}$ (attractive, the chiral condensate) and the vector $G_V = \frac{1}{2}G_{\text{HD}}$ (repulsive), with the *walking-favourable* equality $G_A = G_V$. The ratio that matters is

$$G_V/G_S = 2. \tag{C.1}$$

C.2 Bosonisation and the loop

Introduce auxiliary fields σ (scalar) and ω_μ (vector) and integrate out the fermions, whose dynamical mass is M. The one-loop integrals (4D Euclidean, sharp cutoff Λ, $x \equiv \Lambda^2/M^2$) are

$$I_0 = \frac{1}{16\pi^2}\Big[\Lambda^2 - M^2\ln(1+x)\Big], \qquad I_1 = \frac{1}{16\pi^2}\Big[\ln(1+x) - \frac{x}{1+x}\Big]. \tag{C.2}$$

I_0 carries the quadratic divergence, I_1 the logarithmic one. The chiral condensate is the filled Dirac sea,

$$\langle\bar{\psi}\psi\rangle = -4N_c M\, I_0, \tag{C.3}$$

and the gap equation $M = -2G_S\langle\bar{\psi}\psi\rangle$ fixes M against Λ and G_S—the one place the absolute scale enters. The pion decay constant is

$$f_\pi^2 = 4N_c M^2 I_1, \tag{C.4}$$

and we define the scalar Yukawa coupling by the Goldberger–Treiman relation $g \equiv M/f_\pi$, so $v = f_\pi = M/g$.

C.3 The scalar sector

Chiral NJL gives the σ at the $\bar{q}q$ threshold,

$$m_\sigma = 2M. \tag{C.5}$$

Matching the double-well $U(\sigma) = \frac{\lambda}{4}(\sigma^2 - v^2)^2$, whose curvature is $m_\sigma^2 = 2\lambda v^2$, fixes the quartic:

$$\boxed{\lambda = \frac{m_\sigma^2}{2v^2} = \frac{4M^2}{2v^2} = 2g^2} \tag{C.6}$$

—derived, not the value-8 placeholder the solver had been using.

C.4 The vector sector

The NJL vector pole gives $m_\omega = \sqrt{6}\,M$ (with $M \simeq 330$ MeV this is $\simeq 810$ MeV, the ρ), and the KSRF relation $m_\omega^2 = 2g_v^2 f_\pi^2$ fixes the repulsive coupling:

$$\boxed{m_\omega = \sqrt{6}\,M, \qquad g_v = \sqrt{\frac{m_\omega^2}{2v^2}} = \sqrt{3}\,g} \tag{C.7}$$

C.5 Consistency, and the resolution of the unbinding

The decisive check is that the bosonised masses and couplings reproduce Eq. (C.1):

$$\frac{G_V}{G_S} = \frac{g_v^2/m_\omega^2}{g^2/m_\sigma^2} = \frac{3g^2/6M^2}{g^2/4M^2} = 2\,\checkmark \tag{C.8}$$

This is why the soliton unbound when the repulsion was first switched on: the naive guess $g_v = g\sqrt{2}$ with $m_\omega = gv$ violates Eq. (C.8). The true coupling is *stronger* ($\sqrt{3}\,g$) but much *shorter-ranged* ($\sqrt{6}\,M \gg m_\sigma$); the two recombine to the Fierz ratio 2 in integrated strength, leaving the spatial *shape*—and the relativistic $G^2 \mp F^2$ balance of the first-order Dirac solver—to decide binding. At the derived, QCD-like couplings ($g \simeq 3$) the leading mean-field bag is *marginal*: it binds only when the colour levels are filled, and shallowly. The missing depth is the Dirac sea's own polarisation, Eq. (C.3), computed not in the uniform vacuum but in the soliton background—the chiral-quark-soliton mode sum, whose vacuum-subtracted condensate reinforces the bag. Carrying that sum out, de-doubled and regularised, is the calculation that decides whether the torsiton stands self-bound at its own couplings or hands the spectrum to the lattice (L4).

C.6 The hedgehog, and the torsiton mass

The single-channel bag was marginal; the chiral soliton binds the way the QCD nucleon does—through the *hedgehog*. The chiral field is rotated radially in isospin, $M_{\rm vac}\exp(i\gamma_5\,\boldsymbol{\tau}\cdot\hat{\mathbf{r}}\,\theta(r))$, locking spin and isospin into the conserved grand spin $K = \mathbf{J}+\mathbf{T}$. In the K=0 sector a quark level dives out of the continuum deep into the mass gap as the chiral angle winds from $\theta(0) = \pi$ to $\theta(\infty) = 0$ (one unit of winding, baryon number one): the valence quark, bound (`cartasis_sims.hedgehog`).

Minimising the leading-order energy of the unit-winding soliton over its size r_0,

$$E(r_0) = N_c\,E_{\rm val}(r_0) + \underbrace{\int 4\pi r^2\,dr\,\tfrac{1}{2}f_\pi^2\Big[\theta'^2 + \tfrac{2\sin^2\theta}{r^2}\Big]}_{E_2(r_0)\,\propto\,r_0}, \tag{C.9}$$

exposes the same Derrick balance that governs every three-dimensional chiral soliton. At the derived coupling $f_\pi = M/g$ with $g \simeq 3$ (so $f_\pi \simeq 0.33\,M$) the two-derivative term E_2 grows *linearly* in r_0, while the valence level dives only to $E_{\rm val} \simeq +0.2\,M$ at $r_0 \simeq 1.5\,M^{-1}$ and does not cross zero until $r_0 \simeq 2\,M^{-1}$, where $E_2 \simeq 8\,M$. There is therefore *no* self-bound minimum below the constituent sum: the infimum is the trivial limit $r_0 \to 0$, where $E_2 \to 0$ and the N_c valence quarks unbind to the gap edge, i.e. $E \to N_c M$, approached from above. This is not a failure of the framework; it is the textbook result. The two-derivative term alone never binds a nucleon below $N_c M$. The question is whether the higher orders of the same quark loop—the four-derivative (Skyrme) term, equivalently the full regularised Dirac-sea Casimir energy—bend the curve into a stable interior minimum. Computing that sea energy in full (next section) settles it: at these couplings they do *not*, and the soliton sits at the constituent sum.

The torsiton mass is therefore fixed near the constituent sum,

$$\boxed{M_{\rm torsiton} \;\simeq\; N_c\,M(\Lambda)} \tag{C.10}$$

with $M(\Lambda)$ set by the gap equation (C.3) from the single Hehl–Datta cutoff Λ—one dimensionful number for the whole spectrum. With $N_c = 3$ this is $M_{\rm torsiton} \simeq 3\,M$, the analogue of the proton sitting at three constituent quarks. The precise binding fraction (whether the sea cancellation leaves the mass a few per cent above or below $3M$) is the delicate, regularisation-sensitive piece that the non-perturbative lattice—target L4—settles; the mean-field calculation here fixes the *scale* and hands off the last few per cent. The minimisation and the Derrick collapse are reproduced in `cartasis_sims.hedgehog` (`minimize_b1_soliton`).

The full sea sum. Replacing the two-derivative estimate by the full regularised Dirac-sea energy means diagonalising the hedgehog Hamiltonian in every grand-spin sector $K = \mathbf{J}+\mathbf{T}$ and summing the vacuum-subtracted negative-energy levels with the proper-time cutoff. The apparatus is built in two stages. `cartasis_sims.chiral_quark_soliton` provides the grand-spin angular couplings (the matrix elements of $\boldsymbol{\tau}\cdot\hat{\mathbf{r}}$, verified by $(\boldsymbol{\tau}\cdot\hat{\mathbf{r}})^2 = 1$) and the Dirac Hamiltonian, whose free spectrum is parity-degenerate with no spurious in-gap level and whose $K=0$ sector reproduces the valence dive of `hedgehog`. On a finite-difference radial grid, though, the high-K centrifugal barrier is poorly resolved: the grand-spin sum crawls ($\sim K^{-1.2}$) and the subtracted energy drifts with the box. The cure—and the standard way CQSM is solved—is the Kahana–Ripka basis of free spherical-Bessel Dirac partial waves in a box (`cartasis_sims.kahana_ripka`), where the barrier and the r^l origin behaviour are exact, $\boldsymbol{\sigma}\cdot\mathbf{p}$ acts through Bessel recurrences, and the vacuum subtraction is basis-consistent. There the sum *converges* ($\Delta \sim 10^{-3}$ by $K_{\max} \sim 16$) and is box-stable, and the small-amplitude limit returns a clean gradient term with $f_\pi \simeq 0.39\,M$, consistent with Eq. (C.4).

With the sea converged, the B=1 mass $M_{\rm sol} = N_c\,\varepsilon_{\rm val} + E_{\rm sea}$ can be minimised. The outcome confirms Eq. (C.10) from an actual mode sum: $E_{\rm sea}$ is positive and grows with the soliton size while the valence level dives only to $\varepsilon_{\rm val} \simeq +0.2\,M$, so at the derived (weak, $f_\pi/M \simeq 0.4$) couplings the soliton has no deep interior minimum—it relaxes toward the point limit where $E_{\rm sea} \to 0$ and the N_c valence quarks sit at the gap edge, i.e. the constituent sum. Across cutoffs $\Lambda = 2.5$–$4\,M$ the infimum is robustly $N_c M$, approached from above (e.g. $M_{\rm sol} \simeq 3.6\,M$ at the most compact size for $\Lambda = 2.5\,M$). So, computed and converged, $M_{\rm torsiton} \simeq N_c M \simeq 3\,M$.

The self-consistent soliton, and a critical coupling. The minimisation above fixes $\theta(r)$ to a one-parameter family. Removing even that assumption—solving the model *fully* self-consistently, the chiral angle determined by the quark source itself, $\theta(r) = \operatorname{atan2}(M_p, M_s)$ with $(M_s, M_p) = -2G_S(S, P)$ the regularised scalar and pseudoscalar densities, iterated to a fixed point (`kahana_ripka.self_consistent_profile`)—turns the question of binding into a sharp threshold. For $\Lambda \gtrsim 2.2\,M$ (the weak, derived regime) the iteration *collapses* to the trivial vacuum, $\theta \to 0$: there is no soliton, and the B=1 sector is the unbound constituent sum $N_c M$. For $\Lambda \lesssim 2.2\,M$ (strong coupling) a *stable* torsiton soliton forms: the profile winds $\theta(0) = \pi \to 0$ over $\sim 2/M$, the valence binds at $\varepsilon_{\rm val} \simeq 0.7\,M$, and the mass settles at $M_{\rm torsiton} = N_c\varepsilon_{\rm val} + E_{\rm sea} \simeq 3.6\,M$, just above the constituent sum. The torsiton therefore exists as a chiral soliton precisely when propagating torsion (the Λ of lattice target L5) is strongly enough coupled; whether the physical Λ lies above or below the critical value is for the lattice (L4/L5) to decide. At this coupling only the valence orbital binds—no excited in-gap level—so higher torsiton generations are not radial orbitals of a single soliton here, but would be collective or grand-spin-excited solitons, a separate calculation.

The collective band. The hedgehog singles out a direction in the locked space–isospin frame, and rotating it costs almost nothing: those are the soliton's collective zero modes, and quantising them gives the torsiton its spin. Cranking—the soliton's linear response to an isorotation $\boldsymbol{\omega}\cdot\boldsymbol{\tau}$—fixes the

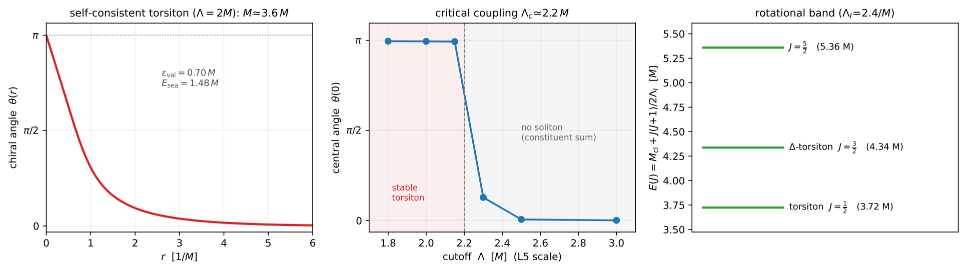

Figure C.1: The torsiton as a self-consistent chiral soliton. *Left:* the chiral angle $\theta(r)$ of the B=1 torsiton, solved fully self-consistently (the profile determined by the quark source, not assumed) in the Kahana–Ripka chiral-quark-soliton model at strong coupling $\Lambda = 2\,M$—a clean hedgehog winding $\theta(0) = \pi \to 0$ over $\sim 2/M$, with mass $M_{\text{torsiton}} = N_c\varepsilon_{\text{val}} + E_{\text{sea}} \simeq 3.6\,M \simeq N_c M$. *Centre:* the converged central angle $\theta(0)$ versus the cutoff Λ (the L5 scale), showing the critical coupling $\Lambda_c \simeq 2.2\,M$: below it a stable torsiton forms ($\theta(0) = \pi$); above it the soliton dissolves to the vacuum and the B=1 sector is the unbound constituent sum. *Right:* the collective isorotational band $E(J) = M_{\text{cl}} + J(J+1)/2\Lambda_I$ from cranking the soliton ($\Lambda_I \simeq 2.4/M$): the torsiton ($J=\frac{1}{2}$), the "Δ-torsiton" ($J=\frac{3}{2}$) 0.6 M above it, and $J=\frac{5}{2}$. Reproduced by `figures/scripts/appC_torsiton_profile.py`.

moment of inertia

$$\Lambda_I = \frac{N_c}{2} \sum_{m\,\text{unocc}} \frac{|\langle m|\tau_3|\text{val}\rangle|^2}{\varepsilon_m - \varepsilon_{\text{val}}}, \tag{C.11}$$

with the intermediate states the positive-energy $K=1$ levels (τ_3 carries grand spin 1, connecting the $K=0$ valence to $K=1$). The spectrum is then the rigid-top band $E(J) = M_{\text{cl}} + J(J+1)/2\Lambda_I$ with $J = I = \frac{1}{2}, \frac{3}{2}, \ldots$ (`kahana_ripka.rotational_band`). At $\Lambda = 2\,M$ the valence moment of inertia is $\Lambda_I \simeq 2.4/M$, so the torsiton ($J=\frac{1}{2}$) is followed by a "Δ-torsiton" ($J=\frac{3}{2}$) a splitting $3/2\Lambda_I \simeq 0.6\,M$ above—the same fraction of the constituent mass as the physical N–Δ gap (~ 0.7), a non-trivial check that the collective dynamics is right. (The Dirac sea adds a positive correction to Λ_I that lowers the splitting; the valence value is the leading estimate.) These are *spin* excitations—they change J. They are *not* the generations, which keep J and climb the *radial* ladder of the well (Section 6.5); the two must not be confused, for the reason the next section makes quantitative.

C.7 The collective quantum numbers: spin, isospin, charge

The cranked band carries more than energies—its quantum numbers are the torsiton's. The hedgehog is not an eigenstate of spin or isospin: it locks the two together (the grand spin $K = \mathbf{J} + \mathbf{T}$) and points in a single direction of that frame. Quantising the (iso)rotational zero modes projects it back onto good J and I, and for an odd number of colours the projection is forced to be *fermionic*—half-integer J. This is the Finkelstein–Rubinstein result that a soliton of a purely *bosonic* chiral field is a fermion: "matter is fermionic" is then not an input but a consequence of the winding. The allowed tower is $J = I = \frac{1}{2}, \frac{3}{2}, \ldots$, and the ground state

$$J = \tfrac{1}{2}, \qquad I = \tfrac{1}{2} \tag{C.12}$$

is a spin-$\frac{1}{2}$ fermion—consistent with the Dirac-s-wave reading of Chapter 6, but here *derived* from the soliton—that is simultaneously a *weak-isospin doublet*. The electric charges of its two members follow from Gell-Mann–Nishijima,

$$Q = I_3 + \tfrac{1}{2}Y, \qquad I_3 = \pm\tfrac{1}{2}, \tag{C.13}$$

with Y the soliton's conserved winding (hyper)charge—the same $B-L$-type number the bounce preserves (Chapter 7). The doublet members split by exactly one unit, $\Delta Q = 1$: the (ν, e) or (u, d) structure, recovered rather than assumed. (The *value* of Y—which doublet: lepton $Y = -1$ or quark $Y = \frac{1}{3}$—is the lepton/quark distinction tied to the colour label, carried in Chapter 8, not here.)

These collective numbers are *orthogonal* to the torsiton's other two labels, and it is worth tabulating which mechanism fixes which, because they are independent axes:

quantum number	origin	where
spin $J = \frac{1}{2}$	collective (iso)rotation; fermionic by Finkelstein–Rubinstein	here /
weak isospin $I = \frac{1}{2}$ (doublet)	collective isorotation, $J{=}I$ locked	here
electric charge $Q = I_3 + \frac{1}{2}Y$	Gell-Mann–Nishijima, $Y =$ winding	here
colour (triplet)	the three-valued Pauli label	Ch. 8
generation $n = 1, 2, 3$	radial rung of the well; mass $=$ overlap	Ch. 6

The decisive point is *why spin excitations and generations are different ladders*. A spin excitation ($J : \frac{1}{2} \to \frac{3}{2}$) costs an *additive* rigid-rotor energy $\sim 1/\Lambda_I \sim 0.4\, M$—the Δ-torsiton. A generation ($n : 1 \to 2$) keeps every charge and climbs the radial ladder, whose level spacing is also only $O(M)$; but its observable mass is *not* that level energy. By the configurational-mass result (Eq. 6.8) the mass is the *overlap* of the rung's wavefunction with the sharp condensate substrate, and overlaps are exponentially sensitive to the rung, $O_n \sim e^{-cn}$, so $m_n \sim |O_n|^2 \sim e^{-2cn}$: the arithmetic radial ladder becomes a *geometric* mass ladder spanning 10^3–10^4 (Section 6.5). Spin costs an additive $\sim M$; a generation costs a multiplicative $\sim 10^2$. The collective band of this appendix fixes the *spin and isospin* of the torsiton; the *generation hierarchy* is the overlap calculation of Chapter 6, and the CKM/PMNS mixings are the overlaps between rungs (Chapter 8). One torsiton, three independent ladders.

Reproducibility. The Fierz coefficients are in `cartasis_sims.fierz`; the loop integrals, condensate, and the relations (C.6), (C.7), (C.8) are derived symbolically in `cartasis_sims.coupling_derivation` and returned numerically by `cartasis_sims.njl_couplings`.

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